

Design of Complex Integrated Circuits

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1 Introduction

Integrated circuits (ICs) are ubiquitous in our modern society. The extremely high density of functions (computation, storage, signal processing, digitization, energy conversion) offered by microelectronics enables all the devices and infrastructure that we depend on: the control of factories and energy generation, computation in server farms, personal computers and mobile devices, communication by wires, optics, or wireless links, and sensing of the environment. All of these are driven by integrated circuits, implemented in various semiconductor technologies.

This course looks at integrated circuits from two complementary angles. In the first part, we discuss how ICs are **built**: the operation of the MOSFET, the CMOS manufacturing technology, the physical layout, packaging, production testing, and the economics of chip making. In the second part, we discuss how complex ICs are **designed**: design methodologies for digital and mixed-signal circuits, protection against electrostatic discharge and latch-up, robust design techniques, and the inner workings of circuit simulators.

It is assumed that readers are familiar with the contents of our Analog Circuit Design course; basic knowledge of digital and analog circuit design is a prerequisite for the later chapters.

! Important

All course material (source code of this document, Python code for all figures, etc.) is made publicly available on GitHub and shared under the Apache-2.0 license.

Please feel free to submit pull requests to fix typos or add content! If you want to discuss something that is not clear, please open an issue.

The production of this document would be impossible without these (and many more) great open-source software products: [VS Code](#), [Quarto](#), [Pandoc](#), [LaTeX](#), [Jupyter Notebook](#), [Python](#), [schemdraw](#), [Numpy](#), [Matplotlib](#), [Pandas](#), [Git](#), ...

1.1 How Small Is a Nanometer?

Modern CMOS technologies pattern structures with dimensions of a few nanometers. To develop a feeling for how small these dimensions really are, Figure 1 compares the feature sizes of integrated circuits with a few familiar objects: a human hair is roughly $70\text{ }\mu\text{ m}$ thick, a red blood cell has a diameter of about $8\text{ }\mu\text{ m}$, a virus measures around 150 nm , and the distance between two silicon atoms in the crystal lattice is only 0.24 nm . The minimum feature sizes of leading-edge CMOS processes are thus much smaller than a virus, and only about one order of magnitude away from the atomic spacing of the silicon crystal itself!

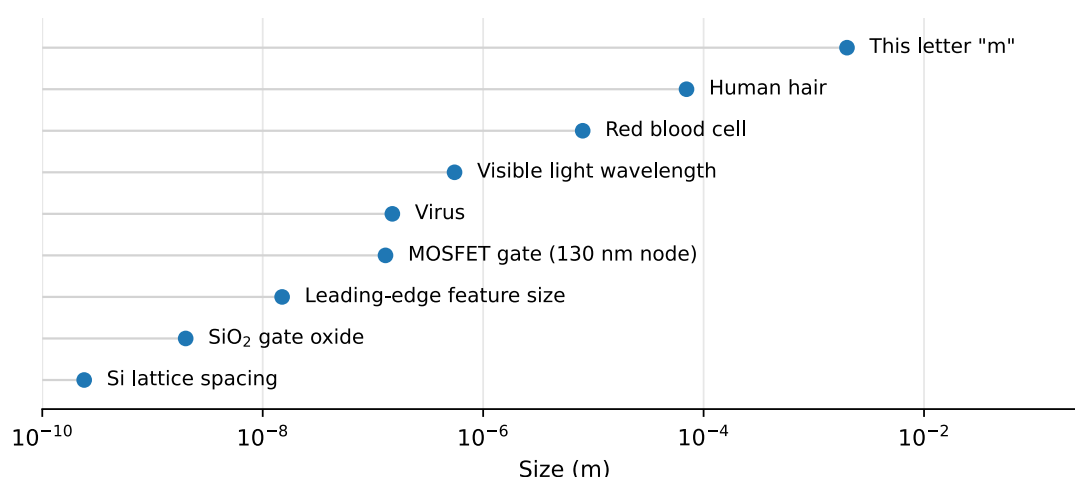


Figure 1: Length scales of integrated-circuit structures compared to familiar objects, on a logarithmic axis.

1.2 A Brief History of Semiconductors

The history of semiconductor electronics spans not even one century from the first observation of semiconductor rectification to today's chips with billions of transistors. Figure 2 summarizes the key milestones which we briefly discuss in the following.

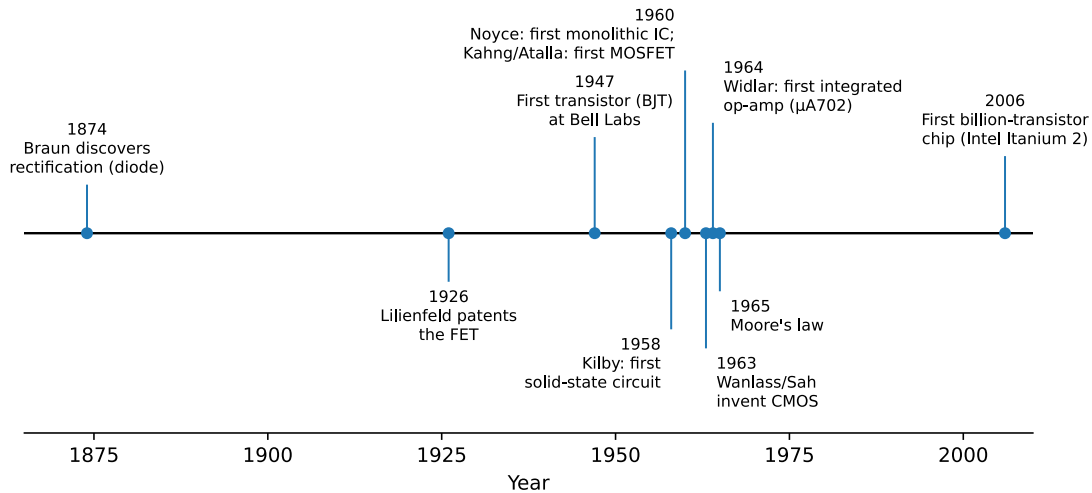


Figure 2: Timeline of key milestones in semiconductor history.

1874, the diode: Karl Ferdinand Braun discovers the **rectifying effect** of a particular type of mineral (galena, “Bleiglanz”). This effect is later exploited in radio receivers in the form of the crystal detector (the “cat’s whisker”), which is the first semiconductor electronic device.

1926, the field-effect transistor (FET): Julius Lilienfeld patents the **field-effect transistor**. Unfortunately, it takes several decades before such a device can actually be manufactured, as surface charges at the semiconductor interface cause malfunction of the early devices. Only in 1960 do Dawon Kahng and Mohamed Atalla finally demonstrate the metal-oxide-semiconductor field-effect transistor (MOSFET).

1947, the bipolar junction transistor (BJT): John Bardeen, William Shockley, and Walter Brattain demonstrate the first **semiconductor transistor** (using germanium) at Bell Labs. This was a big step towards the replacement of vacuum tubes. In the first point-contact transistor, the bottom piece of germanium formed the base terminal, hence the name “base.”

1958/1960, the first integrated circuit: After the first solid-state circuit demonstrated by Jack Kilby in 1958, Robert Noyce and his team at Fairchild build the first **monolithic integrated circuit** with pn-junction isolation in 1960: a flip-flop in transistor-resistor logic, integrated onto a wafer with 3 mm diameter.

1963, complementary MOS (CMOS): In an ISSCC paper, Frank Wanlass and Chih-Tang Sah of Fairchild Semiconductor show that logic built from p-channel and n-channel MOSFETs draws very little standby power—the invention of CMOS [1].

1964, analog integrated circuits: Robert J. Widlar, the “Father of Analog Integrated Circuits,” designs the first integrated operational amplifier, the μ A702.

1965, Moore’s law: Gordon E. Moore, based on only a handful of data points, makes a bold prediction about the growth of the number of components per integrated circuit, which would

eventually become known as “**Moore’s law**” [2]. Figure 3 shows the data available to Moore in 1965 together with his famous extrapolation: a doubling of the number of components per chip every year (later revised to a doubling every two years)—an exponential growth trend that the industry has followed for decades.

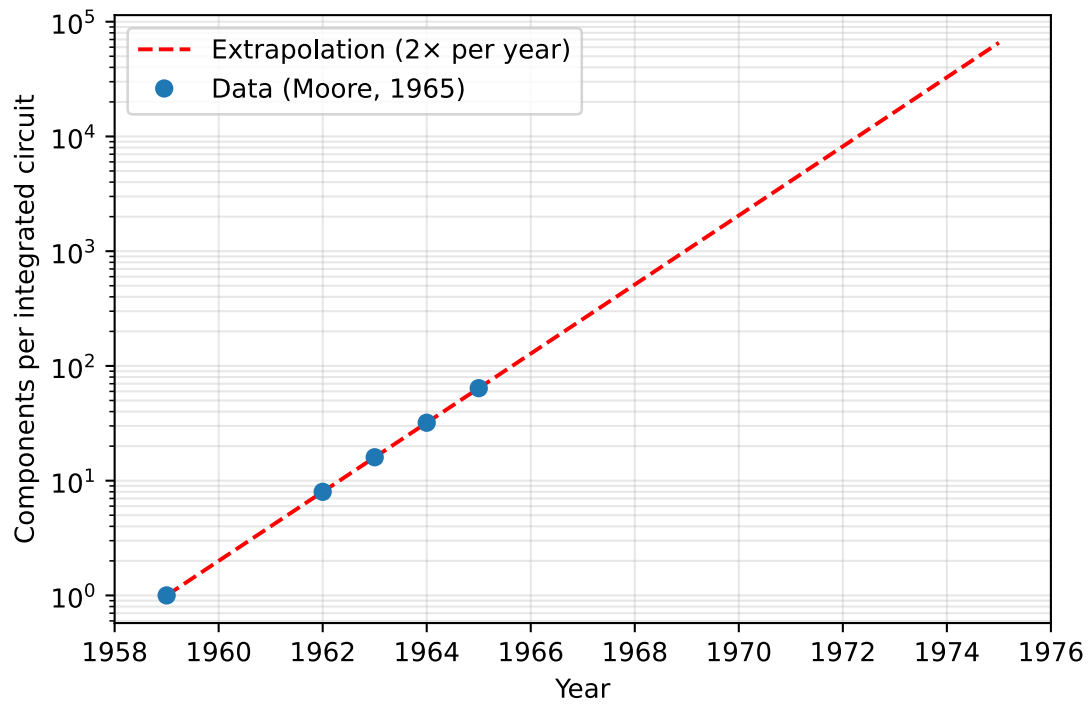


Figure 3: Moore’s law: the number of components per integrated circuit versus time. The blue points reproduce the data available to Moore in 1965; the dashed red line is his extrapolation of a doubling every year (data after Moore, ‘Cramming more components onto integrated circuits,’ Electronics, 1965).

2006, the first billion-transistor logic chip: Intel announces the first microprocessor with more than one billion transistors, the Intel Itanium 2. The chip is manufactured in a 90 nm CMOS process and contains 1.72 billion transistors on a die with an area of 374 mm². The first demonstrated chip with one billion transistors is a 1 Gbit DRAM presented by NEC in 1995, manufactured in 0.25 μm technology [3].

For an excellent, much more detailed account of semiconductor history, visit the Silicon Engine timeline of the Computer History Museum.

2 The MOSFET

This chapter on the metal-oxide-semiconductor field-effect transistor (MOSFET) is largely based on [4], [5], [6], which are also recommended as further reading. In contrast to our Analog Circuit Design course, where we treated the MOSFET essentially as a numerically-characterized black box, we will derive a simple, first-order model of the MOSFET from the physics of the device, and then discuss the most important second-order effects. The goal is to understand the basic operation of the MOSFET and its I/V characteristic, which will be used in later chapters for circuit design.

2.1 MOSFET Operation in a Nutshell

The basic MOSFET structure consists of a gate (terminal G) that is insulated from a doped semiconductor and controls the surface charge in the semiconductor channel below, between the source (terminal S) and the drain (terminal D). The source provides the charge carriers, and an electric field pulls them toward the drain (hence a semiconductor device based on drift current). The electric potential of the semiconductor bulk (terminal B) acts on the channel as well and is the fourth terminal of the MOSFET. Ideally, the effect of the bulk is negligible.

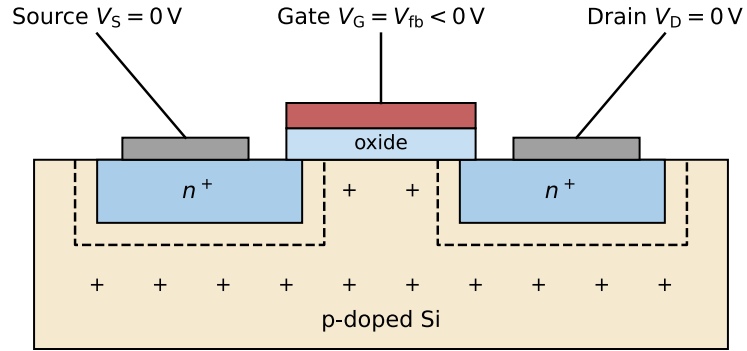
Two complementary flavors of the device exist: the n -type MOSFET (“NMOS”) has n^+ -doped source/drain regions on a p -type substrate and usually an n^{++} -doped gate, while the p -type MOSFET (“PMOS”) has p^+ -doped source/drain regions in an n -type well and usually a p^{++} -doped gate.

i Doping Notation

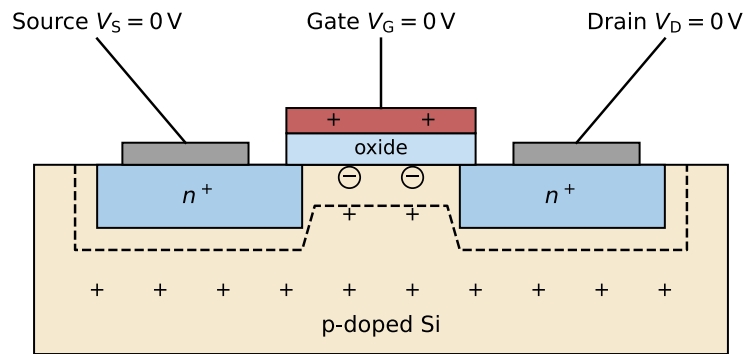
In the doping notation, $+$ indicates a higher doping concentration than the surrounding material, while $++$ indicates an even higher doping concentration. For example, n^+ -doped regions have a higher electron concentration than the n -type substrate, and n^{++} -doped gates have an even higher electron concentration than the n^+ -doped source/drain regions.

To understand the operation of the NMOS transistor, it is instructive to follow the conditions in the semiconductor while the gate voltage is swept upwards, as illustrated in Figure 4:

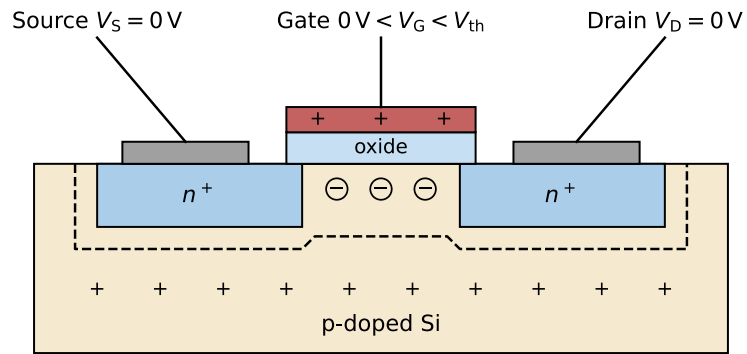
(a)



(b)



(c)



(d)

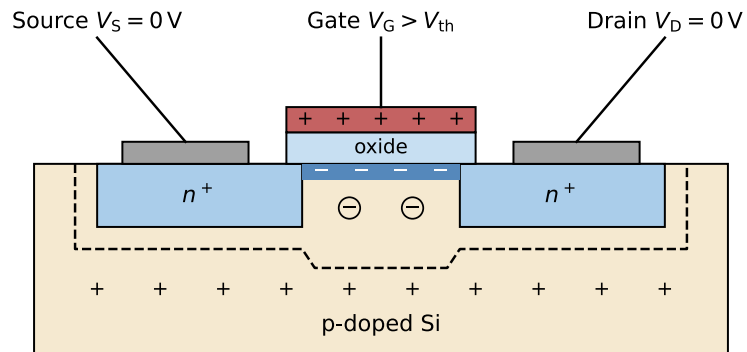
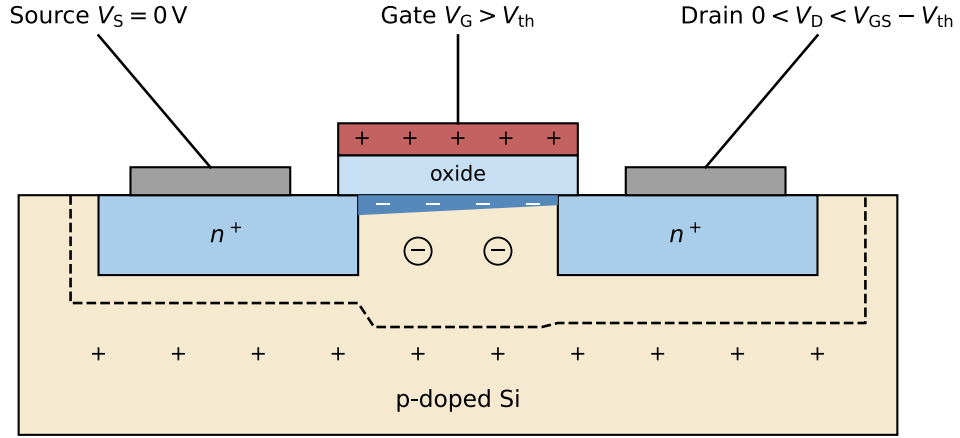


Figure 4: Cross section of an NMOS transistor for increasing gate voltage (drain and source grounded): (a) flatband condition, (b) gate at 0 V , (c) gate below the threshold voltage, and (d) inversion. The dashed line marks the depletion-region boundary, circled minus signs are ionized acceptors, plus signs are holes, and the dark blue layer in (d) is the electron inversion layer.

- **Flatband (a):** When the gate is at a specific negative voltage (the flatband voltage V_{fb}), and all other terminals are at 0 V, a depletion zone forms only around the drain and source junctions. No current can flow, since drain and source are n^+ -doped while the substrate is p -type, so the two junctions form back-to-back diodes.
- **Zero gate voltage (b):** When the gate voltage is increased to 0 V, a depletion zone also forms below the gate. The p -type acceptor ions in this region have captured an electron and are thus immobile, negatively charged sites (illustrated by the circled minus symbols)—still, no current can flow.
- **Below threshold (c):** When the gate voltage increases further (but stays below the threshold voltage), the depletion zone below the gate deepens, reaching a maximum depletion width W_{dep} when $V_{GS} = V_{th}$. There is still no conduction between drain and source, because there are no (or at least very few) free electrons in the channel region.
- **Inversion (d):** When the gate voltage rises above the threshold voltage V_{th} , **inversion** occurs: a thin layer of free electrons forms directly below the gate oxide, creating a conducting channel of electrons between drain and source. The source of the electrons in the inversion layer is the n^+ -doped source region, which is connected to ground. The electrons are pulled from the source into the channel by the positive gate voltage, and they can now flow to the drain if a drain voltage is applied.

Once the channel exists, the drain voltage determines the mode of conduction, as shown in Figure 5:

(a)



(b)

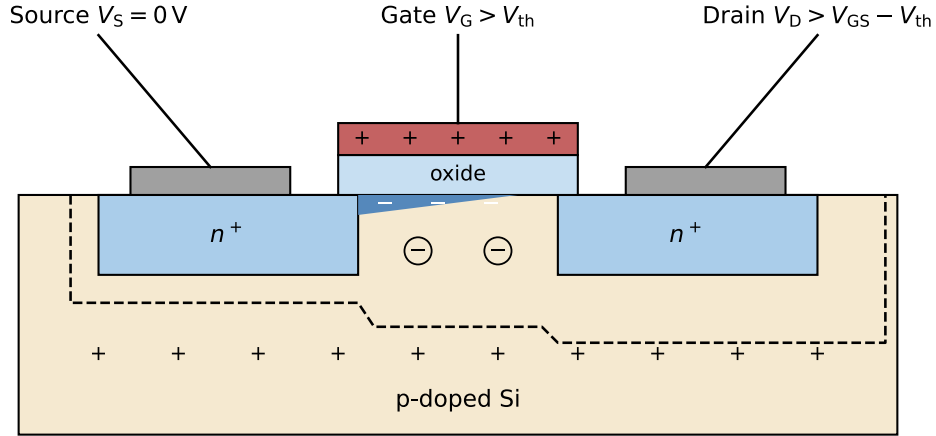


Figure 5: Cross section of an NMOS transistor in inversion for increasing drain voltage: (a) triode region, where the inversion layer extends from source to drain, and (b) saturation, where the channel is pinched off at the drain side.

- **Triode (a):** If a small voltage is applied between drain and source, a current flows, and the depletion zone around the drain widens. The drain current is a function of both V_{GS} and V_{DS} ; the device operates in the **triode** region. The MOSFET operates like a *voltage-controlled resistor* in this region.
- **Saturation (b):** If the drain-source voltage is increased beyond $V_{DS} > V_{GS} - V_{th}$, the channel is pinched off at the drain side, and the drain current is (to first order) no longer a function of V_{DS} , only of V_{GS} —the device is now in **saturation**. The MOSFET operates like a *voltage-controlled current source* in this region.

2.1.1 An Exemplary 130 nm Process Technology

Throughout this text we will use typical parameters of a 0.13 μm CMOS process technology, specifically IHP SG13 (process specification), for exemplary calculations. The key parameters are listed in Table 1.

Table 1: Typical parameters of the IHP SG13 130 nm CMOS technology

Parameter	Symbol	Typical value
Nominal supply voltage	V_{DD}	1.2 V (digital) / 1.5 V (analog)
Minimum gate length	L_{min}	130 nm
Oxide thickness	t_{ox}	2 nm
Threshold voltage NMOS PMOS	$V_{th,n} V_{th,p}$	0.5 V -0.47 V
Maximum voltage	$ V_{GS,max} , V_{DS,max} $	2 V
Gate capacitance	C'_{ox}	17.3 fF/ μm^2
Process constant $K_n K_p$	$\mu_n C'_{ox} \mu_p C'_{ox}$	280 $\mu A/V^2$ 130 $\mu A/V^2$

2.2 Basic I/V Characteristic of the MOSFET

2.2.1 The MOS Capacitor and the Threshold Voltage

If source, drain, and substrate are grounded ($V_S = V_D = V_B = 0$ V), and a positive voltage $V_{GS} = V_G$ is applied to the gate, then positive charge accumulates on the gate and negative charge in the p -type substrate, forming a depletion layer in the substrate.

If we assume a uniform charge density $\rho = -qN_a$ across this layer with N_a being the doping concentration of the p -type substrate (in a practical device, this charge density is not uniform), then we can use Poisson's equation. Note that with the boundary conditions assuming $\Phi(0) = 0$ and $d\Phi(0)/dy = 0$, $y = 0$ is the point where the depletion layer starts in the depth of the substrate, and $y = W_{dep}$ is the point where the depletion layer ends (the depletion width) at the substrate-oxide interface (ϵ_{si} is the permittivity of silicon):

$$\frac{d^2\Phi(y)}{dy^2} = -\frac{\rho(y)}{\epsilon_{si}} = \frac{qN_a}{\epsilon_{si}} \rightarrow \Phi(y) = \frac{qN_a}{2\epsilon_{si}}y^2 \quad (1)$$

The surface potential $\Phi_S = \Phi(W_{dep})$ at the oxide-silicon interface is then related to the depletion width W_{dep} by

$$W_{dep} = \sqrt{\frac{2\epsilon_{si}\Phi_S}{qN_a}}. \quad (2)$$

With W_{dep} we can also calculate the corresponding depletion capacitance (per unit area, indicated with the prime):

$$C'_{dep} = \frac{\epsilon_{si}}{W_{dep}} = \sqrt{\frac{qN_a\epsilon_{si}}{2\Phi_S}} \quad (3)$$

The charge per unit area Q'_{dep} in this region is (negative, because of the captured electrons)

$$Q'_{dep} = -qN_a W_{dep} = -\sqrt{2qN_a\epsilon_{si}\Phi_S} \quad (4)$$

The gate voltage is the result of the sum of the flatband voltage, the surface potential, and the voltage across the oxide

$$V_G = \Phi_{fb} + \Phi_S + V_{ox} = \Phi_{fb0} + \frac{-Q'_{ss}}{C'_{ox}} + \Phi_S + \frac{-Q'_{sub}}{C'_{ox}} \quad (5)$$

where Φ_{fb} is the work-function difference between the gate and the substrate material (the flatband voltage V_{fb} used in Figure 4). Note that the negative sign in the last term arises because voltage and charge are taken from different electrodes of the gate capacitor: the voltage V_{ox} at the top, and the substrate charge Q'_{sub} as well as the interface charge Q'_{ss} at the bottom.

An interface charge Q'_{ss} can be present at the SiO_2 -Si interface due to crystal discontinuities. This interface charge can be accounted for by adapting the flatband voltage Φ_{fb0} to Φ_{fb} (going forward, we will just use Φ_{fb}).

Using Equation 2 to express $\Phi_S = qN_a W_{dep}^2 / (2\varepsilon_{si})$ and Equation 4 for $Q'_{sub} = Q'_{dep}$, Equation 5 can be written as

$$V_G = \Phi_{fb} + \frac{qN_a W_{dep}^2}{2\varepsilon_{si}} + \frac{qN_a W_{dep}}{C'_{ox}} \quad (6)$$

From Equation 6, the depletion width W_{dep} can be calculated as a function of V_G , and from this $\Phi_S = f(V_G)$ using Equation 2.

If the surface potential Φ_S reaches a critical level of twice the Fermi potential Φ_B , then **inversion** occurs: a thin layer of electrons is induced directly under the oxide. The gate voltage that is minimally required to produce this inversion layer is called the **threshold voltage** $V_{th} = V_G(\Phi_S = 2\Phi_B)$.

$$V_{th} = \Phi_{fb} + 2\Phi_B + \frac{-Q'_{dep}}{C'_{ox}} = \Phi_{fb} + 2\Phi_B + \frac{\sqrt{2qN_a\varepsilon_{si}} \cdot 2\Phi_B}{C'_{ox}} \quad (7)$$

with the Fermi potential

$$\Phi_B = \frac{kT}{q} \ln \frac{N_a}{n_i}. \quad (8)$$

Figure 6 illustrates the flatband and the threshold condition in terms of the energy bands of the MOS structure.

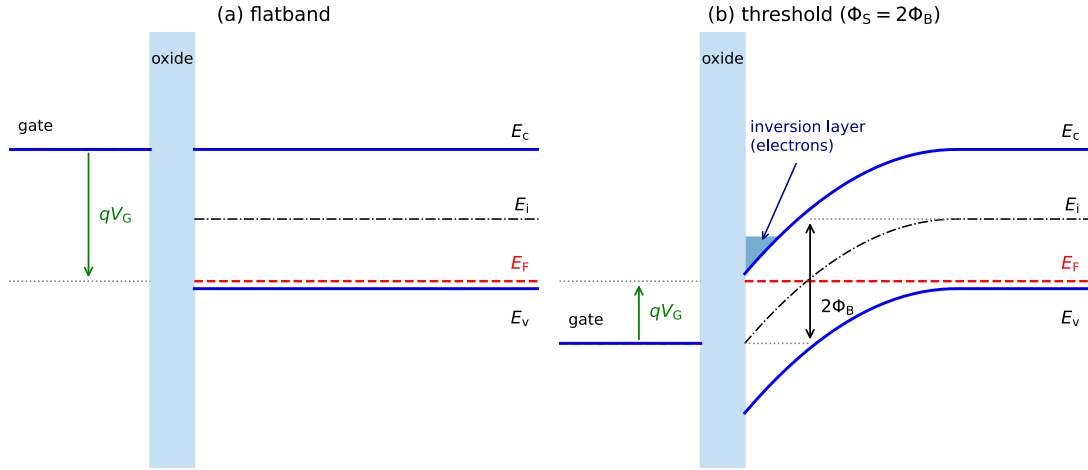


Figure 6: Simplified energy-band diagram of the MOS structure (n^{++} -poly gate, oxide, p -substrate): (a) at the flatband condition, where the bands in the silicon are flat, and (b) at the threshold condition, where the bands at the surface are bent down by $2\Phi_B$ and inversion sets in (band bending shown qualitatively). The green arrow marks the applied gate voltage V_G , which shifts the gate Fermi level by $-qV_G$ against the bulk Fermi level; in (a) $V_G = V_{FB} < 0$, in (b) $V_G = V_{th} > 0$. The blue shading in (b) marks the inversion layer of electrons at the surface.

i Note 1: Example Calculation of the Threshold Voltage

Let us assume a substrate doping level of $N_a = 3 \times 10^{18} \text{ cm}^{-3}$. The Fermi potential can be calculated with Equation 8 (using the intrinsic carrier concentration of silicon $n_i = 1.0 \times 10^{10} \text{ cm}^{-3}$ at room temperature $T = 300 \text{ K}$):

$$\Phi_B = \frac{kT}{q} \ln \frac{N_a}{n_i} = 0.50 \text{ V}$$

We further use $C'_{ox} = \epsilon_{ox}/t_{ox} = 17.3 \text{ fF}/\mu\text{m}^2$ for a $t_{ox} = 2 \text{ nm}$ gate oxide. The flatband voltage for an n^+ -polysilicon gate on the p -substrate with doping N_a is (with the silicon band-gap voltage $\Phi_{bg} = 1.12 \text{ V}$)

$$\Phi_{fb} = -\frac{\Phi_{bg}}{2} - \Phi_B = -1.06 \text{ V}$$

The resulting threshold voltage is thus (*beware of units!*)

$$V_{th} = \Phi_{fb} + 2\Phi_B + \frac{\sqrt{2qN_a\epsilon_{si} \cdot 2\Phi_B}}{C'_{ox}} = 0.53 \text{ V}$$

Using these values, we can also calculate the depletion width W_{dep} in inversion, and the corresponding C'_{dep} (compare W_{dep} to t_{ox} , and C'_{dep} to C'_{ox}):

$$W_{dep} = \sqrt{\frac{2\epsilon_{si} \cdot 2\Phi_B}{qN_a}} = 21.0 \text{ nm} \quad \text{and} \quad C'_{dep} = \frac{\epsilon_{si}}{W_{dep}} = 5.0 \text{ fF}/\mu\text{m}^2$$

Using Equation 6 and Equation 2 we can now visualize the depletion width W_{dep} and the surface potential Φ_S as a function of the gate potential V_G , shown in Figure 7. Note how the surface potential saturates at $2\Phi_B$ once inversion is reached: any additional gate charge is then mirrored by electrons in the inversion layer rather than by a deeper depletion region.

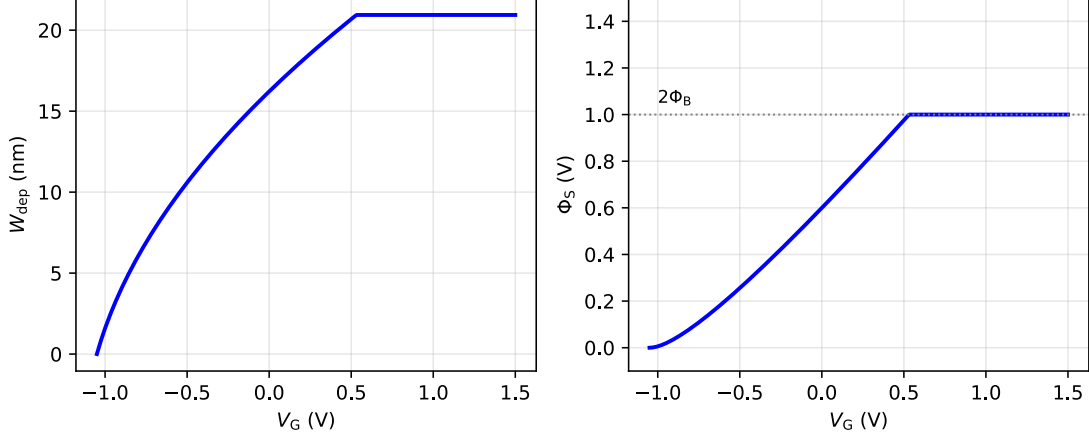


Figure 7: Depletion width W_{dep} (left) and surface potential Φ_S (right) as a function of the gate potential V_G , calculated with the depletion approximation for $N_a = 3 \times 10^{18} \text{ cm}^{-3}$, $t_{\text{ox}} = 2 \text{ nm}$, and $\Phi_{\text{fb}} = -1.05 \text{ V}$. Once inversion is reached at $\Phi_S = 2\Phi_B$ (at $V_G = V_{\text{th}} \approx 0.5 \text{ V}$), both quantities saturate.

2.2.2 Derivation of the I/V Characteristic

If we now apply a small V_{DS} , then a sufficiently large $V_{\text{GS}} > V_{\text{th}}$ builds up the inversion charge $Q'_{\text{inv}} = C'_{\text{ox}}(V_{\text{GS}} - V_{\text{th}})$ in the channel below the gate, which is then moved from source to drain by the electric field caused by V_{DS} , resulting in the drift current I_D (μ_n is the electron mobility):

$$I_D = WQ'_{\text{inv}}v = WQ'_{\text{inv}}\mu_n E = WQ'_{\text{inv}}\mu_n \frac{V_{\text{DS}}}{L} = \mu_n C'_{\text{ox}} \frac{W}{L} (V_{\text{GS}} - V_{\text{th}}) V_{\text{DS}} \quad (9)$$

If V_{DS} increases further, the channel charge $Q'_{\text{inv}}(x)$ becomes a function of the position in the channel, because the drain current causes a potential gradient dV_{cs}/dx along the channel (we assume that V_{DS} is small enough so that the whole channel is still inverted). $x = 0$ is the source end of the channel, and $x = L$ is the drain end.

$$I_D = WQ'_{\text{inv}}(x)v = WC'_{\text{ox}}(V_{\text{GS}} - V_{\text{cs}} - V_{\text{th}})\mu_n \frac{dV_{\text{cs}}}{dx}$$

Integrating along the channel from source to drain,

$$\int_0^L I_D dx = \mu_n C'_{\text{ox}} W \int_0^{V_{\text{DS}}} (V_{\text{GS}} - V_{\text{cs}} - V_{\text{th}}) dV_{\text{cs}},$$

yields the well-known **square-law** characteristic

$$I_D = \mu_n C'_{\text{ox}} \frac{W}{L} \left[(V_{\text{GS}} - V_{\text{th}}) V_{\text{DS}} - \frac{1}{2} V_{\text{DS}}^2 \right]. \quad (10)$$

Note that this simple derivation of the MOSFET I/V characteristic is only valid for long-channel devices ($L > 1 \mu\text{m}$) and does not account for second-order effects such as channel-length modulation, body effect, velocity saturation, mobility degradation with the vertical field, and subthreshold conduction. These effects will be discussed in the next section.

2.2.3 Triode Region

Equation 10 is valid only for classical, long-channel ($L > 1 \mu\text{m}$) devices, and only as long as $V_{DS} \leq V_{GS} - V_{th} = V_{od}$; this range of operation is the so-called **triode region**. The **overdrive voltage**

$$V_{od} = V_{GS} - V_{th} \quad (11)$$

is a frequently used quantity. The ratio W/L is a dimensionless quantity called the **aspect ratio** of the MOSFET—note that only the relative sizing of W and L impacts the transistor's I/V characteristic!

For $V_{DS} \ll V_{GS} - V_{th}$, we can simplify Equation 10 and see that the MOSFET behaves like a voltage-controlled resistor:

$$I_D \approx \mu_n C'_{ox} \frac{W}{L} (V_{GS} - V_{th}) V_{DS} \rightarrow g_{ds} = \frac{1}{r_o} = \frac{\partial I_D}{\partial V_{DS}} = \mu_n C'_{ox} \frac{W}{L} (V_{GS} - V_{th}) \quad (12)$$

Figure 8 shows the resulting output characteristic in the triode region for three different gate voltages.

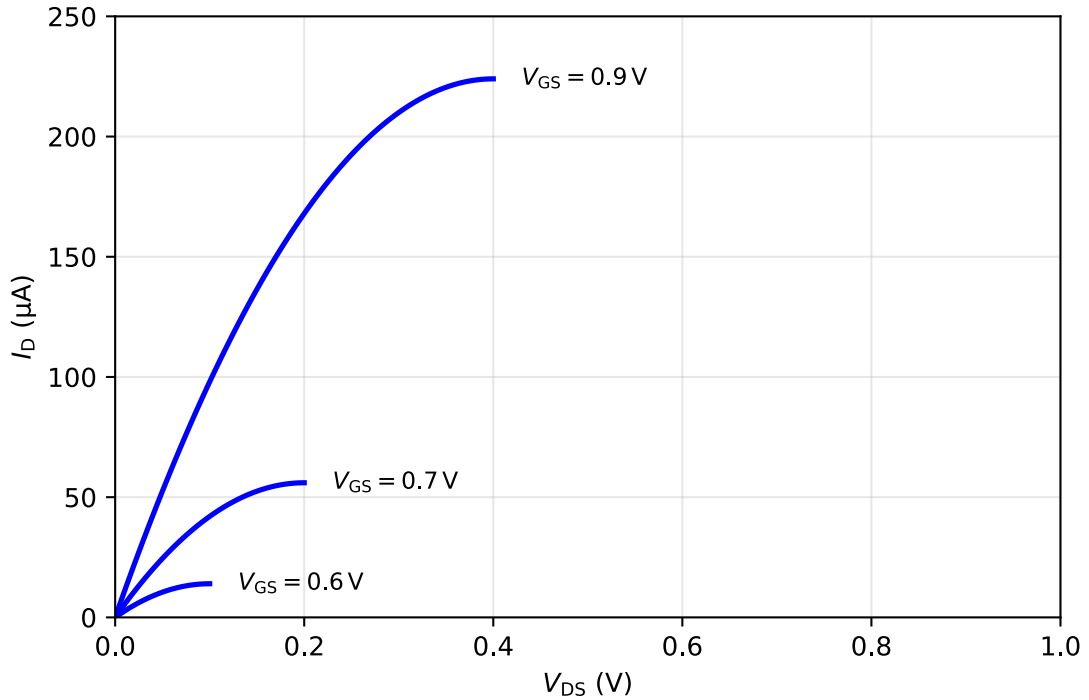


Figure 8: NMOS output characteristic in the triode region according to the square-law model ($\mu_n C'_{ox} = 280 \mu\text{A}/\text{V}^2$, $W/L = 10$, $V_{th} = 0.5 \text{V}$). Each curve ends at the edge of saturation, $V_{DS} = V_{GS} - V_{th}$.

2.2.4 Saturation Region

If $V_{DS} > V_{GS} - V_{th}$, the assumption of a completely inverted channel leading up to Equation 10 is no longer valid. The channel gets **pinched off** at the drain side; this happens at $V_{DS,sat} = V_{GS} - V_{th}$, leading to a maximum (saturated) value of the drain current. Substituting $V_{DS} = V_{DS,sat}$ into Equation 10 gives the drain current in the **saturation region**:

$$I_D = I_{DS,sat} = \frac{1}{2} \mu_n C'_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 \quad (13)$$

In the saturation region, the MOSFET behaves like a voltage-controlled current source, so it is useful to define the transconductance

$$g_m = \frac{\partial I_D}{\partial V_{GS}},$$

which can be expressed in various forms:

$$g_m = \mu_n C'_{ox} \frac{W}{L} (V_{GS} - V_{th}) = \sqrt{2 \mu_n C'_{ox} \frac{W}{L} I_D} = \frac{2I_D}{V_{GS} - V_{th}} \quad (14)$$

We see that g_m is a linear function of the overdrive $V_{od} = V_{GS} - V_{th}$ (leading to a quadratic rise of the drain current), or a square-root function of I_D , or simply the drain current divided by half the overdrive voltage. Note that for a BJT $g_m = I_C/V_T$, so the MOSFET offers more degrees of freedom. Figure 9 shows the complete output characteristic including the boundary between triode and saturation.

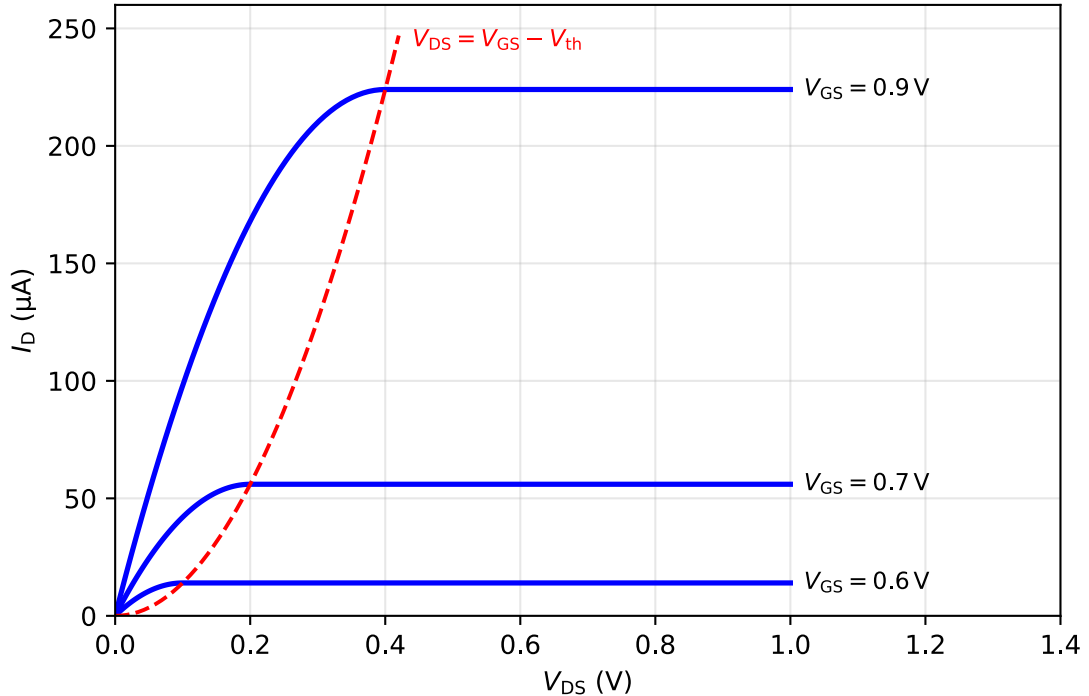


Figure 9: Complete NMOS output characteristic according to the square-law model ($\mu_n C'_{ox} = 280 \mu A/V^2$, $W/L = 10$, $V_{th} = 0.5 V$). The dashed red line marks the boundary $V_{DS} = V_{GS} - V_{th}$ between the triode and the saturation region.

2.2.5 Complementary MOS (CMOS)

In CMOS technologies, both NMOS and PMOS devices are needed and fabricated on the same wafer, as shown in Figure 10. Usually, the PMOS is located in a local n -well, and this n -well is tied to the most positive supply voltage (V_{DD}), while the p -substrate is tied to ground (V_{SS}). The substrate contacts are realized with p^+ -diffusions in the p -substrate (forming an ohmic contact), and the well contacts are realized with n^+ -diffusions in the n -well.

For increased isolation (especially for analog applications), a triple-well (or “deep n -well,” DNW) option is sometimes available; the NMOS is then located in a local p -well which is fully enclosed by an n -well and thus isolated from the common substrate.

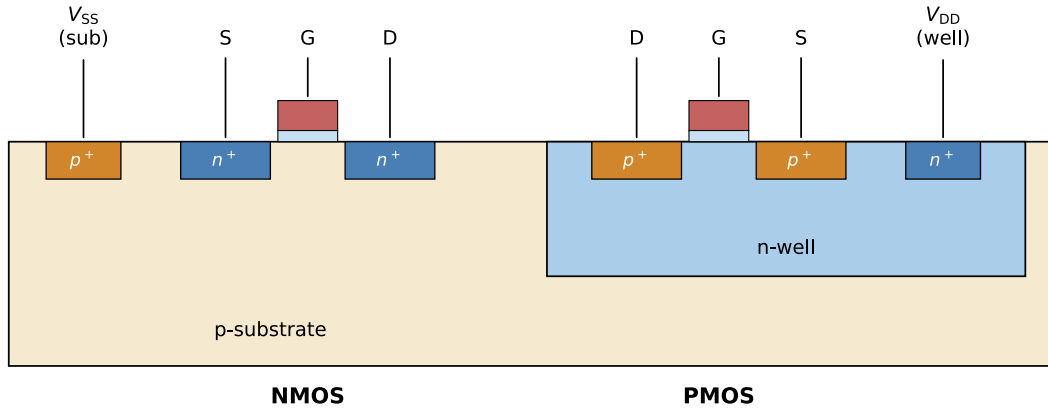


Figure 10: NMOS and PMOS transistors integrated on a single wafer: the NMOS sits directly in the p -substrate, the PMOS in a local n -well. The p^+ substrate contact ties the substrate to V_{SS} , the n^+ well contact ties the n -well to V_{DD} .

In Figure 11, we see a simple CMOS inverter consisting of a PMOS and an NMOS transistor. The PMOS is connected to the positive supply voltage V_{DD} , while the NMOS is connected to ground V_{SS} . The gates of both transistors are tied together and form the input A of the inverter, while the drains are also tied together and form the output Y . When the input is low, the PMOS is on and the NMOS is off, pulling the output high. Conversely, when the input is high, the PMOS is off and the NMOS is on, pulling the output low.

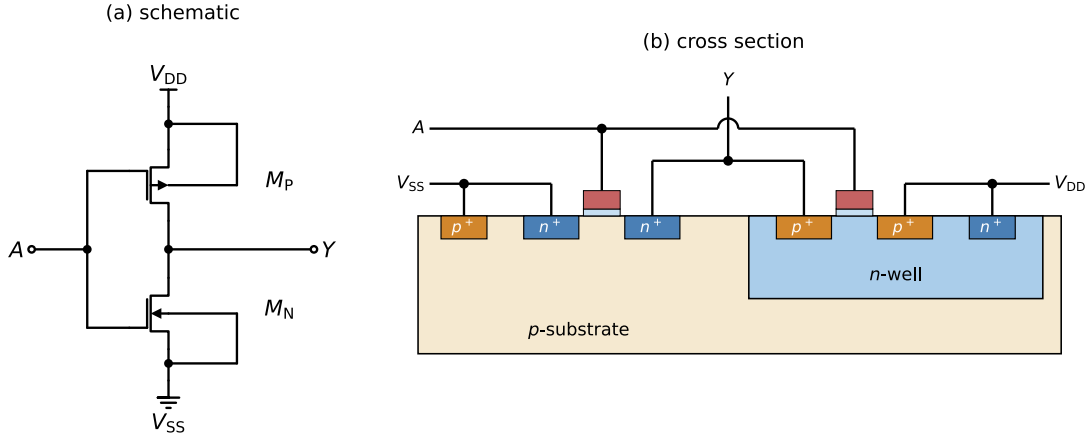


Figure 11: The CMOS inverter: (a) schematic and (b) the same circuit in the cross-section of the wafer. The PMOS pulls the output Y up when the input A is low, the NMOS pulls it down when the input is high, so in the static state no dc current flows from V_{DD} to V_{SS} . The bulk of the PMOS (its n -well) is tied to V_{DD} , the bulk of the NMOS (the p -substrate) to V_{SS} . In (b) the hop marks a wire crossing without contact, which in a real chip is realized on two different metal layers.

Note that in the CMOS technology shown in Figure 11, the bulk nodes of all the NMOS transistors are tied to the common substrate (V_{SS}), while the bulk nodes of all the PMOS transistors are tied to the n -well (V_{DD}). Since the n -wells are separated from the substrate (and can be separated between different PMOS devices), the bulk nodes of selected PMOS can be tied to different potentials than V_{DD} (e.g., to the source of the PMOS), which is called **body biasing**. This can be used to adjust the threshold voltage of the PMOS transistors or to avoid the bulk effect (see Section 2.3.1).

2.3 Second-Order Effects in the MOSFET

The simple MOSFET model developed so far is only valid to first order for long-channel devices ($L_{\min} > 1 \mu\text{m}$) under idealistic conditions. In the following, we discuss the most important second-order effects: the body effect, channel-length modulation, velocity saturation, mobility degradation with the vertical field, and subthreshold conduction.

2.3.1 Body Effect

In the derivation of Equation 7 we tied source, drain, and bulk together. Note that the inversion charge is a function of the bulk (substrate) potential V_B versus the gate potential V_G , so when the source potential V_S differs from V_B , the apparent V_{th} becomes a function of V_{SB} , because inversion is reached at a surface potential of $2\Phi_B + V_{SB}$ instead of $2\Phi_B$.

This is the so-called **body effect** (or “back-gate effect”). The effective threshold voltage is given by (V_{th0} is the threshold voltage with $V_{SB} = 0$)

$$V_{th} = V_{th0} + \gamma \left(\sqrt{2\Phi_B + V_{SB}} - \sqrt{2\Phi_B} \right) \quad (15)$$

with the body-effect coefficient γ (the factor n will be introduced in Section 2.3.5)

$$\gamma = \frac{\sqrt{2qN_a\epsilon_{si}}}{C'_{ox}} = 2(n-1)\sqrt{2\Phi_B}. \quad (16)$$

Note that for $n = 1$ (no body effect), $\gamma = 0$ and thus $V_{th} = V_{th0}$.

i Note 2: Example Calculation of the Body Effect (NMOS)

Starting with the previously used values ($N_a = 3 \times 10^{18} \text{ cm}^{-3}$, $C'_{ox} = 17.3 \text{ fF}/\mu\text{m}^2$, $\Phi_B = 0.50 \text{ V}$), we get

$$\gamma = \frac{\sqrt{2qN_a\epsilon_{si}}}{C'_{ox}} = 0.58 \text{ V}^{1/2}$$

Let us consider the originally calculated $V_{th0} = 0.53 \text{ V}$ and assume the bulk is tied to $V_{SS} = V_B = 0 \text{ V}$ while the source sits at $V_S = 0.5 \text{ V}$ (so $V_{SB} = 0.5 \text{ V}$). Using Equation 15:

$$V_{th} = 0.53 + 0.58 \left(\sqrt{2 \cdot 0.50 + 0.5} - \sqrt{2 \cdot 0.50} \right) = 0.66 \text{ V}$$

In this case, the threshold voltage has been **increased by 130 mV**. Conversely, by tying the bulk to a higher potential than the source, the threshold voltage can also be **decreased**, but beware of the risk of forward-biasing the source-bulk pn-junction (see Figure 10)!

2.3.2 Channel-Length Modulation

In the derivation of the saturation current (Equation 13), we assumed that the channel is pinched off exactly at the drain when $V_{DS} = V_{GS} - V_{th}$. When V_{DS} is increased further, this pinch-off point moves away from the drain towards the source. This can be viewed as an effective **channel-length reduction**, $L_{eff} = f(V_{DS}) < L$, and we have to use L_{eff} instead of L in Equation 13. Avoiding a direct calculation of L_{eff} , it is common to set $1/L_{eff} \approx (1 + \Delta L/L)/L$ with $\Delta L/L = \lambda V_{DS}$, so that

$$I_D = \frac{1}{2} \mu_n C'_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 (1 + \lambda V_{DS}) \quad (17)$$

This so-called **channel-length modulation** leads to a reduced output impedance of the MOSFET, i.e., a nonzero output conductance (ideally, $\lambda = 0$ and thus $g_{ds} = 0$):

$$g_{ds} = \frac{\partial I_D}{\partial V_{DS}} = I_D|_{\lambda=0} \cdot \lambda \quad (18)$$

Since $\lambda \propto 1/L$, the output resistance can be improved by increasing L beyond the minimum length, which is usually done when using the MOSFET as a current source. By scaling W along with L , the aspect ratio can be kept constant. Figure 12 shows the output characteristic including channel-length modulation; note the finite slope of the curves in saturation.

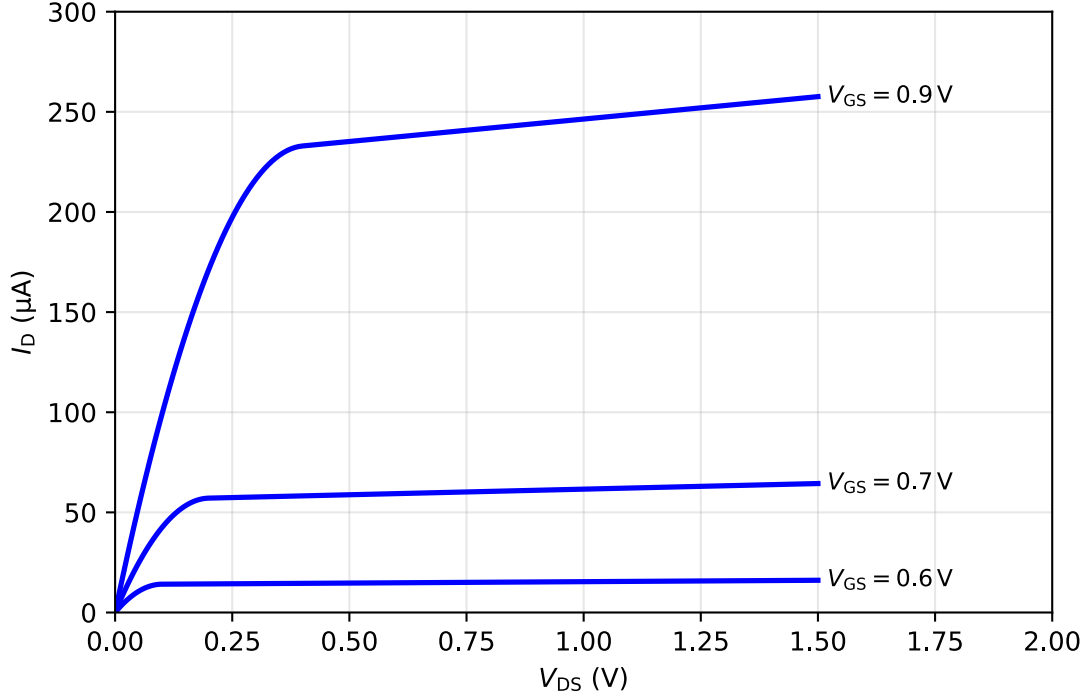


Figure 12: NMOS output characteristic including channel-length modulation ($\mu_n C'_{\text{ox}} = 280 \mu\text{A}/\text{V}^2$, $W/L = 10$, $V_{\text{th}} = 0.5 \text{ V}$, $\lambda = 0.1 \text{ V}^{-1}$). The curves in saturation now exhibit a finite slope, corresponding to a nonzero output conductance.

2.3.3 Velocity Saturation

In the derivation of Equation 9, we assumed that the velocity of the carriers in the channel is given by $v = \mu_n E_{\text{DS}}$. This is only true up to a certain speed limit of the carriers in silicon, which is around $v_{\text{sat}} = 10^7 \text{ cm/s}$. To take this **velocity saturation** into account, the following empirical model for the effective velocity can be used in the derivation:

$$v_{\text{eff}} = \begin{cases} \frac{\mu_n E_{\text{DS}}}{1 + E_{\text{DS}}/E_{\text{sat}}} & \text{if } E_{\text{DS}} \leq E_{\text{sat}} = 2v_{\text{sat}}/\mu_n \\ v_{\text{sat}} & \text{otherwise} \end{cases} \quad (19)$$

If the MOSFET enters the velocity-saturation regime, the square-law relation no longer holds, and the drain current is given by

$$I_{\text{D}} = v_{\text{sat}} C'_{\text{ox}} W (V_{\text{GS}} - V_{\text{th}}). \quad (20)$$

In velocity saturation, I_{D} is not a quadratic but a linear function of the overdrive voltage, and the transconductance hence saturates at

$$g_{\text{m,sat}} = v_{\text{sat}} C'_{\text{ox}} W \quad (21)$$

This result means that increasing the drain current beyond the velocity-saturation point does not result in an increased g_{m} , which is usually not useful.

In sub-micron technologies, it is relatively easy to reach velocity saturation: with $\mu_n = 280 \text{ cm}^2/(\text{V} \cdot \text{s})$, the resulting critical field is $E_{\text{sat}} = 7.1 \text{ V}/\mu\text{m}$, which is reached in a $0.13 \mu\text{m}$ CMOS technology at an overdrive of $V_{\text{od}} \approx 0.9 \text{ V}$ —achievable with a supply voltage of

$V_{DD} = 1.5 \text{ V}$ and $V_{th} = 0.5 \text{ V}$! Note that the stated μ_n is considerably lower than the mobility of pure silicon, because doping reduces mobility—yet another trade-off.¹

2.3.4 Mobility Degradation with Vertical Field

With high V_{GS} , the channel is confined to a narrow region at the surface, leading to more carrier scattering and thus lower mobility (remember that at the SiO_2 -silicon interface, scattering is more pronounced). We can model this effect empirically by setting

$$\mu_{n,\text{eff}} = \frac{\mu_n}{1 + \Theta(V_{GS} - V_{th})} \quad (22)$$

and using this $\mu_{n,\text{eff}}$ instead of μ_n in the saturation current (Equation 13):

$$I_D = \frac{1}{2} \frac{\mu_n C'_{ox}}{1 + \Theta V_{od}} \frac{W}{L} V_{od}^2 \approx \frac{1}{2} \mu_n C'_{ox} \frac{W}{L} (V_{od}^2 - \Theta V_{od}^3) \quad (23)$$

This introduces a *cubic* nonlinearity into an otherwise quadratic relationship, which can be relevant for precision analog and high-frequency circuits, with $\Theta \approx (10^{-9} \text{ m/V})/t_{ox} = 0.5 \text{ V}^{-1}$ for $t_{ox} = 2 \text{ nm}$.

2.3.5 Subthreshold Conduction

In the derivation of Equation 7, we assumed that inversion sets in *abruptly* when the surface potential reaches $2\Phi_B$. However, even before reaching this level, there are already electrons present in the channel (for the NMOS; holes for the PMOS), which lead to conduction between drain and source even for $V_{GS} < V_{th}$.

This mode of operation is called **subthreshold conduction**, and it leads to a leakage current based on diffusion when the MOSFET should be off. The drain current in this regime is described by

$$I_D = I_0 \frac{W}{L} e^{\frac{V_{GS}}{nV_T}} \left(1 - e^{-\frac{V_{DS}}{V_T}}\right) = \mu_n C'_{ox} \frac{W}{L} 2nV_T^2 e^{\frac{V_{GS}-V_{th}}{nV_T}} \left(1 - e^{-\frac{V_{DS}}{V_T}}\right). \quad (24)$$

The nonideality factor (or slope factor) n is defined by (see Equation 16)

$$n = 1 + \frac{C'_{dep}}{C'_{ox}} \quad (25)$$

and is on the order of 1 (perfect MOSFET) to 2 (pretty bad). Using the values of C'_{ox} and C'_{dep} from the previous examples, we get $n = 1.29$. $V_T = kT/q$ is the well-known thermal voltage of 25.8 mV at 300 K, and

$$I_0 = \mu_n C'_{ox} 2nV_T^2 e^{-\frac{V_{th}}{nV_T}}.$$

Note that the depletion capacitance C'_{dep} is given by Equation 3 evaluated at $\Phi_S = 2\Phi_B$:

¹The mobility value used here is not fully consistent with the process constant $K_n = \mu_n C'_{ox} = 280 \mu\text{A/V}^2$ in Table 1, which (with $C'_{ox} = 17.3 \text{ fF}/\mu\text{m}^2$) would imply an effective $\mu_n \approx 160 \text{ cm}^2/(\text{V} \cdot \text{s})$. The effective mobility depends strongly on the vertical field and thus on bias; such spread between simple hand-calculation parameters is normal and is one more reason to verify all sizing choices by simulation.

$$C'_{\text{dep}} = \sqrt{\frac{qN_a\epsilon_{\text{si}}}{2 \cdot 2\Phi_B}}$$

The inversion condition used in the derivation of the threshold voltage in Equation 7 is the point where *the electron concentration at the surface equals the bulk doping concentration* N_a —but even before this condition is met, there are electrons in the channel. Another way to look at this is that the surface potential effectively forward-biases the channel-source junction (since the channel region is p -type and the source is n^+), which can be seen in the plot of Φ_S in Figure 7. When the forward bias of this junction is gradually increased, electrons are injected into the channel area (similar to a BJT), moving across the channel by diffusion, and collected by the drain. At the point where the channel switches from p - to n -type due to inversion, this “diode” approaches an ohmic connection linking drain and source.

Although this **weak inversion** mode of the MOSFET is generally unwanted in digital circuits (as it leads to leakage currents), it is very useful for low-power analog (and digital) operation. Assuming $V_{\text{DS}} > 4V_T \approx 100 \text{ mV}$, then

$$I_D \approx I_0 \frac{W}{L} e^{\frac{V_{\text{GS}}}{nV_T}}$$

and the resulting transconductance is

$$g_m = \frac{\partial I_D}{\partial V_{\text{GS}}} = \frac{I_D}{nV_T} \quad (26)$$

This transconductance in weak inversion is similar to that of a BJT, except for the reduction by the factor n !

2.3.6 MOSFET I/V Characteristic Including the Body Effect

In the earlier derivation of the square law (Equation 10), we made a deliberate error: when the channel potential V_{cs} departs from V_S towards V_D , we need to take the body effect along the channel into account by using nV_{cs} instead of V_{cs} , so $Q'_{\text{inv}}(x) = C'_{\text{ox}}(V_{\text{GS}} - nV_{\text{cs}} - V_{\text{th}})$. The corrected derivation is then

$$\int_0^L I_D dx = \mu_n C'_{\text{ox}} W \int_0^{V_{\text{DS}}} (V_{\text{GS}} - nV_{\text{cs}} - V_{\text{th}}) dV_{\text{cs}}$$

resulting in

$$I_D = \mu_n C'_{\text{ox}} \frac{W}{L} \left[(V_{\text{GS}} - V_{\text{th}}) V_{\text{DS}} - \frac{n}{2} V_{\text{DS}}^2 \right] \quad (27)$$

In the deep-triode region, there is no difference from the previous result, as the term $nV_{\text{DS}}^2/2$ vanishes for small V_{DS} . The saturation voltage is now $V_{\text{DS,sat}} = (V_{\text{GS}} - V_{\text{th}})/n$, and the drain current in saturation becomes (compare to Equation 13)

$$I_D = I_{\text{DS,sat}} = \frac{1}{2n} \mu_n C'_{\text{ox}} \frac{W}{L} (V_{\text{GS}} - V_{\text{th}})^2. \quad (28)$$

The corrected expressions for the transconductance are

$$g_m = \mu_n C'_{ox} \frac{W}{L} \frac{V_{GS} - V_{th}}{n} = \sqrt{\frac{2}{n}} \mu_n C'_{ox} \frac{W}{L} I_D = \frac{2I_D}{V_{GS} - V_{th}}. \quad (29)$$

2.4 MOSFET AC Operation

So far, we have only considered the dc operation of the MOSFET, but the physical construction of the transistor leads to unwanted capacitances and resistances, which need to be accounted for in dynamic operation. The most important capacitive elements are:

- The oxide capacitance between the gate and the channel, $C_{gg} = C'_{ox} WL$.
- The gate-drain and gate-source overlap capacitances $C_{ov} = C'_{ov} W$.
- The junction capacitance between drain/source and substrate, $C_j = C_{j0}/(1 + V_{DB|SB}/\Phi_B)^m$ with $m \approx 0.3...0.4$. C_{j0} is defined by the area and perimeter of the drain/source regions.
- The depletion capacitance between the channel (the inversion layer) and the substrate, $C_{dep} = C'_{dep} WL$.

To identify the dominant capacitive elements between gate-source (C_{GS}) and gate-drain (C_{GD}), we need to distinguish between the triode and the saturation region, see Table 2.

Table 2: Dominant gate capacitances of the MOSFET in the triode and saturation regions

Operating region	C_{GS}	C_{GD}
Triode	$\frac{1}{2}C_{gg} + C_{ov}$	$\frac{1}{2}C_{gg} + C_{ov}$
Saturation	$\frac{2}{3}C_{gg} + C_{ov}$	C_{ov}

In addition to these capacitances, the resistance of the gate connection is usually dominant among the parasitic resistances. The end-to-end resistance of the gate stripe is $R_s \cdot W/L$ (with the sheet resistance R_s set by the gate material). Due to the distributed nature of the gate along the width of the transistor, a reasonable lumped model for a single-sided gate contact is

$$R_G = \frac{1}{3}R_s \frac{W}{L} \quad (30)$$

Using a **double-sided gate contact**, the effective gate resistance is reduced to

$$R_G = \left(\frac{1}{3}R_s \frac{W}{2 \cdot L} \right) \parallel \left(\frac{1}{3}R_s \frac{W}{2 \cdot L} \right) = \frac{1}{12}R_s \frac{W}{L} \quad (31)$$

The impact of R_G on performance can be further minimized by constructing a large transistor out of m smaller transistors connected in parallel (“fingers”). If the total width is $W = mW'$ (with each finger having width W'), the gates of all unit transistors are connected in parallel, and the effective single-sided gate resistance becomes

$$R_{G,eff} = \frac{R'_G}{m} = \frac{\frac{1}{3}R_s \frac{W}{m \cdot L}}{m} = \frac{1}{m^2}R_G \quad (32)$$

A typical analog/RF layout therefore splits a wide transistor into several fingers of limited length (a few μm), uses double-sided gate contacts, and surrounds the device with a closed well-contact ring; the multi-finger layout technique is discussed in detail in Section 4.5.

2.5 MOSFET Small-Signal Model

The complete small-signal model of a MOSFET in saturation is shown in Figure 13. The transconductance g_m is modeled by a voltage-controlled current source, as is the body effect, which contributes a back-gate transconductance

$$g_{mb} = g_m \frac{\gamma}{2\sqrt{2\Phi_B + V_{SB}}}.$$

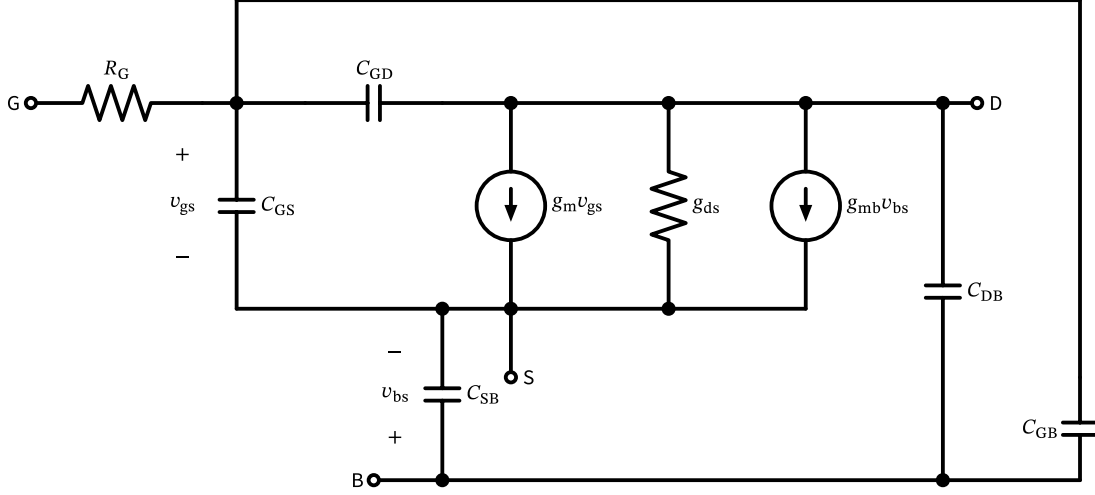


Figure 13: Complete small-signal model of the MOSFET in saturation, including the gate resistance, the capacitances to all terminals, the transconductance g_m , the back-gate transconductance g_{mb} , and the output conductance g_{ds} .

2.5.1 High-Frequency Operation

For high-frequency operation, the definition of the transit frequency f_T is useful; this is the frequency where the magnitude of the current gain $|I_{ds}|/|I_{gs}|$ falls to one. The ac drain current can be calculated based on the small-signal model:

$$I_{ds}(j\omega) = I_{gs}(j\omega) \frac{g_m - j\omega C_{GD}}{j\omega(C_{GS} + C_{GD})}$$

The transit frequency is thus (extrapolating the low-frequency transfer function)

$$\omega_T = 2\pi f_T \approx \frac{g_m}{C_{GS} + C_{GD}} \approx \frac{g_m}{C_{GG}} \approx \frac{3}{2n} \frac{\mu}{L^2} (V_{GS} - V_{th}). \quad (33)$$

In addition, the maximum oscillation frequency f_{max} is a useful metric; it is the frequency where the maximum available power gain drops to one (MAG = 0 dB):

$$f_{max} \approx \sqrt{\frac{f_T}{8\pi R_G C_{GD}}} \quad (34)$$

2.5.2 Simplified Small-Signal Models

In most cases, a simplified model is sufficient for hand calculations. Neglecting most model elements, the two models shown in Figure 14 can be used in the saturation region. The left π -model is usually better suited for circuits with the signal input at the gate (like in a common-

source or common-drain amplifier), whereas the right τ -model is advantageous when the input is at the source (like in a common-gate amplifier).

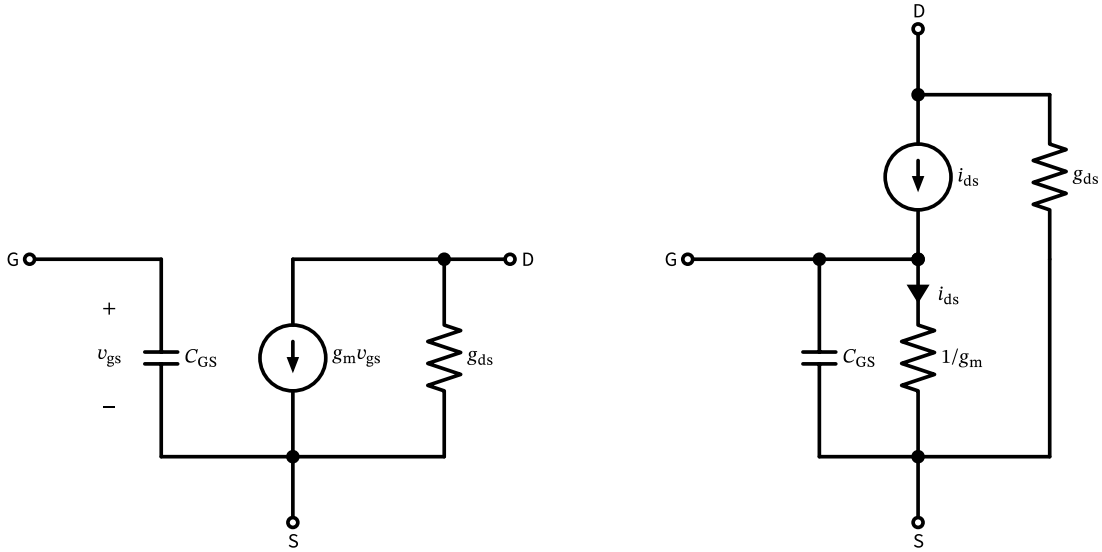


Figure 14: Simplified small-signal models of the MOSFET in saturation for hand calculations: π -model (left), preferred when the input is at the gate, and τ -model (right), preferred when the input is at the source. In the τ -model, the current through the resistor $1/g_m$ equals i_{ds} , which is mirrored by the controlled source into the drain.

2.6 MOSFET Scaling

Making MOSFETs smaller has benefits for power, performance, and area (PPA). Traditionally, every 2–3 years a scaled (= smaller) CMOS technology node has been introduced. In Figure 15, the technology nodes are plotted versus their year of first volume production; note the logarithmic axis!

In the beginning, the node name stated the minimum feature size of the given technology, which is usually the minimum gate length ($L_{\min}$): in a 130 nm technology, $L_{\min} = 130$ nm. This holds well down to about the 90 nm node; from there on the printed gate length is already shorter than the node name (the 65 nm node used a physical gate length of roughly 35 nm), and from the 22/20 nm node onwards the technology name is merely a “label.” Since traditionally a scaling factor of $\sqrt{2}$ has been used, the node names reflect this: 350 $\rightarrow$ 250 $\rightarrow$ 180 $\rightarrow$ 130 $\rightarrow$ 90 $\rightarrow$ 65 $\rightarrow$ 45 $\rightarrow$ 32/28 $\rightarrow$ 22/20 $\rightarrow$ 16/14 $\rightarrow$ 10 $\rightarrow$ 7 $\rightarrow$ 5 $\rightarrow$ 3 $\rightarrow$ 2 $\rightarrow$ 1.6 nm.

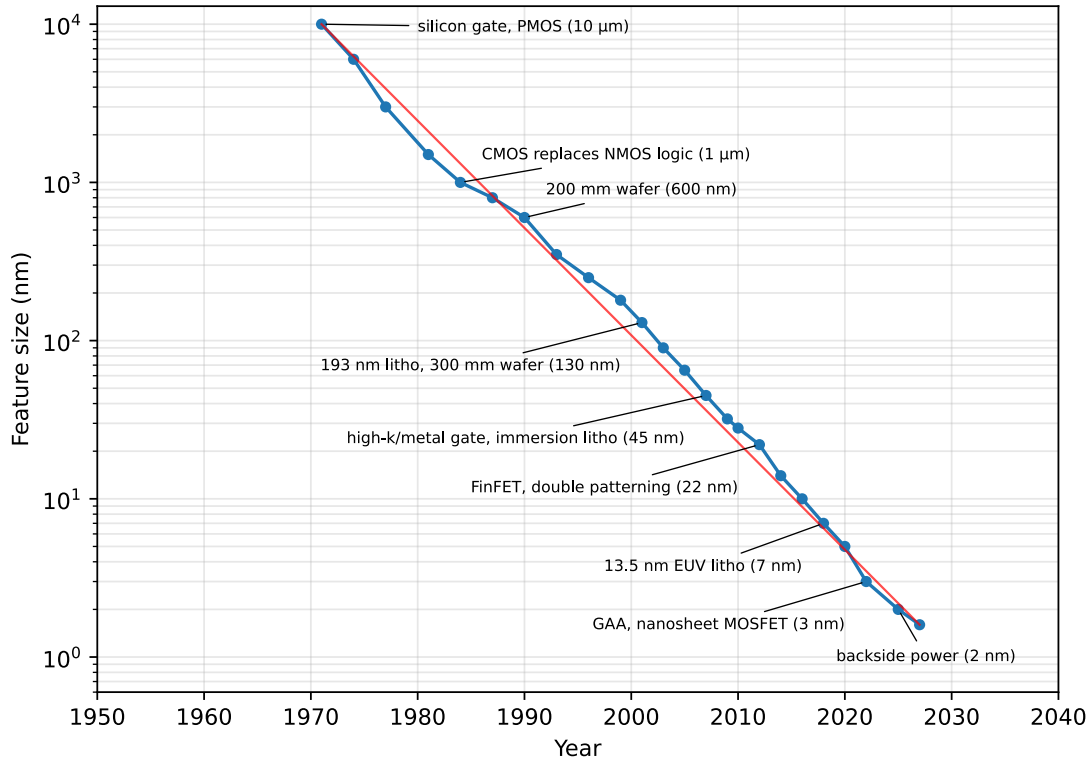


Figure 15: CMOS technology scaling trend: minimum feature size (node name) versus year of first volume production, with key technology innovations annotated (data compiled from public sources, e.g., wikichip.org).

2.6.1 Scaling Theory

The ideal scaling theory follows three rules, formulated by R. Dennard in 1974 [7], to generate the next CMOS technology node from the current one:

1. Reduce all lateral and vertical dimensions by α ($\alpha = \sqrt{2}$ for classical scaling; the area of a MOSFET thus *shrinks* by $\alpha^2 = 2$).
2. Reduce the threshold voltage V_{th} and the supply voltage V_{DD} by α .
3. Increase all doping levels by α .

This scaling is called **constant-field scaling**. If we apply these rules, the MOSFET parameters in the shrunken technology (marked by *) scale as follows. The drain saturation current is reduced by α :

$$I_D^* = \frac{1}{2} \mu_n (\alpha C'_{ox}) \frac{W/\alpha}{L/\alpha} \left(\frac{V_{GS}}{\alpha} - \frac{V_{th}}{\alpha} \right)^2 = \frac{1}{\alpha} I_D$$

This is offset by the fact that the input capacitance is also reduced:

$$C_{gg}^* = \frac{W}{\alpha} \frac{L}{\alpha} (\alpha C'_{ox}) = \frac{1}{\alpha} C_{gg}$$

To estimate the scaling of the gate delay of logic gates built from scaled devices, we use the simple model $T_d \propto C_{gg} V_{DD} / I_D$, which relates the gate delay to the time it takes to charge or discharge the load capacitance of a driven gate between the supply rails. The scaled delay is

$$T_d^* = \frac{C_{gg}/\alpha}{I_D/\alpha} \frac{V_{DD}}{\alpha} = \frac{1}{\alpha} T_d$$

The speed of digital circuits thus benefits from scaling, as the gate delay shrinks in each new technology node! Using a simple model for the dynamic power consumption of CMOS logic, $P_{\text{dyn}} = f_{\text{clk}} C_{\text{dyn}} V_{DD}^2$ (where f_{clk} is the clock frequency and $C_{\text{dyn}} \propto C_{\text{gg}}$ is the total switched capacitance), we see that

$$P_{\text{dyn}}^* = f_{\text{clk}} \frac{C_{\text{dyn}}}{\alpha} \left(\frac{V_{DD}}{\alpha} \right)^2 = \frac{1}{\alpha^3} P_{\text{dyn}}$$

so the dynamic power consumption of a scaled logic circuit benefits dramatically. Note that the power per function, clocked at the same frequency, scales by α^{-3} —but since we can also increase the clock frequency by α , and can integrate α^2 more gates per area, the total power consumption would stay constant if we fully exploited the scaling benefit. To improve power, performance, *and* area (PPA) simultaneously, a trade-off has to be found.

2.6.2 Analog Scaling

While there is a clear scaling benefit for digital circuits, the benefits for analog circuits are less clear. Generally, the reduction of V_{DD} is a drawback, since it likely reduces the dynamic range when processing voltage signals. The transconductance scales like

$$g_m^* = \mu_n (\alpha C'_{\text{ox}}) \frac{W/\alpha}{L/\alpha} \frac{V_{GS} - V_{th}}{\alpha} = g_m$$

so it seems to be scale-invariant. However, there is a benefit in device speed, as the transit frequency scales like

$$f_T^* = \frac{1}{2\pi} \frac{g_m}{C_{gg}/\alpha} = \alpha f_T$$

so the speed of the MOSFET increases, opening the door to better analog and RF performance, albeit at lower supply voltages.

2.6.3 Scaling Summary

Table 3 summarizes the constant-field scaling factors derived above. Scaling equations fitted to real technology data from 180 nm down to 7 nm can be found in [8].

Table 3: Constant-field (Dennard) scaling of MOSFET parameters

Parameter	Symbol	Scaling factor
Gate length	$L_{\min}$	α^{-1}
MOSFET area	WL	α^{-2}
Gate oxide thickness	t_{ox}	α^{-1}
Substrate doping	N_{a}	α
Drain current in saturation	I_{D}	α^{-1}
Gate capacitance	C_{GS}	α^{-1}
Supply voltage	V_{DD}	α^{-1}
Threshold voltage	V_{th}	α^{-1}
Gate delay	T_{d}	α^{-1}
Clock frequency	f_{clk}	α
Dynamic power consumption per gate	P_{dyn}	α^{-3}

2.6.4 Interconnect Scaling

As the MOSFETs get smaller, the metal wiring also needs to scale to finer dimensions. We assume that, along with the lateral dimensions, the metal thickness and the dielectric inter-metal insulation scale down as well. The resistance per length of a scaled metal wire (with thickness t and width w) scales as

$$\frac{R^*}{l} = \rho \frac{1}{(t/\alpha)(w/\alpha)} = \alpha^2 \frac{R}{l}$$

so the wires get significantly more resistive! The wire-to-wire fringe capacitance per length for a minimum-width wire with minimum spacing s scales as

$$\frac{C_{\text{f}}^*}{l} = \epsilon_{\text{ox}} \frac{t/\alpha}{s/\alpha} = \frac{C_{\text{f}}}{l}$$

and the vertical coupling capacitance per length (with insulator thickness h) scales as

$$\frac{C_{\text{p}}^*}{l} = \epsilon_{\text{ox}} \frac{w/\alpha}{h/\alpha} = \frac{C_{\text{p}}}{l}$$

so both capacitance values are, to first order, *invariant to scaling*. The RC wire delay per squared length is thus

$$\frac{T_{\text{wire}}^*}{l^2} \propto \frac{R^*}{l} \left(\frac{C_{\text{f}}^*}{l} + \frac{C_{\text{p}}^*}{l} \right) = \alpha^2 \frac{T_{\text{wire}}}{l^2}$$

For short routes, this increased relative delay is offset by the length scaling, so that

$$T_{\text{wire}}^* \propto \left(\frac{l}{\alpha}\right)^2 \frac{T_{\text{wire}}^*}{l^2} = T_{\text{wire}}$$

For long routes, however, the wire delay increases significantly. To cope with this delay, other solutions are needed: using multiple wiring levels (lower metals levels scaled for density, higher ones un-scaled for speed), or adapting the architecture, e.g., pipelining long-range data transfers.

2.7 MOSFET Short-Channel Effects

As the MOSFET gate length is scaled below $1 \mu\text{m}$, various imperfections kick in. We discuss subthreshold conduction and leakage, gate oxide leakage, and drain-induced barrier lowering (DIBL). Other effects, like random dopant fluctuation (RDF), poly depletion, or quantum-mechanical effects, are not covered here.

2.7.1 Subthreshold Leakage

Revisiting Equation 24 and neglecting the V_{DS} -dependent term, we can re-formulate the subthreshold drain current as

$$I_{\text{D}} = 100 \text{ nA} \cdot \frac{W}{L} e^{\frac{V_{\text{GS}} - V_{\text{th}}}{n V_{\text{T}}}} \quad (35)$$

Here, V_{th} has been re-defined as the V_{GS} at which $I_{\text{D}} = 100 \text{ nA} \cdot W/L$, which is a quite practical definition for leakage current discussions. The nonideality factor can also be written as

$$n = 1 + \frac{C'_{\text{dep}}}{C'_{\text{ox}}} = 1 + \underbrace{\frac{\varepsilon_{\text{si}}}{\varepsilon_{\text{ox}}}}_{\approx 3} \frac{t_{\text{ox}}}{W_{\text{dep}}} \quad (36)$$

Using Equation 35, we see that the leakage current in the off-state ($V_{\text{GS}} = 0$) is impacted by only *two* parameters (W/L is usually fixed by the circuit design):

$$I_{\text{leak}} = 100 \text{ nA} \cdot \frac{W}{L} e^{-\frac{V_{\text{th}}}{n V_{\text{T}}}} \quad (37)$$

1. The threshold voltage V_{th} : a lower V_{th} increases the overdrive when the MOSFET is turned on and thus reduces the gate delay, but it exponentially increases the leakage current.
2. The nonideality factor n (ideally, $n = 1$).

2.7.2 Subthreshold Slope

To quantify the “turn-off” behavior of a MOSFET, the **subthreshold slope** S is defined, starting from Equation 35, and consisting of the two only parameters V_{T} and n :

$$S = \left[\frac{\partial(\log_{10} I_{\text{D}})}{\partial V_{\text{GS}}} \right]^{-1} = \frac{n V_{\text{T}}}{\log_{10}(e)} = 2.3 n V_{\text{T}} \quad (38)$$

The subthreshold slope expresses which change of V_{GS} lowers (or increases) the drain current by a factor of ten. Ideally, S should be as small as possible, meaning that the MOSFET can be turned off rapidly with a small change of V_{GS} :

- The lower bound is $S = 2.3 V_{\text{T}} = 59.4 \text{ mV/dec}$ at 300 K.

- Using $n = 1.29$ from the previous example, $S = 76.6 \text{ mV/dec}$, which is a good value; S for bulk CMOS is in the range of 80 to 120 mV/dec.

The impact of the subthreshold slope essentially stopped the scaling of V_{DD} and V_{th} , as the leakage current would otherwise become impractically large. This issue is exacerbated by the temperature dependence of the threshold voltage of $\approx -1 \text{ mV/K}$ and its process variation on the order of $\pm 50 \text{ mV}$. A threshold-voltage shift of a hundred millivolts may not sound like much, but it enters the leakage exponentially and thus has a drastic impact! Figure 16 shows how the supply voltage stopped scaling at around 0.8 V.

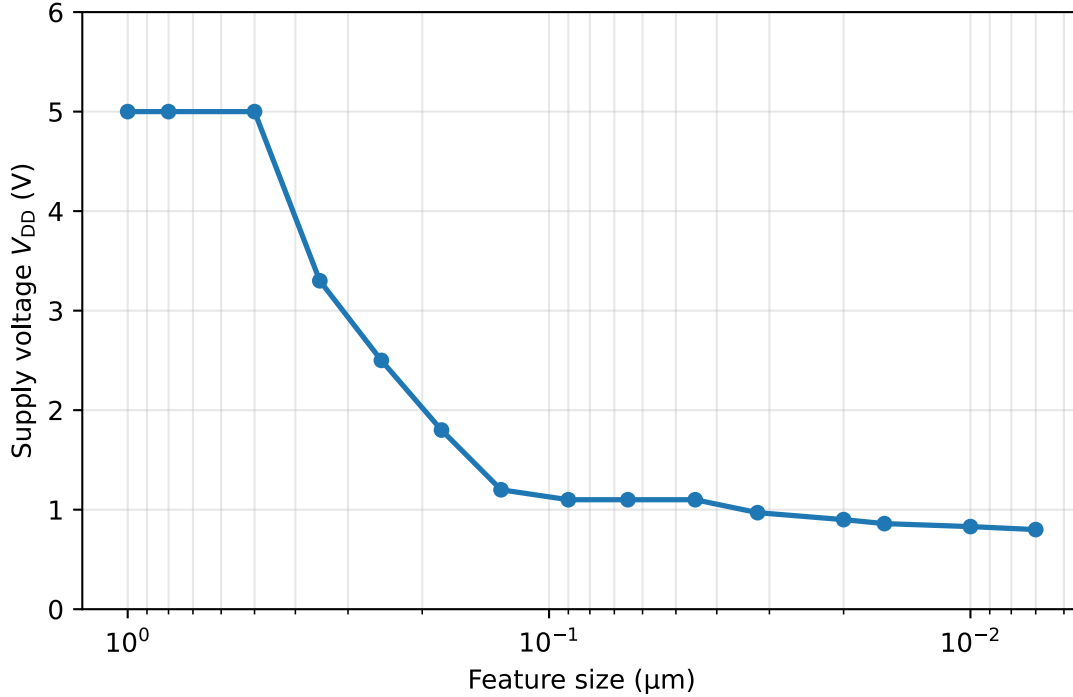


Figure 16: Nominal core supply voltage versus CMOS technology node. Below roughly 90 nm, the supply voltage no longer scales with the feature size but flattens out around 1 V, limited by the subthreshold leakage (data after Stillmaker and Baas, 2017).

2.7.3 Gate Oxide Leakage

As can be seen in Equation 36, to minimize n (and thus the subthreshold slope), and also to increase the drain current in saturation, C'_{ox} needs to be maximized. This has driven the scaling down of the gate oxide thickness, as discussed in the scaling rules. Over the years, the oxide thickness has been reduced from 300 nm to 1.2 nm, but severe issues appear once the oxide is only a few nanometers thick:

- The breakdown voltage of the oxide is reduced, mandating a low V_{DD} .
- The gate leakage current increases dramatically due to quantum-mechanical tunneling.

An alternative is needed to increase C'_{ox} without further reducing t_{ox} : the introduction of alternative gate dielectrics with increased permittivity, the so-called **high-k gate dielectrics**. One suitable material is hafnium dioxide: while SiO_2 has $\epsilon_r = 3.9$, HfO_2 offers $\epsilon_r = 24$. However, there are potential manufacturing issues, like chemical reactions with the silicon

substrate and the gate material, reliability concerns, and reduced surface mobility. Still, high- κ dielectrics are now widely used in modern CMOS processes.

2.7.4 Drain-Induced Barrier Lowering (DIBL)

In scaled MOSFETs, there exists another parasitic effect that lowers V_{th} as a function of L , called **drain-induced barrier lowering (DIBL)**. (Remember that lowering V_{th} has an exponential effect on the leakage current!) This lowering of V_{th} with *decreasing* L is also called the short-channel effect (SCE). Note that there is also a *reverse* short-channel effect (RSCE), which *increases* V_{th} with decreasing L , caused by halo implants around drain and source in modern CMOS processes. Effectively, $V_{th} = f(L)$ can be a wild function of the MOSFET length, so devices will only match if constructed with the same L ! Note that V_{th} also depends on V_{DS} due to DIBL, and on device W , as well as temperature, and other second-order effects, so ultimately $V_{th} = f(L, W, V_{DS}, T)$!

The impact of L and V_{DS} on V_{th} can be modeled by

$$\Delta V_{th} = -(V_{DS} + 0.4 \text{ V}) e^{-L/l_d} \quad (39)$$

and is illustrated in Figure 17. The DIBL characteristic length is $l_d \propto \sqrt[3]{t_{ox} W_{dep} X_j}$. To reduce l_d , one or all of these parameters must be reduced:

- The oxide thickness t_{ox} , which has already been discussed in the context of gate leakage.
- The depletion width W_{dep} .
- The drain junction depth X_j .

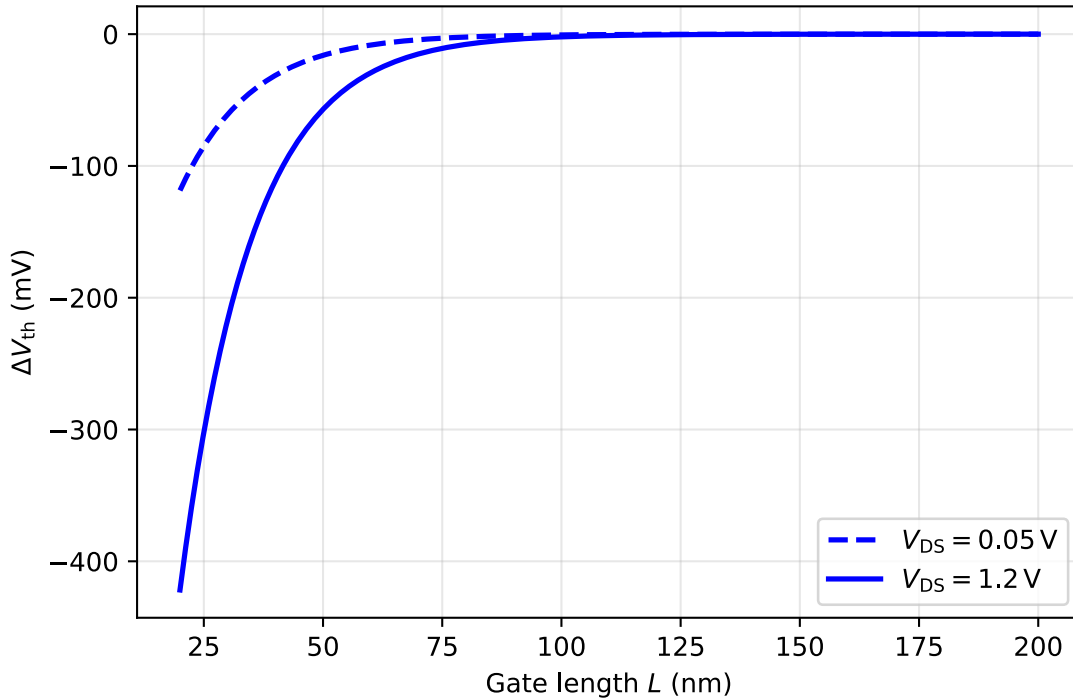


Figure 17: Threshold-voltage shift ΔV_{th} due to DIBL as a function of gate length, calculated with the model of Equation 39 for a characteristic length $l_d = 15 \text{ nm}$. For short gate lengths and high drain voltages, the threshold voltage drops rapidly (compare, e.g., the measured data for a 65 nm technology in [9]).

Reducing the depletion width and the junction depth requires increasing N_a (see Equation 2). However, increasing N_a also increases V_{th} (see Equation 7), which must be counteracted by increasing C'_{ox} —but there are limits due to gate leakage and breakdown, as discussed before. In addition, increasing N_a increases the subthreshold slope (see Equation 36) and lowers the mobility due to dopant scattering. So an alternative to ever-increasing substrate doping had to be found.

2.7.5 Solving DIBL: The Ultra-Thin-Body MOSFET

As scaling up N_a to constrain the *vertical* dimensions has reached its limits, an alternative solution is to *remove the silicon* under the channel to *physically* constrain W_{dep} and X_j . In the ultra-thin-body (UTB) MOSFET, the silicon channel is realized on top of a layer of SiO_2 , so the depletion and junction depths are limited to the silicon film thickness $t_{si} \leq L/4$. This is called **silicon-on-insulator (SOI)** technology; the structure is shown in Figure 18 (a). Additional advantages are:

- Since much less body doping N_a is needed, the carrier mobility is higher.
- There is no (or only a small) body effect, since the body is floating and fully depleted.
- The drain/source-to-bulk capacitance is largely reduced and almost **constant**.

The disadvantage is the higher cost of the SOI wafers, as manufacturing the Si- SiO_2 -Si stack is expensive.

2.7.6 Solving DIBL: FinFET and Nanosheet

Another alternative is to tilt the silicon channel by 90° , so that it sticks out of the wafer like a “fin,” and to wrap the gate around this fin on three sides: the **FinFET** (or tri-gate MOSFET), shown in Figure 18 (b). FinFETs have a few important differences compared to bulk MOSFETs:

- The device width is quantized, with $W = W_F + 2H_F$ per fin (W_F is the width and H_F the height of the fin), so the circuit design style must be adapted (the design parameter is the number of fins N_F instead of W).
- Drain and source are contacted at the ends of the fin, so the series resistances R_S and R_D are higher, degrading the transconductance due to series feedback: $g_{m,eff} = g_m / (1 + g_m R_S)$.
- Since the subthreshold slope is improved (smaller), the leakage current is lower, which allows lowering V_{th} and V_{DD} .
- DIBL and thus g_{ds} are lower, so the MOSFET self-gain g_m/g_{ds} is high.
- There is practically no body effect, and due to the low channel doping the mobility is high, leading to improved g_m/I_D and $\mu_n \approx \mu_p$.
- The silicon fins are isolated and surrounded by SiO_2 , so self-heating of the fin can be an issue.
- The 3D structure increases the coupling capacitance C_{ov} .
- A drawback is that SRAM and digital standard cells often use minimum-size transistors, and for yield reasons a minimum-sized FinFET requires two fins—a strong constraint when, e.g., the PMOS needs to be sized differently from the NMOS (the next step up is three fins, a large quantization step).

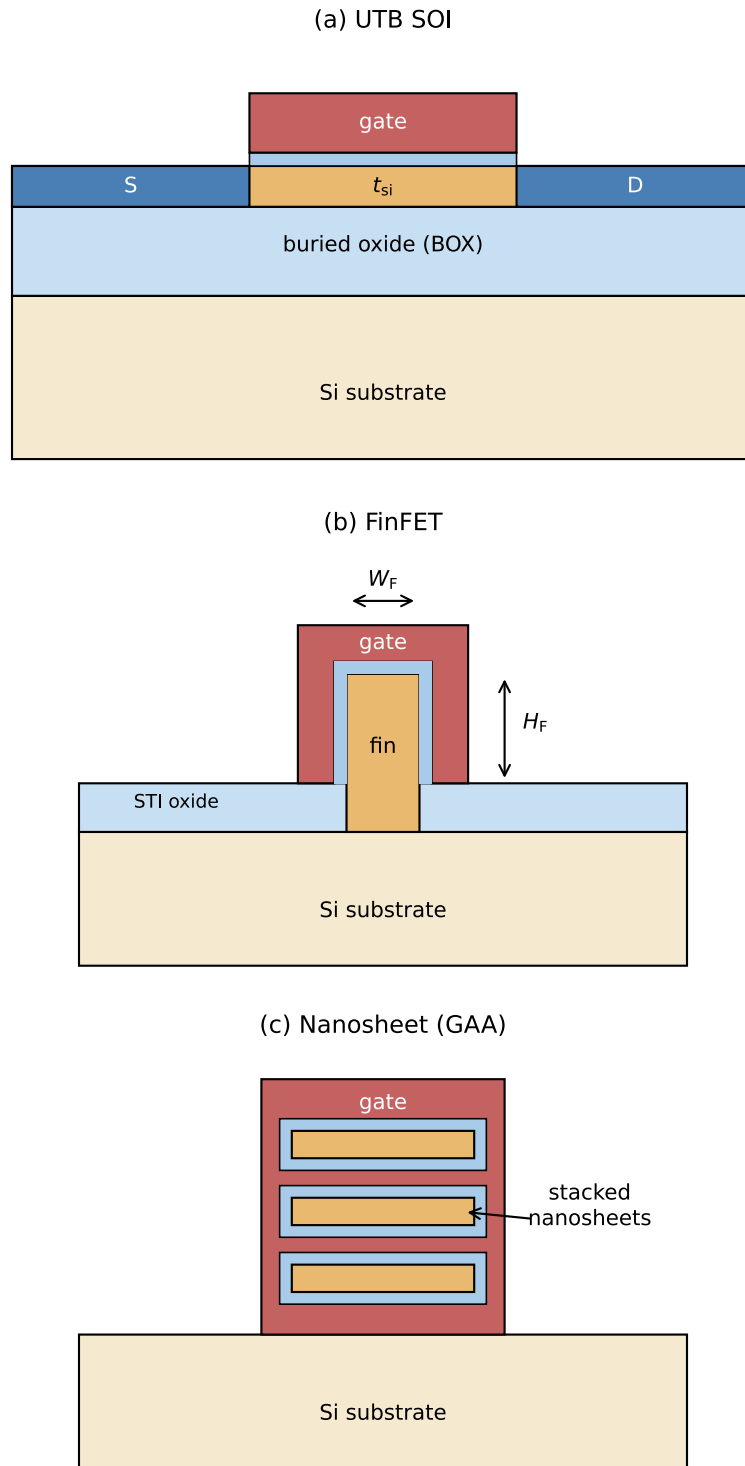


Figure 18: Advanced MOSFET structures with improved electrostatic gate control: (a) ultra-thin-body (UTB) SOI MOSFET shown along the channel, (b) FinFET shown perpendicular to the channel (the gate wraps around the fin on three sides), and (c) nanosheet (gate-all-around) MOSFET, where stacked channels are completely surrounded by the gate.

The evolution of the FinFET is the **nanosheet MOSFET**, first introduced at the 3 nm node (and used by all foundries at 2 nm) and shown in Figure 18 (c). One advantage is the even

better electrostatic control of the gate over the channel, as the stacked channels are completely wrapped by the gate—hence also called gate-all-around (GAA) MOSFET. In addition, the stacked channels can be either *n*- or *p*-type (or mixed), and could even use different materials, like SiGe or GaAs, opening the door to 3D integration.

2.8 Simulated Characteristics in IHP SG13

To connect the simple models we derived earlier to reality, Figure 19 to Figure 22 show simulated I/V characteristics of NMOS and PMOS devices ($W = 1.3 \mu\text{m}$, $L = 0.13 \mu\text{m}$, $V_{\text{SB}} = 0$) in the IHP SG13 130 nm CMOS technology, using the foundry's PSP models. The output characteristics show the triode and saturation regions including channel-length modulation, while the logarithmic transfer characteristics clearly show the exponential subthreshold region and the drain-current dependence on V_{DS} caused by DIBL.

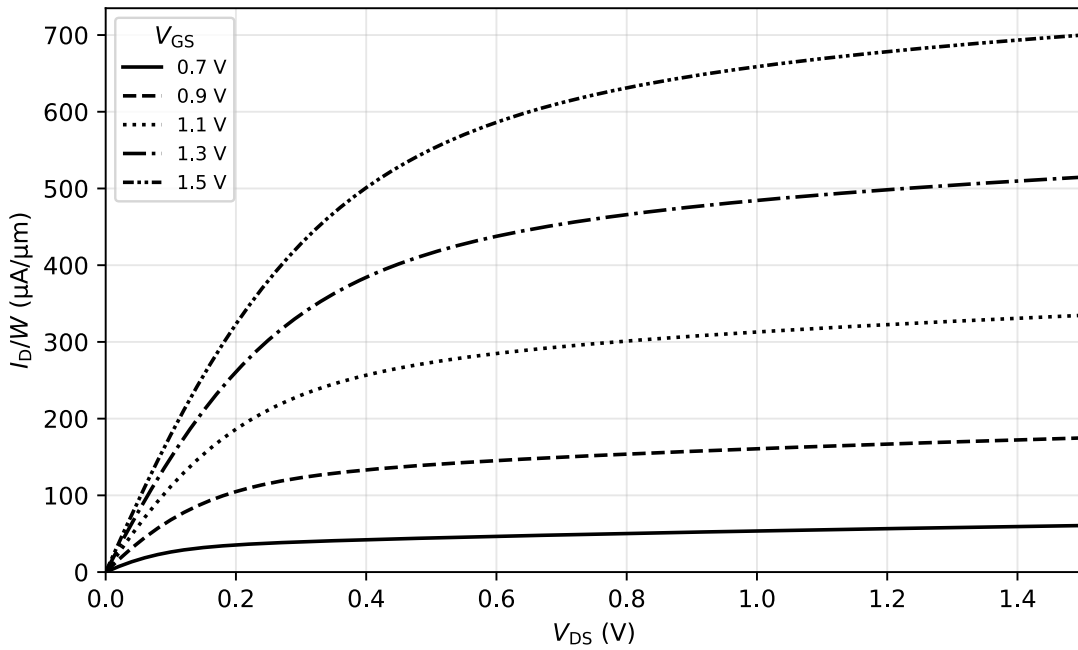


Figure 19: Simulated output characteristic of an NMOS in IHP SG13 ($W = 1.3 \mu\text{m}$, $L = 0.13 \mu\text{m}$, $V_{\text{SB}} = 0$) for different gate-source voltages.

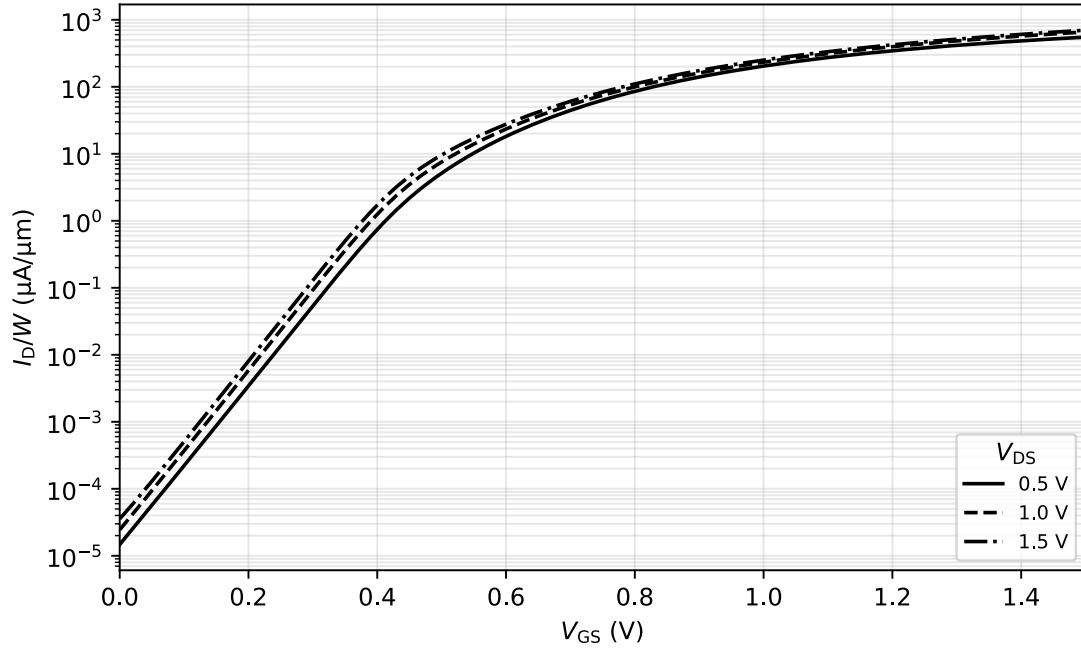


Figure 20: Simulated transfer characteristic of an NMOS in IHP SG13 ($W = 1.3\ \mu\text{m}$, $L = 0.13\ \mu\text{m}$, $V_{SB} = 0$) on a logarithmic current axis. The exponential subthreshold region is clearly visible, as is the drain-voltage dependence caused by DIBL.

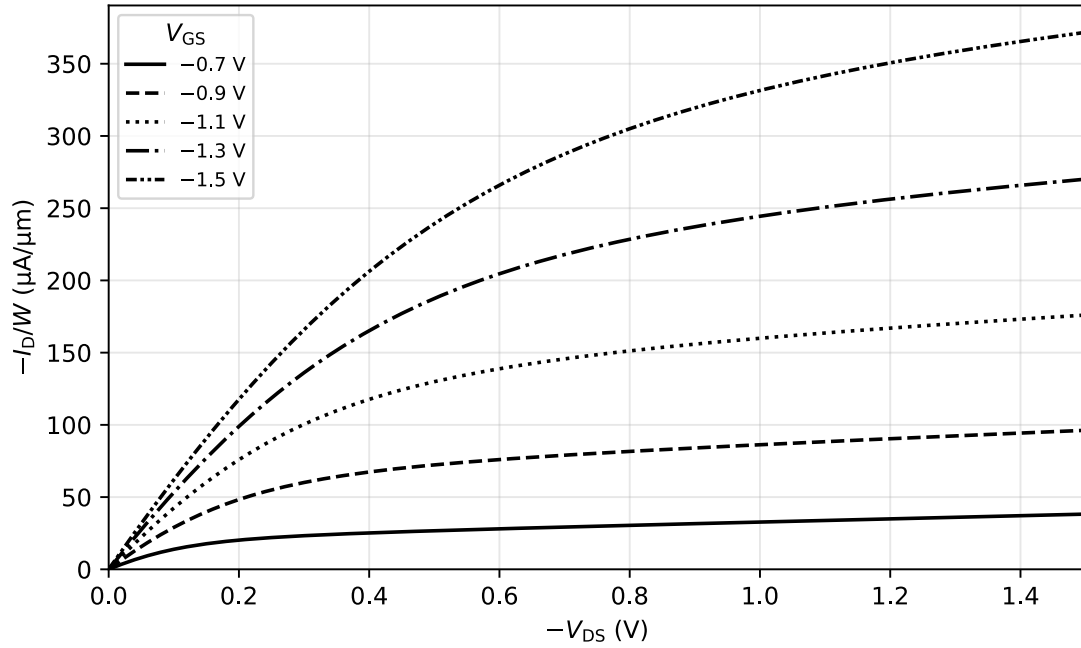


Figure 21: Simulated output characteristic of a PMOS in IHP SG13 ($W = 1.3\ \mu\text{m}$, $L = 0.13\ \mu\text{m}$, $V_{SB} = 0$) for different gate-source voltages. Note the lower current level compared to the NMOS due to the lower hole mobility.

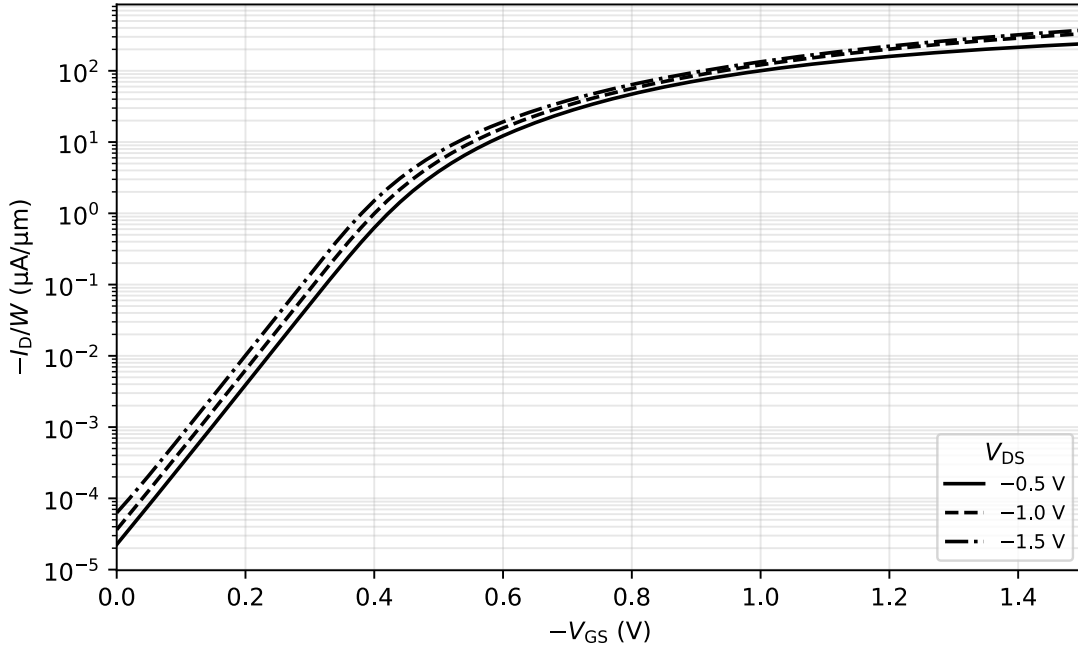


Figure 22: Simulated transfer characteristic of a PMOS in IHP SG13 ($W = 1.3 \mu\text{m}$, $L = 0.13 \mu\text{m}$, $V_{\text{SB}} = 0$) on a logarithmic current axis.

2.9 MOSFET Sizing

2.9.1 The g_m/I_D Method

When designing a circuit with MOSFETs, how should one select the parameters like I_D , V_{GS} , W , and L ? For digital designs, this is relatively straightforward: L is (usually) set to the minimum length, W is sized according to the driving requirements of the logic circuit, V_{GS} swings between V_{SS} and V_{DD} , and no static drain current flows. For analog, mixed-signal, and RF designs, the sizing question is more involved.

In the Analog Circuit Design course, the g_m/I_D method is used to find the proper sizing of a MOSFET depending on the circuit requirements. This method is the “go-to” method, as it uses numerical data found by evaluating the PDK models in a circuit simulator, and it describes the MOSFET correctly in all operating regions.

2.9.2 The Inversion-Coefficient Method

Despite the practical advantages of the g_m/I_D method, alternative approaches exist, and they can be helpful by providing relatively simple and unified analytical equations which are valid in all MOSFET operating regions. One such method is based on the **inversion coefficient**, which has been popularized as a design methodology by W. Sansen [10], [11]. Let us define a specific current

$$I_S = \mu C'_{\text{ox}} \frac{W}{L} \cdot 2nV_T^2. \quad (40)$$

Then, the inversion coefficient IC is defined as

$$\text{IC} = \frac{I_D}{I_S} = \left[\ln \left(1 + e^{\frac{V_{\text{GS}} - V_{\text{th}}}{2nV_T}} \right) \right]^2 \quad (41)$$

The beauty of this approach is that these equations work all the way from weak inversion (subthreshold operation, $IC < 0.1$) through moderate inversion ($IC \approx 1$) to strong inversion ($IC > 10$). Based on the selected IC , the important device properties can be calculated. The drain current is

$$I_D = I_S \cdot IC \quad (42)$$

the transconductance is given by

$$g_m = \frac{I_D}{nV_T} \frac{1 - e^{-\sqrt{IC}}}{\sqrt{IC}} \quad (43)$$

and the speed of the MOSFET is

$$f_T = \frac{1}{2\pi} \frac{3}{2} \frac{\mu}{L^2} V_T (\sqrt{1 + 4IC} - 1) \quad (44)$$

Let us check the asymptotic behavior of these equations against the earlier results (see Table 4).

Table 4: Asymptotic behavior of the inversion-coefficient equations in weak and strong inversion

Equa- tion	$V_{GS} \ll V_{th}$ (weak inversion)	$V_{GS} \gg V_{th}$ (strong inversion)
Equa- tion 41	$IC = e^{\frac{V_{GS}-V_{th}}{nV_T}}$	$IC = \left(\frac{V_{GS}-V_{th}}{2nV_T}\right)^2$
Equa- tion 42	$I_D = \mu C'_{ox} \frac{W}{L} 2nV_T^2 e^{\frac{V_{GS}-V_{th}}{nV_T}}$	$I_D = \frac{1}{2n} \mu C'_{ox} \frac{W}{L} (V_{GS} - V_{th})^2$
Equa- tion 43	$g_m = \frac{I_D}{nV_T}$	$g_m = \frac{2I_D}{V_{GS}-V_{th}}$
Equa- tion 44	$f_T = \frac{1}{2\pi} \frac{3}{2} \frac{\mu}{L^2} 2V_T e^{\frac{V_{GS}-V_{th}}{nV_T}}$	$f_T = \frac{1}{2\pi} \frac{3}{2n} \frac{\mu}{L^2} (V_{GS} - V_{th})$

The asymptotes match the subthreshold model (Equation 24) and the corrected square-law model (Equation 28), respectively.

Plotting Equation 43 and Equation 44 versus IC , as done in Figure 23, reveals the classical trade-off between g_m -efficiency (g_m/I_D) and speed (f_T). Note that velocity saturation is not represented in this simple model, so f_T does not saturate at high IC in the plot.

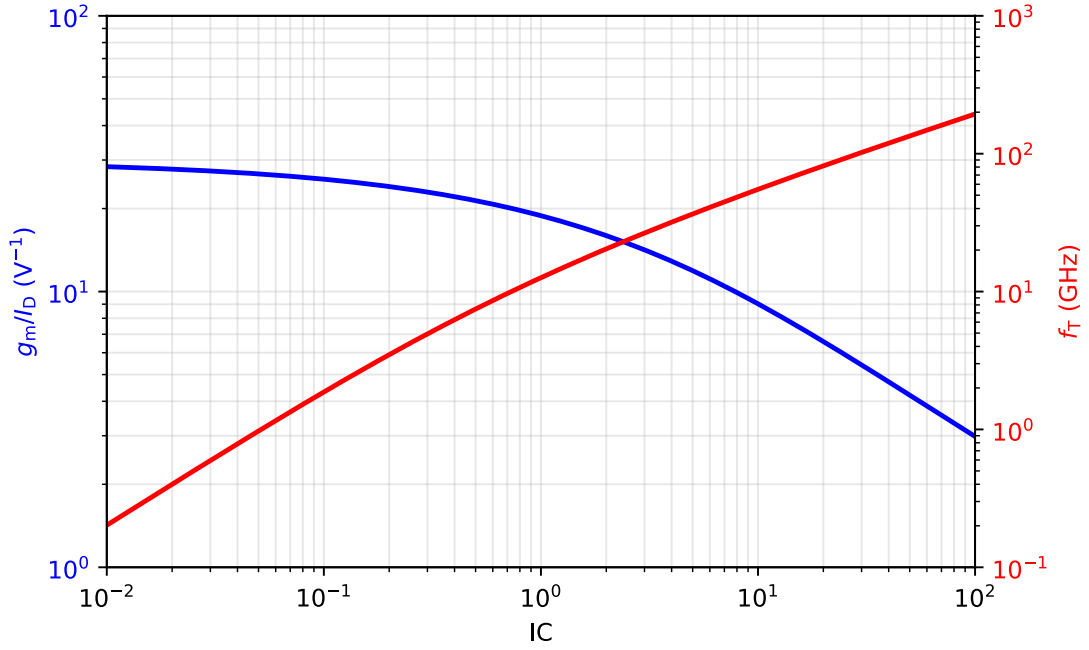


Figure 23: Transconductance efficiency g_m/I_D (blue, left axis) and transit frequency f_T (red, right axis) versus the inversion coefficient IC, calculated with the analytical model ($n = 1.3$, $L = 0.13 \mu\text{m}$, $\mu = 280 \text{ cm}^2/(\text{V s})$). Weak inversion offers the best efficiency, strong inversion the highest speed.

Figure 24 shows the same trade-off extracted from simulation data of an NMOS ($W/L = 1.3 \mu\text{m}/0.13 \mu\text{m}$) in IHP SG13. The simulation matches the analytical model well for g_m/I_D ; the simulated f_T comes out higher than the simple model predicts, since the model pessimistically assumes the full $\frac{2}{3}C'_{\text{ox}}WL$ gate capacitance at all bias points, while at high IC velocity saturation causes the simulated f_T to flatten.

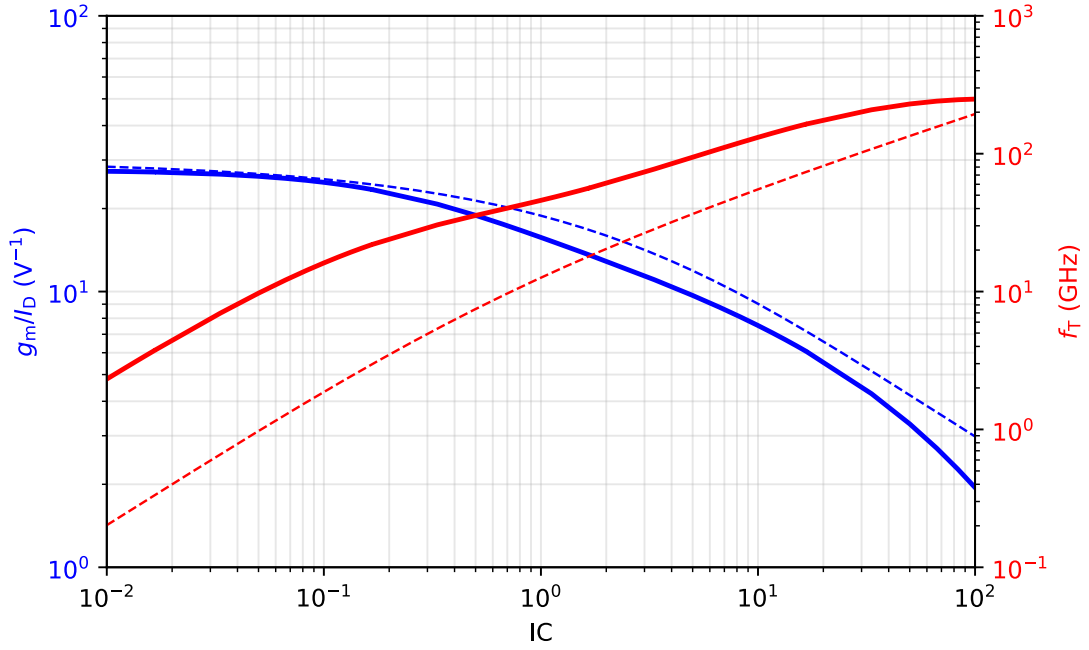


Figure 24: Transconductance efficiency g_m/I_D (blue, left axis) and transit frequency $f_T = g_m/(2\pi C_{gg})$ (red, right axis) versus inversion coefficient, extracted from a simulation of an NMOS ($W/L = 1.3 \mu\text{m}/0.13 \mu\text{m}$) in IHP SG13, using a specific current of $I_S = 6 \mu\text{A}$ extracted from the simulation data. The thin dashed lines show the analytical model (Equation 43 and Equation 44) with $n = 1.3$, $L = 0.13 \mu\text{m}$, and $\mu = 280 \text{ cm}^2/(\text{V s})$.

2.9.3 Sizing Procedure

The design procedure using the inversion coefficient is as follows:

1. Depending on the trade-off between low power (high g_m/I_D) and high speed (high f_T), select the IC (often, $IC \approx 1$ is a reasonable compromise).
2. Choose either the bias current or the required g_m , and calculate the other one using Equation 43.
3. Using Equation 42, calculate I_S from the known I_D .
4. Depending on the requirements for transistor speed (minimum L), matching (large $W \cdot L$), or small g_{ds} (large L), choose the appropriate L ; use Equation 40 to calculate the corresponding W .
5. Finally, estimate f_T using Equation 44. From f_T , the required bandwidth of the circuit can be estimated, and the sizing can be iterated if necessary.

Now the most important transistor parameters (W , L , I_D , g_m , V_{GS} , and f_T) are selected and known.

2.10 Summary

The MOSFET is a very versatile device, which can, depending on the bias conditions, work as a:

- **Switch** (for small V_{DS} and large V_{GS}).
- **Voltage-controlled resistor** (since $g_{ds} = f(V_{GS})$ in the triode region).
- **Voltage-controlled current source** (since $I_D = f(V_{GS})$ in the saturation region).

- **Diode** (since the MOSFET is off for $V_{GS} < V_{th}$ and on otherwise).
- **Variable capacitor** (using the gate oxide or drain/source junction capacitance).

This versatility is nicely illustrated by the collection of two-transistor circuits in [12]; the paper, plus posters showing all fifty circuits, is available at the Fifty Nifty Variations of Two-Transistor Circuits homepage.

When making the MOSFET physically small by scaling $L \ll 1 \mu\text{m}$ to improve power, performance, and area, various imperfections kick in (like DIBL), causing increased leakage currents and degraded output conductance—so alternative device constructions like SOI, FinFET, and nanosheet MOSFETs have been introduced.

2.11 Appendix: Silicon Material Properties

Table 5 collects the physical properties of silicon used throughout this text.

Table 5: Physical material properties of silicon

Physical parameter of Si	Symbol	Typical value
Lattice constant		0.357 nm
Density		$2.33 \times 10^3 \text{ kg/m}^3$
Relative permittivity of Si	$\epsilon_{r,\text{si}}$	11.9
Relative permittivity of SiO_2	$\epsilon_{r,\text{ox}}$	3.9
Band gap at 300 K	E_{bg}	1.12 eV
Intrinsic carrier concentration at 300 K	$n_i = p_i$	$1.0 \times 10^{10} \text{ cm}^{-3}$
Linear coefficient of thermal expansion	CTE	$2.6 \times 10^{-6} \text{ K}^{-1}$
Melting point		1415 °C
Mobility of electrons at 300 K	μ_n	$1500 \text{ cm}^2/(\text{V s})$
Mobility of holes at 300 K	μ_p	$450 \text{ cm}^2/(\text{V s})$
Thermal conductivity		$1.5 \text{ W}/(\text{cm K})$
Breakdown field	E_{crit}	$3 \times 10^5 \text{ V/cm}$

3 IC Technology

This chapter walks through the manufacturing technology of integrated circuits: from the raw silicon wafer through the individual processing steps (doping, photolithography, etching, thin-film deposition, and interconnect formation) to a complete CMOS process flow. The treatment largely follows [4].

3.1 The Silicon Wafer

The starting material for integrated-circuit production is, quite literally, sand: quartz sand (SiO_2) is reduced with coal in an electric arc furnace at about 2000 °C to roughly 98 % purity, following the reaction $\text{SiO}_2 + 2 \text{C} \rightarrow \text{Si} + 2 \text{CO}$. This “metallurgical-grade” silicon is then chemically purified via trichlorosilane ($\text{Si} + 3 \text{HCl} \rightleftharpoons \text{SiHCl}_3 + \text{H}_2$), which can be distilled and decomposed back into ultra-pure silicon.

From the purified silicon, a large single crystal—the **ingot**—is grown by the **Czochralski method**, illustrated in Figure 25: a seed crystal is dipped into molten silicon and slowly pulled upwards while rotating, so that the melt crystallizes at the interface, continuing the crystal structure of the seed. Further purification can be achieved by zone refining, where a small molten zone is moved along the ingot, dragging impurities along; the final purity reaches 99.999999 %, which is equivalent to one impurity atom per billion silicon atoms.

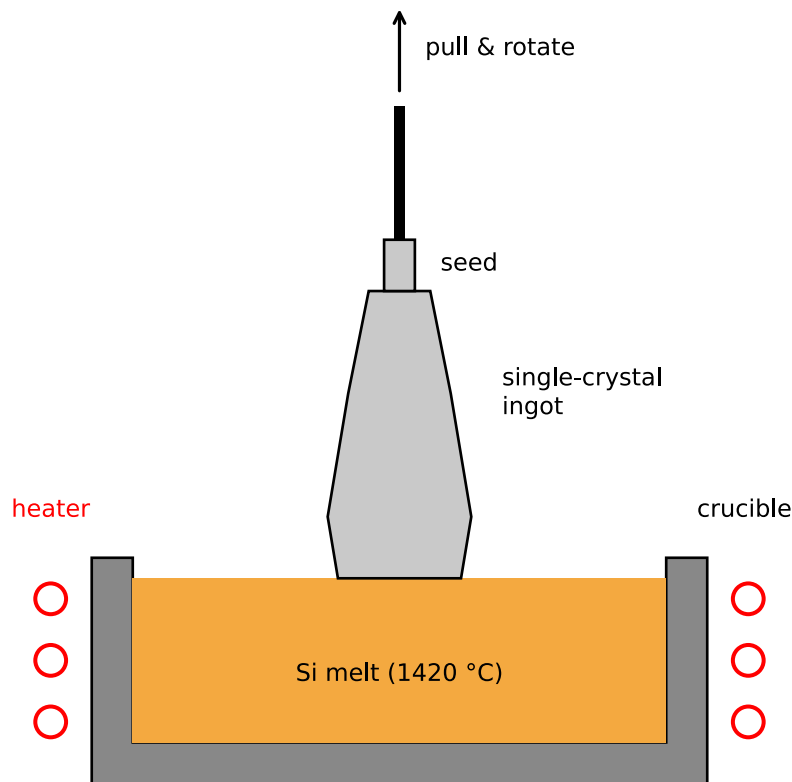


Figure 25: Czochralski crystal growth: a rotating seed crystal is slowly pulled out of the silicon melt, growing a large single-crystal ingot.

The ingot is then cut into **wafers** using a saw, to a thickness of 0.3 to 0.8 mm, and the wafers are polished. The standard wafer diameters today are 150 mm (6"), 200 mm (8"), and 300 mm (12"). The resistivity of the wafer material, set by doping, is 0.1 to 20 Ω cm, with up to 1000 Ω cm for specialty processes. Wafers can be *n*- or *p*-type, and often epitaxial ("epi") wafers are used.

3.1.1 Doping

The silicon atoms are spaced roughly 0.24 nm apart in the crystal lattice, resulting in about 5×10^{22} atoms/cm³. The intrinsic carrier concentration is $n_i = p_i \approx 1.0 \times 10^{10}$ cm⁻³ at 300 K, with an exponential temperature dependence (older textbooks often quote 1.5×10^{10} cm⁻³; the accepted measured value is 9.65×10^9 cm⁻³ [13]). By introducing doping materials in small quantities, the carrier concentration can be modulated strongly; see Table 6 (N_d and N_a are given in cm⁻³).

Table 6: Overview of n -type and p -type doping of silicon

n -type (donor materials)	p -type (acceptor materials)
Effect: Introduces additional free electrons	Introduces a lack of electrons ("holes")
Elements: Phosphorus (P), Arsenic (As)	Boron (B), Gallium (Ga), Indium (In)
Carrier concentrations: $n_0 \approx N_d, p_0 \approx n_i^2/N_d$	$p_0 \approx N_a, n_0 \approx n_i^2/N_a$

The resulting resistivity of doped silicon is shown in Figure 26 as a function of the doping concentration.

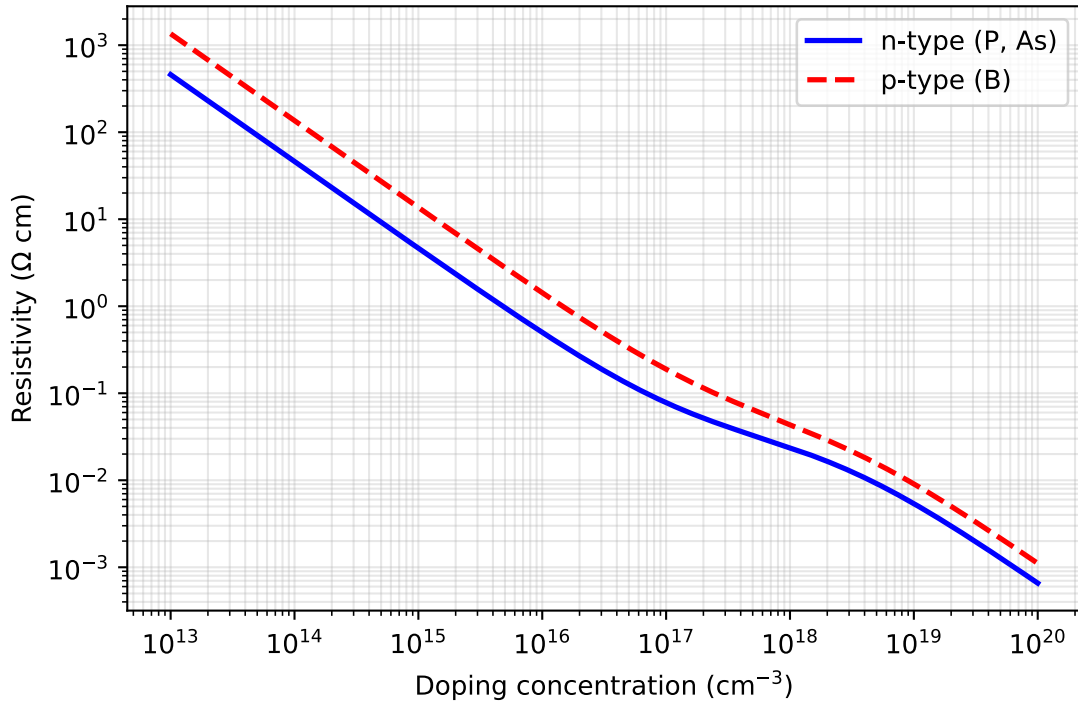


Figure 26: Resistivity of silicon at 300 K as a function of the doping concentration, calculated as $\rho = 1/(q\mu N)$ with doping-dependent carrier mobilities (Arora mobility fits). p -type silicon has a higher resistivity for the same doping level because of the lower hole mobility.

i Note 3: Example: Wafer Doping

Let us assume a p -type wafer with $50\ \Omega\text{ cm}$ resistivity (as used in IHP SG13 CMOS). Using Figure 26, we find $N_a = p_0 \approx 2.7 \times 10^{14}\text{ cm}^{-3}$, which results in $n_0 \approx n_i^2/N_a = 3.7 \times 10^5\text{ cm}^{-3}$. This doping concentration equates to about one acceptor atom per 180 million silicon atoms!

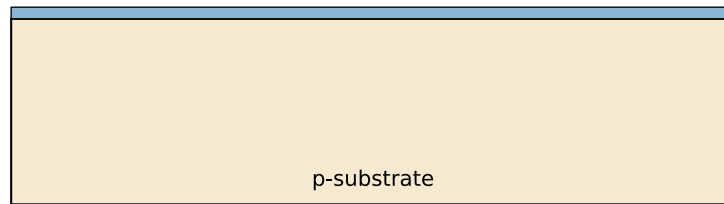
3.2 Selective Doping

To introduce dopants into the silicon wafer selectively, a masking process is used:

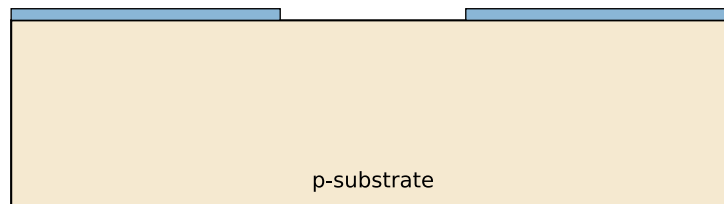
- The picture from a photomask is optically transferred into a photoresist on the wafer, using a process called **photolithography**.
- This photoresist is developed and used to selectively etch a masking layer, like SiO_2 (a “hard mask”).
- After this step, the silicon surface is opened at specific locations and covered otherwise, allowing the introduction of dopants into defined areas of the wafer.

The complete sequence—hard-mask formation, lithography, etching, dopant introduction, and annealing (which removes defects from the crystal structure while the dopants diffuse into the silicon)—is shown in Figure 27.

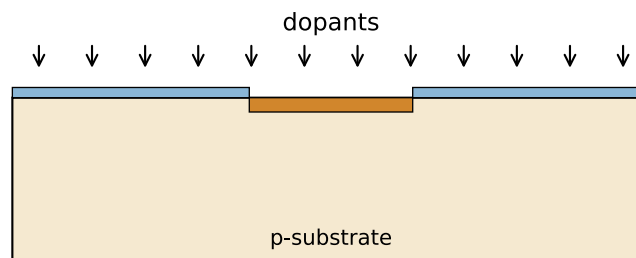
(a) grow/deposit hard mask



(b) open hard mask by lithography and etching



(c) introduce dopants



(d) anneal (dopants diffuse into the Si)

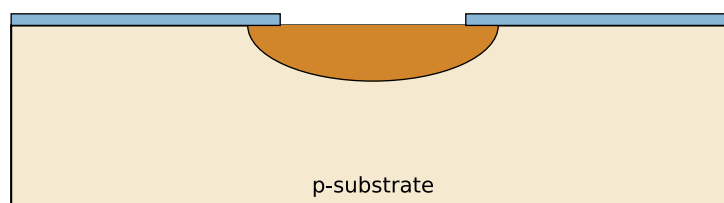


Figure 27: Selective doping using a hard mask: (a) the wafer is covered by a hard mask (e.g., SiO_2), (b) the hard mask is opened by photolithography and etching, (c) dopants are introduced through the opening (shown here by implantation), and (d) after annealing, the dopants have diffused into the silicon and the crystal damage is repaired.

3.2.1 Oxidation of Silicon

Oxidation of silicon is done at elevated temperatures in a furnace, using either “dry” ($\text{Si} + \text{O}_2 \rightarrow \text{SiO}_2$) or “wet” ($\text{Si} + 2 \text{H}_2\text{O} \rightarrow \text{SiO}_2 + 2 \text{H}_2$) oxidation. The oxide growth follows the **Deal-Grove model** [14]; the resulting growth curves are shown in Figure 28. Wet oxidation

is much faster, while dry oxidation yields denser, higher-quality oxides (used, e.g., for gate oxides).

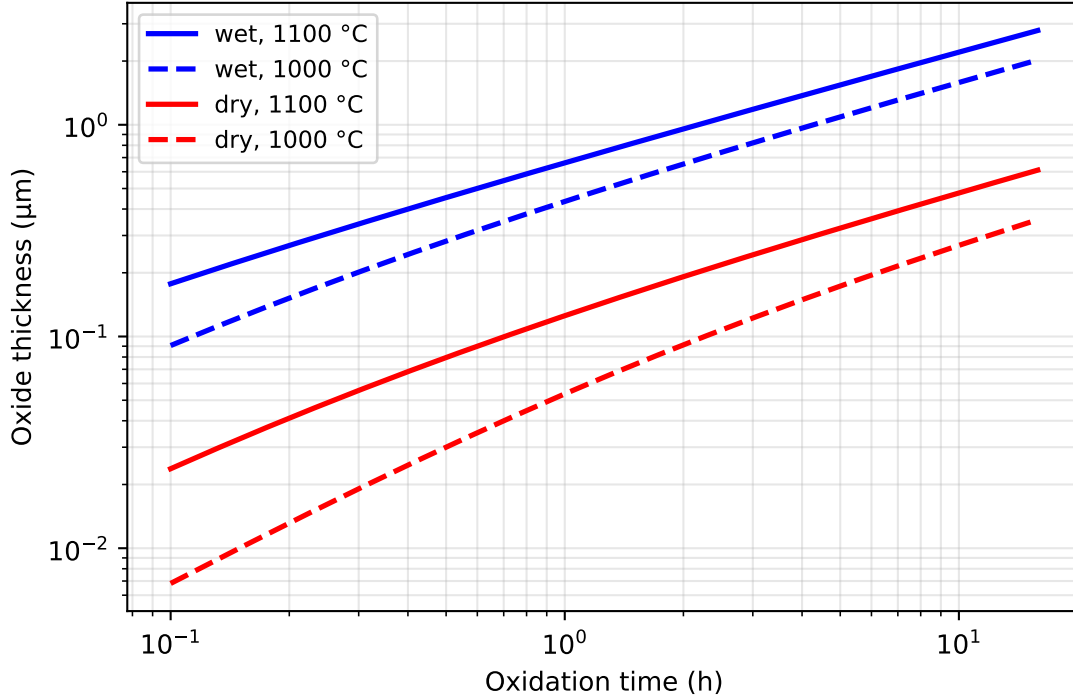


Figure 28: Thermal oxide growth on silicon calculated with the Deal-Grove model, $x_{\text{ox}}^2 + Ax_{\text{ox}} = Bt$, for wet and dry oxidation at different temperatures. Wet oxidation is much faster and used for thick isolation oxides, dry oxidation for thin, high-quality gate oxides.

The SiO_2 is used for:

- The gate dielectric of the MOSFET.
- Hard masks for implantation or doping.
- Device isolation and isolation of wiring.

The growth of SiO_2 consumes silicon; the silicon thickness is reduced by 44 % of the grown oxide thickness. Alternatively, chemical vapour deposition (CVD) can be used for depositing SiO_2 without consuming silicon. Local oxidation of silicon (LOCOS) can be used to laterally isolate active regions of an integrated circuit, but in modern CMOS technologies shallow trench isolation (STI) is used instead, for higher transistor density.

3.3 Photolithography

In photolithography, the wafer is first oxidized to create a hard-mask layer and then coated with photoresist. The photomask is imaged onto the photoresist with a projection lens system, the resist is developed and baked, and the hard mask is etched through the resist openings. There are two types of resist: with a **positive** resist, the exposed areas dissolve in the developer, transferring the mask image directly; with a **negative** resist, the exposed areas remain, creating an inverse image. After etching, the remaining photoresist is removed. The process is illustrated in Figure 29.

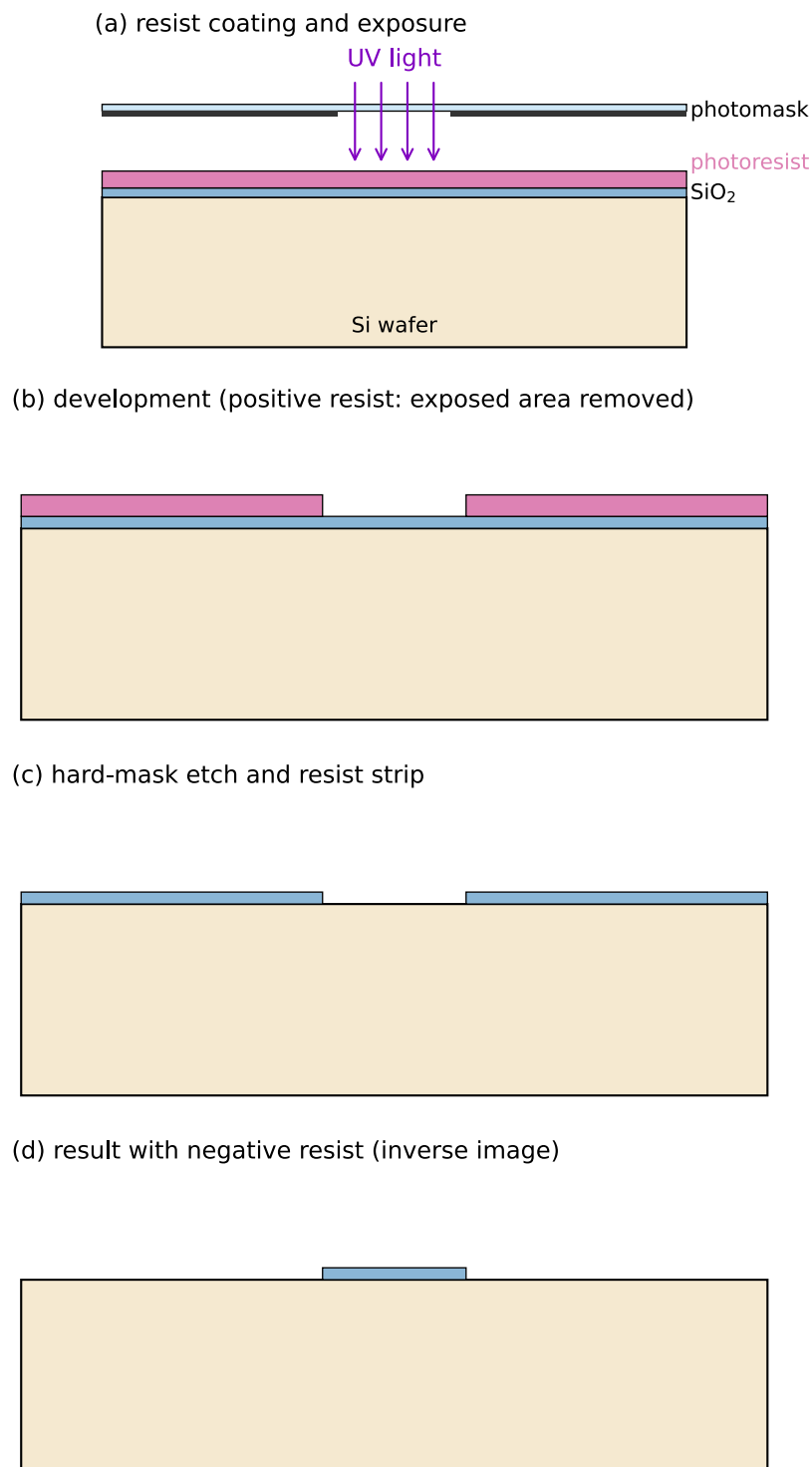


Figure 29: Patterning by photolithography: (a) the oxidized wafer is coated with photoresist and exposed through the photomask, (b) with a positive resist, the exposed resist dissolves during development, (c) the hard mask is etched through the resist opening and the resist is removed, and (d) a negative resist creates the inverse image.

3.3.1 Photomasks

Instead of a photomask which exposes the whole wafer at once, a **reticle** is mostly used, which is smaller than the wafer, with a field size of ca. 26 mm × 33 mm (limiting the maximum die size to 858 mm²). A **stepper** exposes the complete wafer field by field with the reticle. The photomask/reticle is usually made of a glass carrier with a structured chromium layer.

The reticle must be protected from contamination, so a thin transparent membrane (called a **pellicle**) keeps particles out of the focal plane, away from the mask structures. The reticle can be the same size (1×) or larger (2× or 4×) than the structures on the wafer, and the photomask can touch the wafer (“contact printing”) or be kept at a distance (“proximity printing”)—modern lithography uses projection printing through a reduction lens.

The complete exposure system is shown in Figure 30. The light passes from the source through the **illuminator**, which shapes the angular distribution of the illumination (e.g. into a dipole or quasar pattern), onto the reticle; the projection optics then image a demagnified copy of the reticle field into the resist, and the scanner steps this field across the wafer, one exposure at a time.

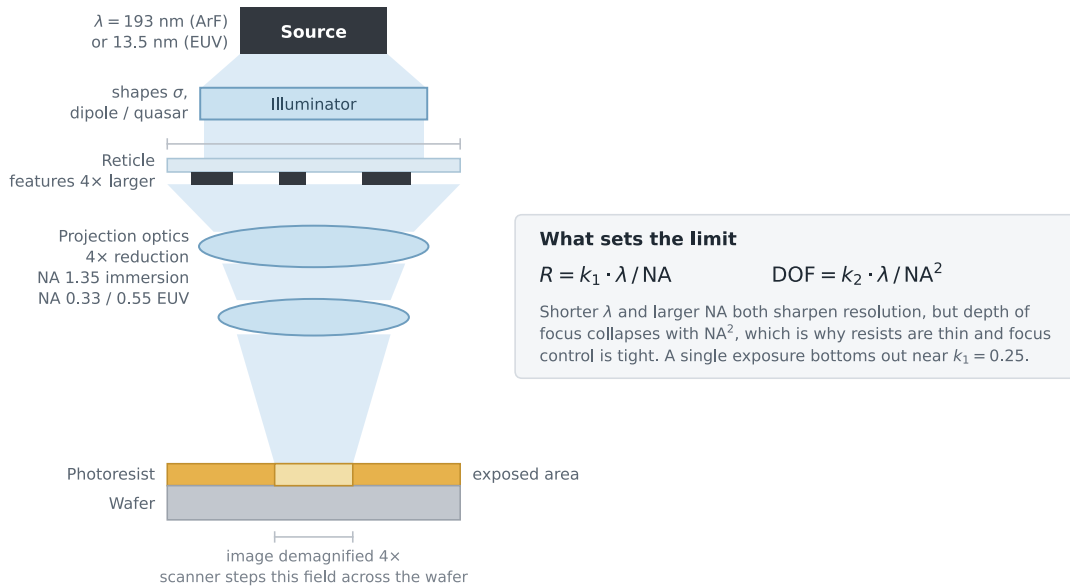


Figure 30: The optical column of a projection lithography scanner: light from the source passes through the illuminator, which shapes its angular distribution, onto the reticle, and the projection optics image a demagnified copy of the reticle field into the photoresist. The scanner exposes the wafer field by field. Resolution and depth of focus both follow from the wavelength and the numerical aperture of the optics.

3.3.2 Resolution Limit

As in any other optical system, the resolution limit of photolithography is given by the Rayleigh criterion:

$$R = k_1 \frac{\lambda}{n \sin \Theta} = k_1 \frac{\lambda}{NA} \quad (45)$$

Here, λ is the wavelength of the light source, and k_1 is a factor that captures various resolution-enhancement measures, like phase-shift masks or off-axis illumination ($0.25 \leq k_1 \leq 1$). NA is the numerical aperture of the lens system (in air, $\text{NA} \leq 0.93$). The resulting half-pitch R is the smallest pattern that can be printed on a wafer: $R = 100$ nm means that lines with a width of 100 nm and a spacing of 100 nm can be printed.

Resolution is not free. The **depth of focus**, i.e. the range over which the projected image stays sharp, shrinks with the *square* of the numerical aperture:

$$\text{DOF} = k_2 \frac{n\lambda}{\text{NA}^2} \quad (46)$$

Here, k_2 is the depth-of-focus counterpart of k_1 : it captures how much image degradation the process still tolerates, set by resist contrast and thickness, feature type and illumination scheme ($0.5 \leq k_2 \leq 1$). The two factors pull against each other—the same off-axis illumination that buys a small k_1 also shrinks k_2 , so pushing resolution costs focus budget twice. Note that the refractive index n does not cancel here: raising n recovers depth of focus at a given NA.

This is why resists are kept thin and why the wafer surface must be held in focus to within a few tens of nanometers while the field is scanned.

3.3.3 Light Sources

In the 1980s, mercury lamps with $\lambda = 436$ nm (g-line) or $\lambda = 365$ nm (i-line) were used. Starting with 0.25 μm CMOS (until 180 nm), excimer lasers with $\lambda = 248$ nm were introduced, called deep-ultraviolet (DUV) light. Since 130 nm CMOS, ArF lasers with $\lambda = 193$ nm are used, which can be combined with multi-patterning to break the half-pitch limit of $R \approx 40$ nm. Finally, extreme-ultraviolet (EUV) lithography was introduced at the 7 nm CMOS node, offering a huge reduction in wavelength to $\lambda = 13.5$ nm—accompanied by an extreme cost impact due to the need for all-reflective optics.

Note that there is only one company in the world that currently manufactures EUV lithography machines: ASML from the Netherlands. This gives ASML a unique position in the semiconductor equipment market, as EUV is critical for advanced nodes. Zeiss from Germany produces the high-precision optics required for these machines. Further, there is only one company in the world that makes the multi-beam mask writers needed for creating the EUV masks: IMS Nanofabrication from Austria.

3.3.4 Resolution Enhancements

Various methods are used to reduce the half-pitch in Equation 45 by compensating distortion due to diffraction and interference, modifying either the amplitude or the phase of the light:

- **Optical proximity correction (OPC)** enhances the contrast of critical layout features like line ends and corners by pre-distorting the mask shapes.
- **Phase-shift masks (PSM)** vary the thickness of the mask, so that phase changes of the light can be used for contrast enhancement.
- **Immersion lithography** uses the refractive index of water ($n = 1.44$) between lens and wafer to achieve $\text{NA} > 1$; per Equation 46, the higher n also restores part of the depth of focus lost to the larger NA.

Ultimately, using these techniques, structures considerably smaller than the wavelength of the used light can be printed.

3.3.5 EUV Lithography

EUV light at 13.5 nm is created by firing laser pulses at molten tin droplets. There are no transmissive lens materials for EUV, so the optics are constructed from reflective mirrors with about 30 % loss per mirror. A first-generation EUV scanner is a 180-tonne machine using 0.5 MW of electrical power to create roughly 200 W of optical power at the wafer, processing about 200 wafers per hour. Comparing the achievable resolution:

- ArF immersion lithography: $\lambda = 193 \text{ nm}$, $\text{NA} = 1.35$, $k_1 = 0.27$ (with PSM) $\Rightarrow R \approx 40 \text{ nm}$.
- First-generation EUV (NXE): $\lambda = 13.5 \text{ nm}$, $\text{NA} = 0.33$, $k_1 = 0.32 \Rightarrow R \approx 13 \text{ nm}$.
- Current high-NA EUV (EXE): $\lambda = 13.5 \text{ nm}$, $\text{NA} = 0.55$, $k_1 = 0.32 \Rightarrow R \approx 8 \text{ nm}$.

3.4 Etching

Etching is used to transfer the image from the photoresist into the hard mask (or from one layer into another layer) by selectively removing material. Etching can be isotropic or anisotropic, characterized by the anisotropy A ; the selectivity S measures the ratio of the etch rate of the film to the etch rate of the mask. An optimal etch has $A = 1$ (only vertical etching, no lateral etch) and $S = \infty$ (the mask is not attacked). Two types of etches are used:

- **Wet etching**, using liquid chemicals (typically isotropic).
- **Dry etching**, using chemically active ionized gases in a plasma reactor (can be highly anisotropic).

3.5 Doping Techniques

Three methods of introducing dopants into the silicon can be used:

- **Diffusion**: dopants move (by diffusion) from the wafer surface into the wafer material; the source of the dopants is a gas (e.g., POCl_3 for phosphorus doping) or a solid film on the wafer surface.
- **Ion implantation**: dopants are ionized, accelerated, and shot into the wafer material.
- **Epitaxy**: additional silicon is grown on the wafer surface and doped in situ while growing the silicon layer.

3.5.1 Diffusion

Diffusion is the movement of impurity atoms from the surface into the bulk of the silicon, performed at high temperatures (900 to 1200 °C). There are two limiting cases:

- **Infinite source** of impurities at the surface: the resulting doping profile follows a complementary error function (erfc).
- **Finite source** of impurities at the surface: the resulting doping profile is a Gaussian, $N(x, t) = N_0 / \sqrt{\pi D t} \cdot \exp(-x^2 / (4 D t))$, with x the distance from the wafer surface, N_0 the initial dopant dose per area, D the diffusivity, and t the diffusion time.

The drawback of diffusion is that the doping concentration is always highest at the wafer surface and constrained to these profile shapes.

3.5.2 Ion Implantation

Ion implantation lodges impurity ions into the target material at high velocity; implantation can even be done through surface layers (e.g., through the gate oxide for threshold-voltage adjustment). The resulting doping depth profile is

$$N(x) = \frac{N_i}{\sqrt{2\pi} \Delta R} e^{-\frac{(x-R)^2}{2\Delta R^2}} \quad (47)$$

which peaks at the projected range R ; the straggle ΔR is tied to R , so the achievable profiles are also constrained (N_i is the implanted dose per area). A subsequent anneal (typically a rapid thermal anneal at 900 to 1100 °C) is required to activate the impurities and to repair the damage to the crystal lattice. Figure 31 compares typical diffusion and implantation profiles.

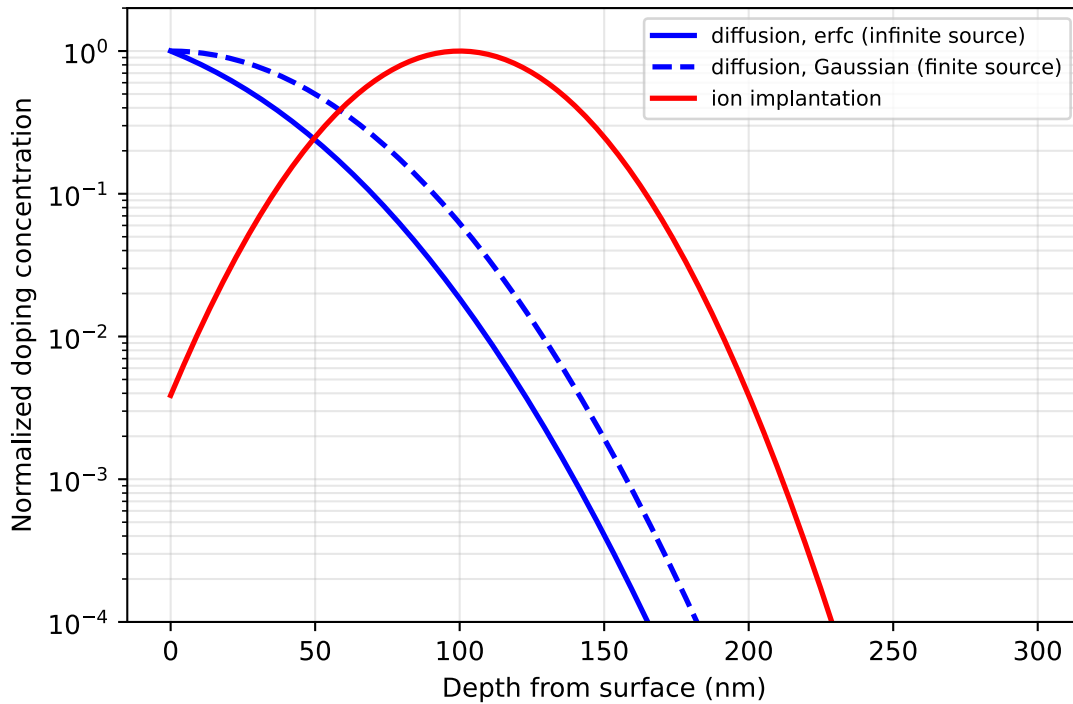


Figure 31: Doping depth profiles (normalized): diffusion always peaks at the wafer surface, following an erfc profile (infinite source) or a Gaussian profile (finite source), while ion implantation places the doping peak at the projected range R below the surface (shown for $R = 100$ nm, $\Delta R = 30$ nm).

3.5.3 Epitaxy

Epitaxial growth forms a layer of single-crystal silicon on the surface of the wafer, such that the crystal structure is continuous across the interface. The epi layer can be doped differently from the material where it is grown, with a typical thickness of 1 to 20 μm (patterned epitaxial growth using a mask is also possible).

3.6 Thin-Film Deposition

Various thin films are used in IC processing:

- The gate dielectric of the MOSFET, like SiO_2 (either thermally grown or deposited via CVD).
- The gate material, polysilicon, which is also used for local interconnects and resistors.

- Metal films for interconnects (metals like Al or Cu, and silicides like TiSi_2 , WSi_2 , TaSi_2 , NiSi_2).
- Contact and via materials, like W, Al, or Cu.
- Dielectrics between metal layers and spacers, like SiO_2 and Si_3N_4 .

These films are either **sputtered** or deposited from chemical gases in **chemical vapour deposition (CVD)**; both methods are sketched in Figure 32. Exemplary chemical reactions for CVD are:

- Depositing polysilicon: $\text{SiH}_4 \rightarrow \text{Si} + 2 \text{H}_2$
- Depositing SiO_2 (as an alternative to thermal oxidation): $\text{SiH}_4 + \text{O}_2 \rightarrow \text{SiO}_2 + 2 \text{H}_2$ or $\text{SiH}_2\text{Cl}_2 + 2 \text{N}_2\text{O} \rightarrow \text{SiO}_2 + 2 \text{HCl} + 2 \text{N}_2$
- Depositing Si_3N_4 : $3 \text{SiH}_2\text{Cl}_2 + 4 \text{NH}_3 \rightarrow \text{Si}_3\text{N}_4 + 6 \text{HCl} + 6 \text{H}_2$

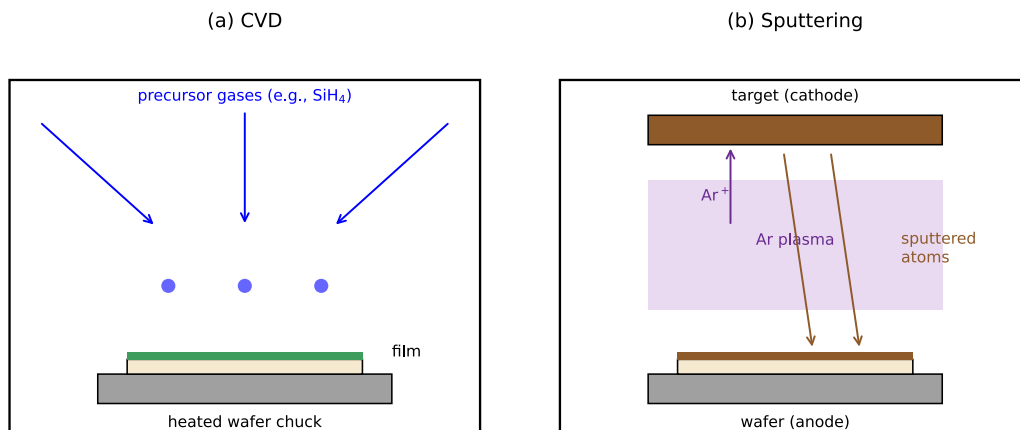


Figure 32: Thin-film deposition methods: (a) in chemical vapour deposition (CVD), precursor gases react at the heated wafer surface and deposit a thin film; (b) in sputtering, ions from a plasma knock atoms out of a target, which then condense on the wafer surface.

3.7 Interconnects

In the so-called **back-end** of the process, the devices formed in the **front-end** (like MOSFETs and resistors) are connected by a sandwich of metallic and dielectric thin films. The metal layers are connected through the dielectric layers by **vias** (between metals) and **contacts** (from the lowest metal layer to the front-end devices).

Historically, aluminum has been used as the interconnect metal, as it can be deposited by sputtering and then patterned by etching. However, Al is prone to void formation by electromigration. In newer technologies, copper is used: its resistance is 40 % lower, and it has excellent electromigration reliability. However, dry etching of Cu is difficult, so a **damascene** process is used, shown in Figure 33: the dielectric is structured, a liner and Cu are deposited into the trenches, and the excess metal is removed by **chemical-mechanical polishing (CMP)**. The liner prevents the Cu from diffusing into the dielectric.

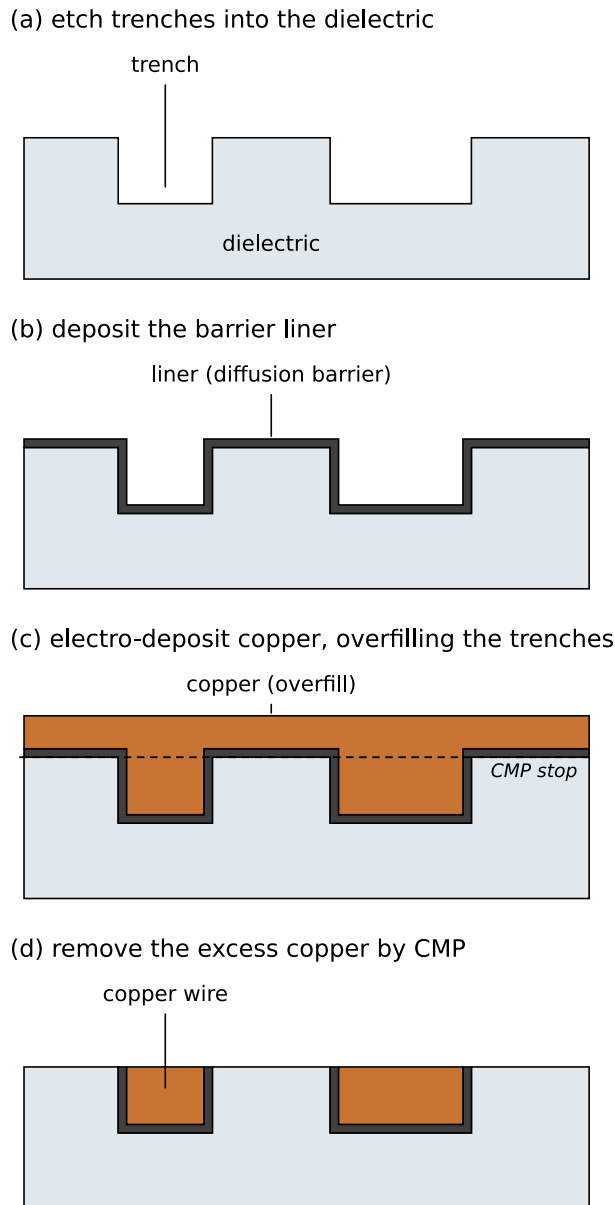


Figure 33: The damascene process for copper interconnects: (a) trenches are etched into the dielectric, (b) a conformal diffusion-barrier liner is deposited, (c) copper is (electro-)deposited, overfilling the trenches, and (d) the excess copper and the liner on the field are removed by chemical-mechanical polishing (CMP), leaving metal only in the trenches.

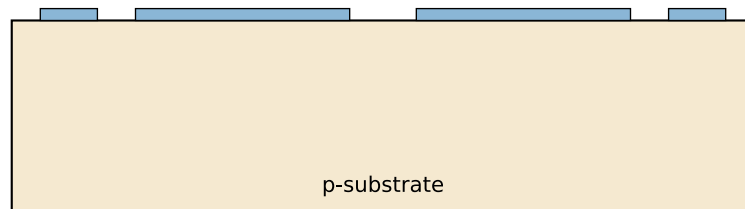
The classic dielectric between the metal layers is SiO_2 , but in order to reduce capacitive parasitics, low-k dielectrics (e.g., F- or C-doped SiO_2) are used in modern processes. Between processing steps, CMP is used to planarize the surfaces.

3.8 An Exemplary CMOS Process Flow

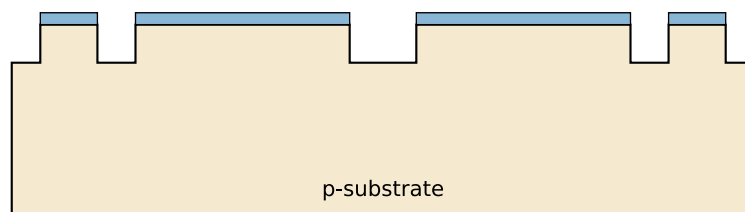
The following figures illustrate a simplified CMOS process flow, inspired by the openly documented LibreSilicon process; the cross sections show an NMOS (in the p-well) and a PMOS (in the n-well), each with its well tap on the outside, evolving through the process steps.

Step 1—Shallow trench isolation: After an initial cleaning, a hard mask (SiO_2 or Si_3N_4) is grown/deposited and patterned by photolithography. The isolation trenches are etched anisotropically and filled with oxide (Figure 34).

(a) hard mask grown and patterned



(b) anisotropic etch of the isolation trenches



(c) trenches filled with oxide, hard mask removed

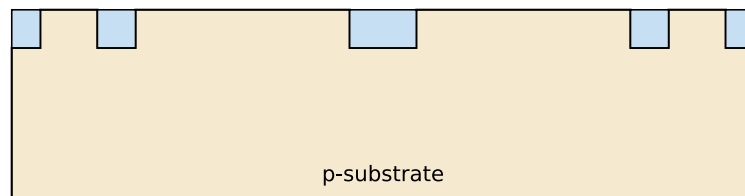


Figure 34: CMOS process, step 1—shallow trench isolation: (a) a hard mask is grown and patterned by photolithography, (b) the isolation trenches are etched anisotropically into the silicon, and (c) the trenches are filled with oxide and the surface is planarized.

Steps 2 and 3—Well formation: A hard mask is formed and opened over the p-well region, boron is implanted, and the wafer is annealed. The sequence is repeated with phosphorus for the n-well (Figure 35).

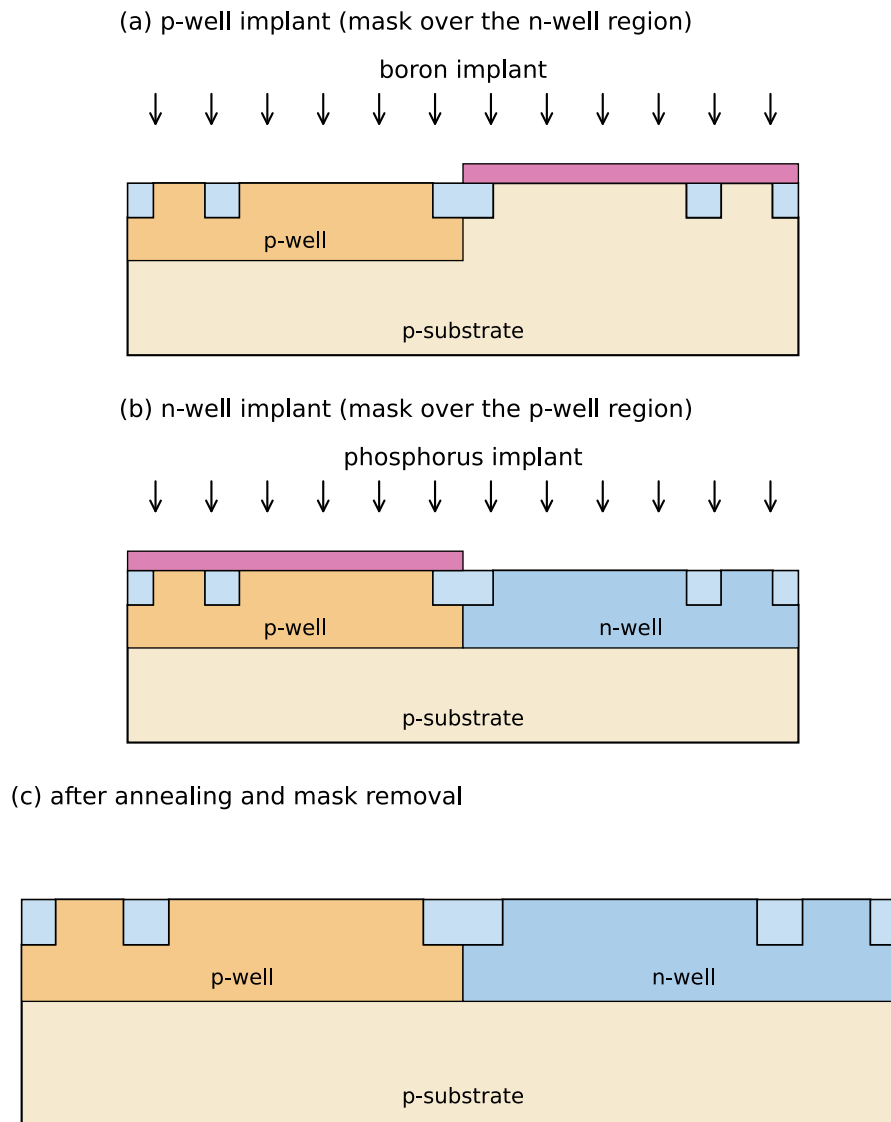
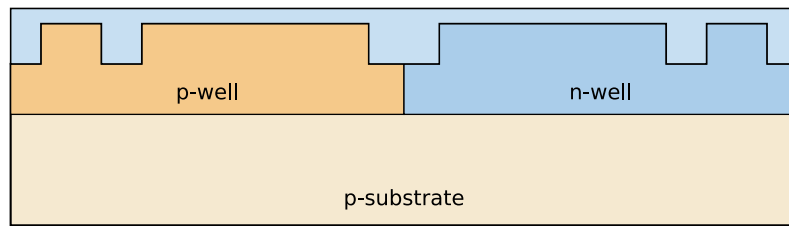


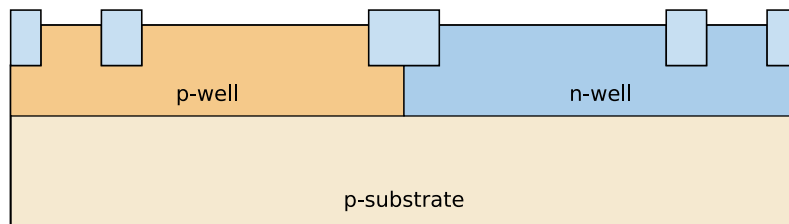
Figure 35: CMOS process, steps 2 and 3—well formation: (a) a mask covers the right half of the wafer while boron is implanted to form the p-well, (b) the complementary mask is used for the phosphorus implant of the n-well, and (c) after annealing, both wells are formed.

Steps 4 and 5—Gate formation: A field oxide is grown and removed over the active areas; then the thin gate oxide is grown by dry oxidation, polysilicon is deposited by CVD, and the gate stack is patterned (Figure 36).

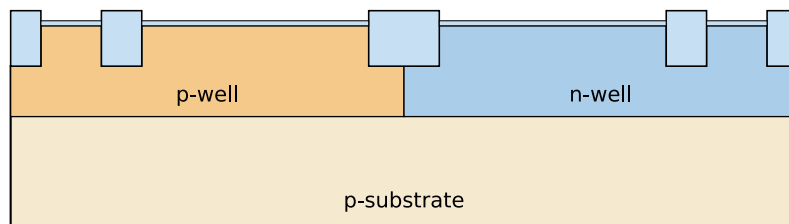
(a) field-oxide growth over the whole wafer



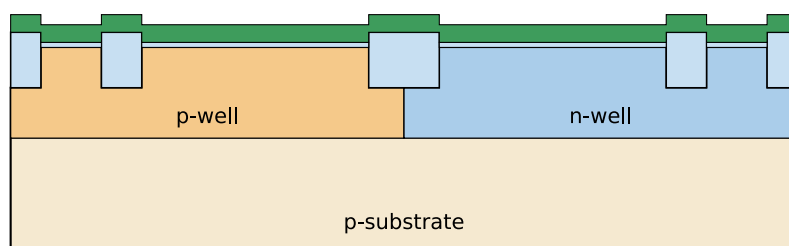
(b) field oxide removed over the active areas



(c) gate-oxide growth (dry oxidation)



(d) polysilicon deposition (CVD)



(e) gate stack patterned

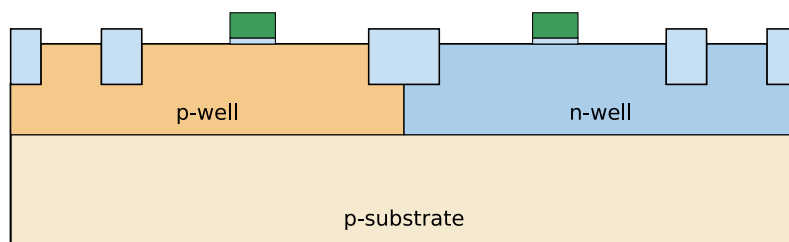


Figure 36: CMOS process, steps 4 and 5—gate formation: (a) the field oxide is grown over the whole wafer, merging with the oxide already filling the trenches, (b) it is removed again over the active areas, (c) the thin gate oxide is grown by dry oxidation on the exposed active areas, (d) polysilicon is deposited by CVD over the whole wafer, following the topography of the field oxide, and (e) the gate stack is patterned by photolithography and etching.

Steps 6 and 7—Source/drain implants: With photoresist masking the PMOS area, phosphorus is implanted to form the NMOS source/drain regions (and the n-well contacts); the implant is **self-aligned** to the gate, which acts as an implantation mask for the channel region. The complementary boron implant forms the PMOS source/drain regions (and the p-well contacts). Each implant is followed by an anneal (Figure 37).

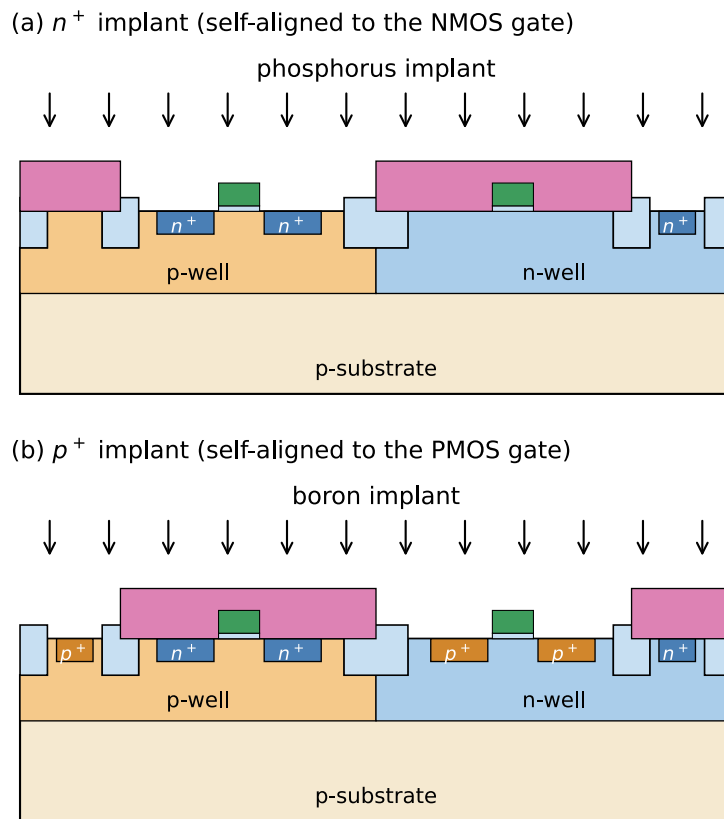
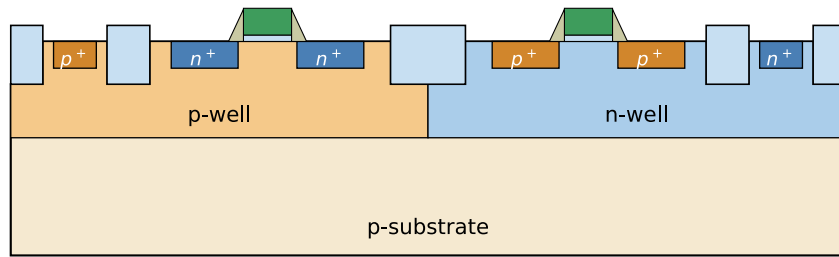


Figure 37: CMOS process, steps 6 and 7—self-aligned source/drain implants: (a) with the PMOS and the p-well tap covered by resist, phosphorus is implanted; the polysilicon gate itself masks the channel, so the n^+ regions are self-aligned to the gate, and the same implant forms the n^+ tap contacting the n-well; (b) the complementary boron implant forms the p^+ source/drain regions of the PMOS and the p^+ tap contacting the p-well.

Step 8—Silicide formation: An oxide/nitride layer is deposited and anisotropically etched, leaving **spacers** at the gate edges. Titanium is sputtered onto the wafer and reacts where it touches silicon, forming low-resistance TiSi_2 on the source/drain areas and the gate; the unreacted metal is removed (Figure 38).

(a) spacer formation at the gate edges



(b) silicide (TiSi_2) on source/drain and gate

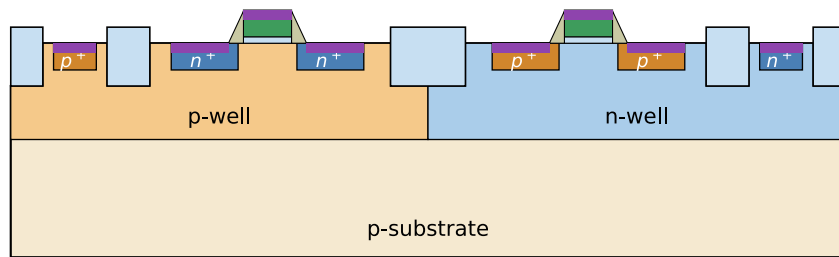
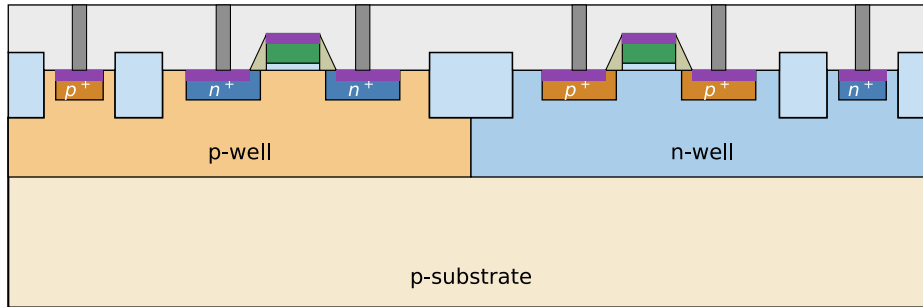


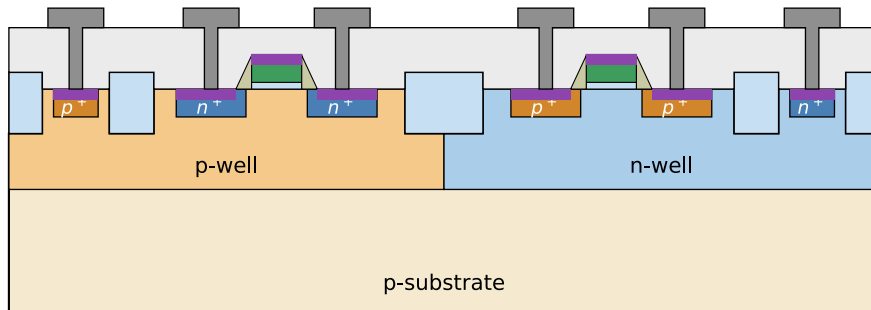
Figure 38: CMOS process, step 8—spacer and silicide formation: (a) an oxide layer is deposited and anisotropically etched, leaving spacers at the gate edges, and (b) sputtered titanium reacts where it touches silicon, forming a low-resistance silicide (dark) on source/drain and gate; the unreacted metal is then removed.

Steps 9 to 11—Contacts and metal stack: An inter-layer dielectric is deposited and planarized by CMP, contact holes are etched and filled, and the first metal layer is deposited and patterned. Further levels of dielectric, vias, and metal follow; finally, the passivation is deposited and opened at the pads (Figure 39).

(a) inter-layer dielectric, CMP, and contacts



(b) metal-1 sputtered (Al) and patterned



(c) inter-metal dielectric, via-12, and metal-2

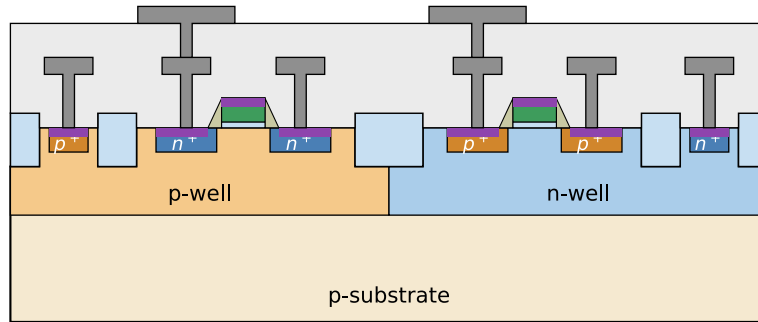


Figure 39: CMOS process, steps 9 to 11—back-end: (a) inter-layer dielectric deposition, CMP, and contact formation, (b) first metal layer sputtered and patterned, and (c) inter-metal dielectric, vias, and second metal layer; the finished wafer is then passivated and opened at the pads.

3.9 The Open-Source PDK IHP SG13

IHP has open-sourced the process design kit (PDK) for a 130 nm BiCMOS technology (the current generation is named SG13G2, referred to here simply as IHP SG13) under the Apache-2.0 license. The relevant files, including documentation, can be found at github.com/IHP-GmbH/IHP-Open-PDK. This technology has a lot of interesting features:

- Support for internal 1.2 V/1.5 V core MOSFETs and 3.3 V I/O MOSFETs

- An aluminum metal stack with 5 thin and 2 thick levels of metal (see Figure 40)
- Silicided, standard, and high-sheet-rho poly resistors
- A metal-insulator-metal (MIM) capacitor
- A Schottky diode
- Optional high-speed SiGe:C heterojunction bipolar transistors (HBTs)

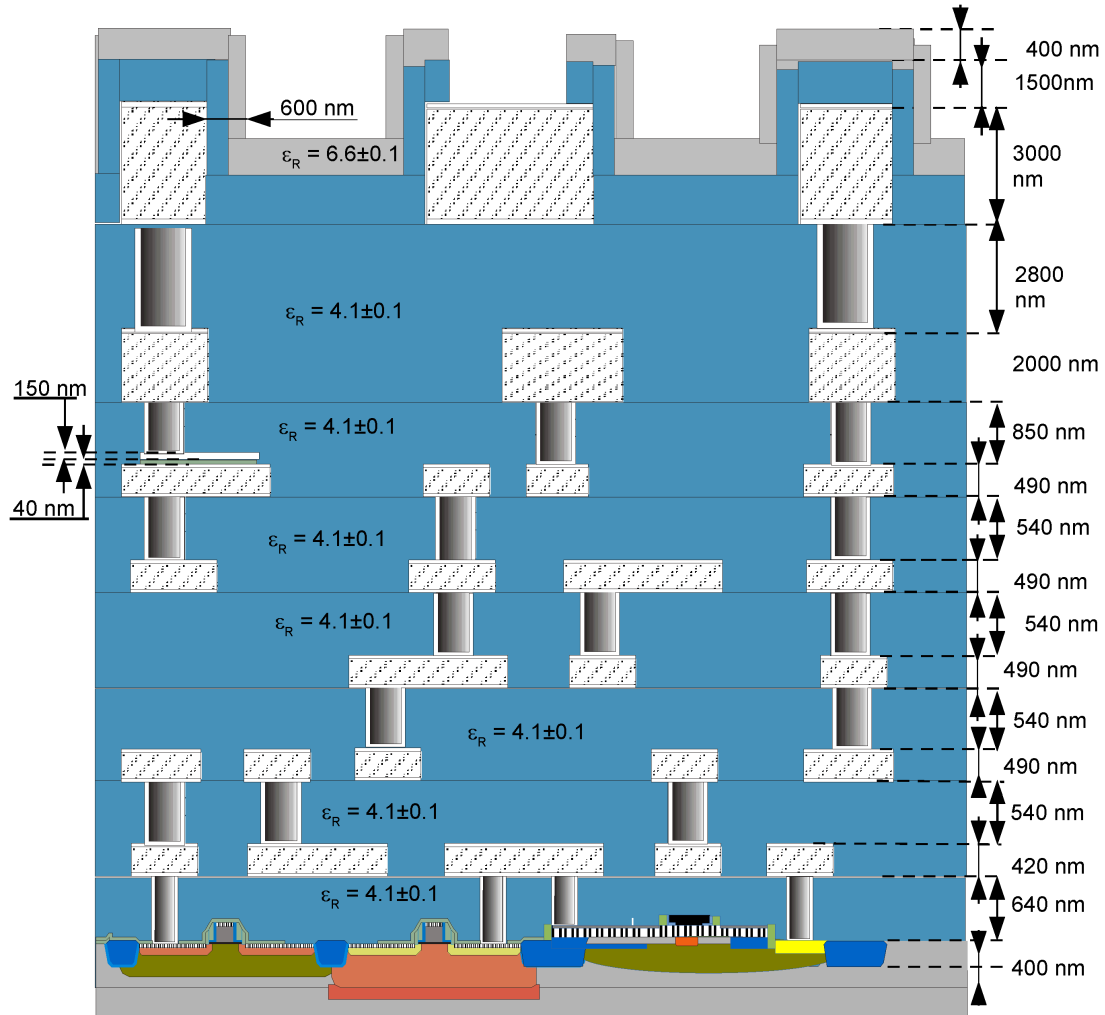


Figure 40: Cross-section of the IHP SG13 technology (not to scale): five thin metal layers (Metal1 to Metal5) for dense routing and two thick top metal layers (TopMetal1, TopMetal2) for power routing, inductors, and low-loss RF interconnects, with the MIM capacitor embedded between Metal5 and TopMetal1. The FEOL devices are shown at the bottom, and the numbers give the layer and dielectric thicknesses. Source: Fig. 1.1.1 of the *SG13G2 Process Specification Rev. 1.2*, Copyright 2023 IHP PDK Authors, licensed under Apache-2.0, taken from IHP-Open-PDK and cropped out of the original page.

3.10 Technology Qualification

Before a product in a certain semiconductor technology is shipped to customers, a qualification procedure has to be completed. Its purpose is to demonstrate, on a statistically meaningful sample and before the first customer shipment, that the product will meet its datasheet over the intended lifetime and operating conditions. The overall framework is set by JEDEC

JESD47 (*Stress-Test-Driven Qualification of Integrated Circuits*), which references the individual test methods; automotive products additionally follow AEC-Q100, which defines temperature grades and larger sample sizes. Table 7 lists typical tests.

Table 7: Typical technology qualification tests

Test (examples)	Standard	Motivation & check
HTOL (high-temperature operating life)	JESD22-A108	Subject the IC to high temperature under operation over an extended duration.
ESD HBM	JS-001-2023	ESD test, human-body model.
ESD CDM	JS-002-2022	ESD test, charged-device model.
Latch-up	JESD78	Latch-up test.
Electrical test (ED)	JESD86	Check conformance to the datasheet across conditions and lots.

These tests fall into three groups: **lifetime tests** that accelerate wear-out, **robustness tests** that check survival of a single electrical stress event, and **electrical verification** against the datasheet.

HTOL stresses a sample of packaged devices—typically three lots of 77 parts, so that zero failures give a meaningful confidence bound—at an elevated junction temperature, usually 125 °C and often combined with a supply voltage above nominal, while the circuit is electrically exercised, for a duration of typically 1000 h. It accelerates exactly those wear-out mechanisms that limit the life of a MOSFET: **time-dependent dielectric breakdown (TDDB)** of the gate oxide, **hot-carrier injection (HCI)**, **bias temperature instability (BTI)**, and electro-migration in the interconnect. Since these mechanisms are thermally activated, an Arrhenius acceleration factor—commonly in the range of 20 to 100—translates the 1000 h of stress into years of operation at the application temperature. A device fails the test if it stops working, or if any parameter has drifted outside the datasheet limits.

ESD HBM models the electrostatic discharge of a charged person touching a pin. The **human-body model** discharges a 100 pF capacitor through a 1.5 kΩ resistor, which produces a comparatively slow pulse (rise time of a few nanoseconds, decay time constant of 150 ns) with a peak current of about 0.67 A per kV. Every pin is stressed against every supply, in both polarities; a common target is 2 kV. The protection consists of diodes at each pad plus a clamp across the supply, arranged so that there is a low-impedance path between *any* two pins—which is why ESD protection is a full-chip problem, not a per-pad one.

ESD CDM models the opposite situation: the device itself has been charged, for example by sliding through a shipping tube, and discharges when one of its pins touches a grounded surface. The **charged-device model** pulse is far faster than HBM, with a rise time below 500 ps and a total duration of about 1 ns, reaching peak currents of several amperes. Because the pulse is largely over before the protection structures have fully turned on, CDM is usually the harder of the two to pass, and it is the more relevant one in modern automated assembly, where a human rarely touches the parts. The stored charge grows with package size, so large

packages are stressed hardest. A typical target is 500 V, and interfaces between separate supply domains are the classic weak spot.

Latch-up targets the parasitic thyristor that is inherent to every CMOS well structure: the source of a PMOS, the n-well, the p-substrate, and the source of a neighboring NMOS form a PNP structure, equivalent to a pair of cross-coupled parasitic bipolar transistors. If enough current is injected into a well—by an I/O pin driven beyond a supply rail, or by a supply overshoot—the structure latches into a low-impedance state that shorts supply to ground and persists until the power is removed, usually destroying the chip. JESD78 therefore injects ± 100 mA into the I/O pins and applies an overvoltage to the supplies, at the maximum operating temperature. The countermeasures are all layout measures: a generous number of well and substrate taps, which is why the flow above places one next to each transistor (Figure 37), guard rings around injecting circuits, and minimum-spacing rules between NMOS and PMOS.

Electrical test (ED) is not a stress test. It assesses the electrical parameters of devices from the manufacturing and test process being qualified, verifying that they meet every datasheet parameter across the specified supply-voltage and temperature range, and that they do so consistently across several production lots. This is what catches a systematic process shift that leaves the technology inside its own control limits, but the product outside its specification.

A complete qualification also covers the package rather than the silicon, with temperature cycling, highly accelerated stress testing (HAST) for moisture resistance, and solder-reflow sensitivity (see Section 5.6).

4 The Layout

The **layout** is the 2D representation of the IC manufacturing data. While the schematic captures the electrical intent of a design, the layout captures its physical realization: in the layout tools, the design is drawn on several **layers**, which are the foundation for the generation of the photomasks used in IC production. Commercial layout tools are, e.g., Cadence Virtuoso, Siemens L-Edit, and open-source tools are Magic and KLayout. The layers are usually manipulated before the geometries are transferred to the masks (shrink, enlarge, invert, combination of several layers into one mask layer, optical proximity correction, etc.).

4.1 Files and Tapeout

The layout is checked for compliance with the wafer-fab design rules in the **design rule check (DRC)**, and for conformance with the intended circuit in the **layout-versus-schematic (LVS)** check, where a netlist extracted from the layout is compared to the netlist of the original schematic. If all checks pass, the layout data is sent to mask production—this step is called **tapeout**. The mask house processes the layout data and produces the photomasks used by the wafer fab (depending on technology complexity, between 10 and 80 masks).

The standard file format for layout data is **GDSII**, a hierarchical database in a binary stream format. A newer format with more efficient storage is **OASIS**, but GDSII is still very common.

4.2 Generic Layer Definition

Different foundries use different layer names. The layer names defined in Table 8 are quite typical for a basic CMOS process technology and are used for the remainder of the text; the corresponding IHP SG13 names are given for reference.

Table 8: Generic layer definition used in this text

Layer name	IHP SG13	Purpose
RX	Activ	n^+ (RX) or p^+ (RX in BP) diffusion regions (e.g., S/D of MOSFETs)
PC	GatPoly	Polysilicon gate
NW	NWell	n -well region
BP	pSD	p -implant marker (i.e., blocks the n -implant)
OP	SalBlock	Silicide blocking over PC and RX
CA	Cont	Contact (connects RX or PC to M1)
M1	Metal1	First-level metal
V1	Via1	Via connecting M1 to M2
M2	Metal2	Second-level metal
V2	Via2	Via connecting M2 to M3
M3	Metal3	Top metal, mostly aluminum, to facilitate wire-bond packages
PAD	Passiv	Opening in the passivation to contact the top metal for pads

4.3 Devices

4.3.1 NMOS and PMOS Transistors

Figure 41 shows the layout (mask top view) of an NMOS and a PMOS transistor. The following observations can be made:

- The MOSFET is created by the intersection of RX and PC; this intersection defines W and L of the transistor. The areas of RX not covered by PC become the source and drain regions.
- Whether a device is NMOS or PMOS is decided by whether the RX shape is contained inside a BP region (RX plus BP is p^+ , RX without BP is n^+).
- A p^+ substrate contact is created by drawing RX covered by BP (outside the n -well).
- An n^+ well contact is created by drawing plain RX located inside NW.
- The NW must surround the PMOS with sufficient margin.
- The layer BP is a pure marking layer, assisting mask generation and device extraction from the layout.
- CA contacts connect source/drain/gate as well as the p^+/n^+ contacts to M1.

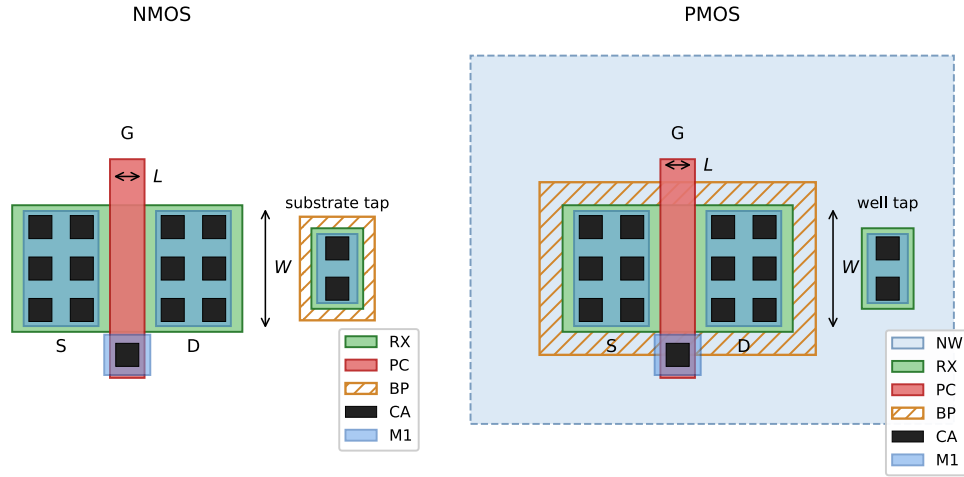


Figure 41: Layout (mask top view) of an NMOS (left) and a PMOS transistor (right). The transistor channel is formed where **PC** crosses **RX**; the **BP** marker turns the diffusion p^+ , and the **NW** well surrounds the PMOS. The area where **PC** overlaps **RX** is the actual transistor channel, defining its W and L . Next to each transistor sits the corresponding substrate or well contact. Note that **BP** acts as a marking layer, whereas **RX**, **PC**, and **NW** define the actual device regions on the silicon.

4.3.2 Polysilicon Resistor

The polysilicon resistor (Figure 42) uses the gate material (**PC**) as a resistive device. Since the gate poly is normally covered by silicide to lower its resistance, the silicide is suppressed with an **OP** shape; the intersection of **OP** with **PC** precisely defines the L and W of the resistor. Note that the existence of **OP** is a process option and might not be available in every technology. The gate polysilicon can be either p - or n -type; p -type poly resistors are often preferred, as they offer a lower temperature coefficient (thus the **PC** is marked with **BP**).

The resistance value of the polysilicon resistor is given by (R_s is the sheet resistance in $\Omega/\square$)

$$R = \rho_t \frac{L}{W} = R_s \frac{L}{W} \quad (48)$$

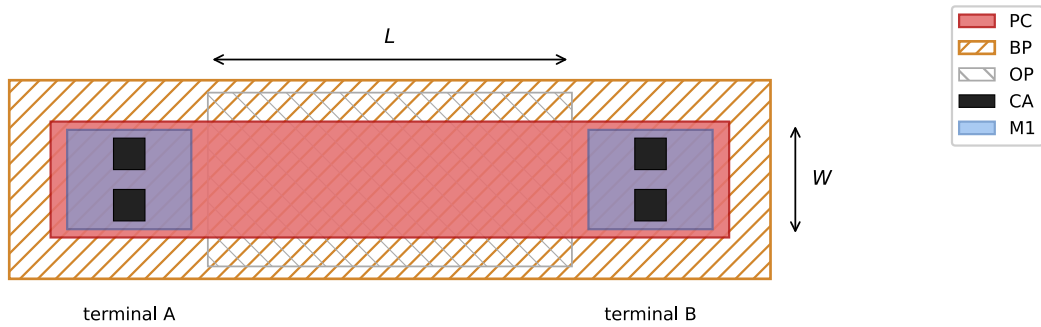


Figure 42: Layout of a polysilicon resistor: the silicide on the **PC** strip is blocked by the **OP** shape, whose intersection with **PC** defines the resistor body of length L and width W ; the **BP** marker selects p -type poly. The silicided ends are contacted with **CA** and **M1**.

Note that modern FinFET technologies might not support traditional planar polysilicon resistors, as the gate material is used to form the fins rather than a continuous sheet. In this case, often a metal resistor implemented in the back-end layers is used as an alternative.

4.3.3 Diffusion Resistor

Diffusion resistors (Figure 43) use source/drain diffusions or wells (with or without silicide) and are available in basically all silicon technologies, but they are usually inferior to polysilicon resistors: they have larger tolerances and a relatively high voltage coefficient, which can cause nonlinearity, and the nonlinear parasitic junction capacitance to the substrate can be critical as well. In older technology nodes ($> 0.25 \mu\text{m}$), the matching of diffusion resistors was better than that of poly resistors. One clear advantage of diffusion resistors is reduced self-heating, as the resistor body is embedded in the silicon substrate, whereas poly resistors are isolated from the substrate by SiO_2 . So resistor with increased power dissipation can benefit from this property.

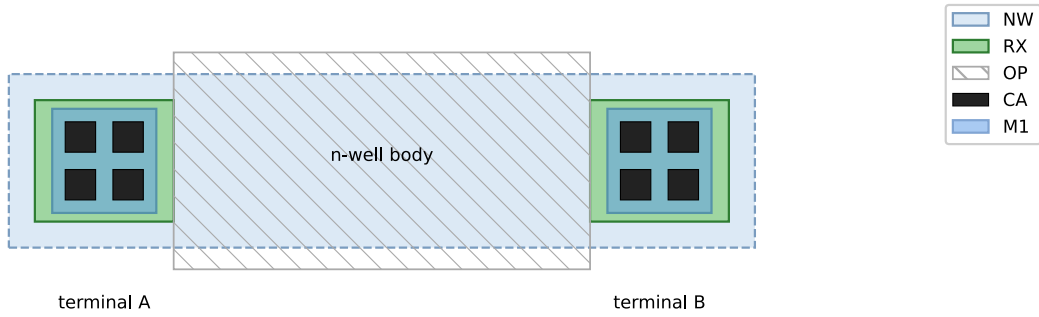


Figure 43: Layout of a diffusion (n-well) resistor: the resistive body is the **NW** strip, contacted at both ends by n^+ diffusions (**RX**); the **OP** shape blocks the silicide over the body. The parasitic junction to the substrate makes this resistor nonlinear with the applied voltage.

The resistor types available in IHP SG13 are summarized in Table 9.

Table 9: Resistor types available in IHP SG13

Type	R_s ($\Omega/\square$)	Process tolerance ($\pm 4\sigma$)	Matching	Temp. coefficient
Silicided n^{++} -poly (rsil)	7	$\pm 12 \%$	$0.6 \mu\text{m}$	$0.3 \%/K$
Blocked p^+ -poly (rppd)	260	$\pm 10 \%$	$1.5 \mu\text{m}$	$0.017 \%/K$
Blocked compensated poly (rhigh)	1360	$\pm 15 \%$	$4.8 \mu\text{m}$	$-0.23 \%/K$

4.3.4 Capacitors

Linear capacitors can be implemented using the metal stack, either as **vertical sandwich capacitors** (parallel plates on adjacent metal layers) or as **finger capacitors** (exploiting the lateral coupling between interdigitated fingers on the same layers); both are shown in Figure 44. Depending on the design rules, the vertical or the lateral coupling capacitance

might be more area-effective; such metal-oxide-metal (MOM) capacitors come for free in any technology. Due to the very tight lateral pitch in nm CMOS technologies, finger capacitors are often preferred for capacitor implementation (when a linear capacitor is required).

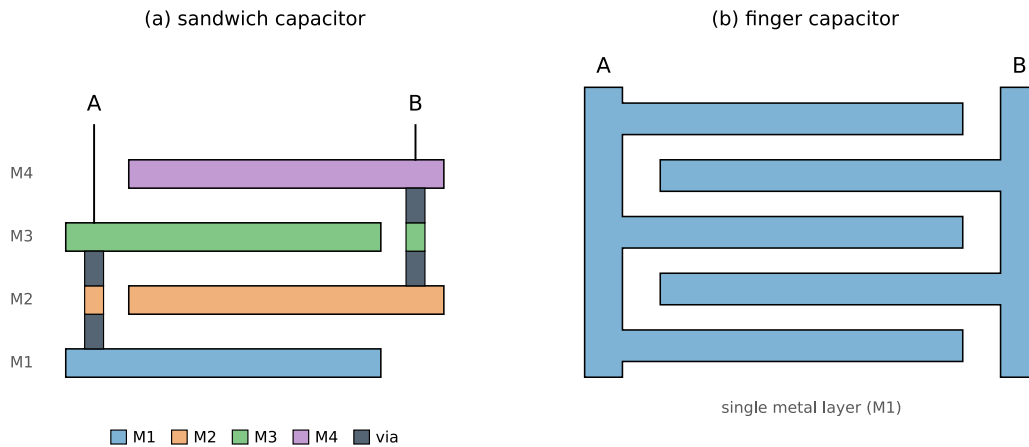


Figure 44: Metal-oxide-metal (MOM) capacitors: (a) vertical sandwich capacitor (cross section) using the plate capacitance between stacked metal layers, with alternating layers connected to the two terminals by via stacks; (b) finger capacitor (top view) using the lateral coupling capacitance between interdigitated fingers drawn in a single metal layer.

A few additional considerations for capacitors:

- The density of MOM capacitors scales with smaller design rules and the number of metal layers. The lower metal layers should be skipped if the parasitic capacitance to the substrate is a concern.
- The process may offer a specialized **metal-insulator-metal (MIM)** capacitor (density on the order of $\text{fF}/\mu\text{m}^2$) using extra masks and processing steps, offering higher specific capacitance and low parasitics at extra cost.
- For high capacitance density, or where nonlinearity is not critical, the **MOSFET gate capacitance** can be used, as the gate oxide is tightly controlled in manufacturing.
- In IHP SG13, a MIM capacitor between `Meta15` and `TopMeta11` is available (ca. $1.5 \text{ fF}/\mu\text{m}^2$); the MOM finger capacitor has ca. $0.3 \text{ fF}/\mu\text{m}^2$, so when using 4 metal layers the effective capacitance is comparable to the MIM capacitor.

4.3.5 Diodes

Diodes are formed by the available pn-junctions (n^+ in substrate, p^+ in n-well, n-well in substrate). They are intended mainly for reverse-biased operation, as the forward operation is usually not modeled accurately and should be avoided. Note that forward operation might inject charge into parasitic structures and cause latch-up (see [?@sec-latchup](#))! Applications of reverse-biased diodes include ESD protection structures, antenna diodes, and varactors.

4.3.6 Bipolar Junction Transistors

High-performance BJTs are available in bipolar and BiCMOS technologies, but at increased wafer cost compared to pure CMOS. In standard CMOS, a parasitic **vertical substrate PNP** can be implemented (Figure 45): the p^+ diffusion in the n-well forms the emitter, the n-well the base, and the common p -substrate the collector. The performance is limited ($\beta \approx 1 \dots 10$), and the collector is tied to the substrate potential. Such a structure is mainly used in bandgap

reference circuits where collector and base are shorted to V_{SS} (ground), and the foundry usually supplies a fixed layout template that shall not be modified.

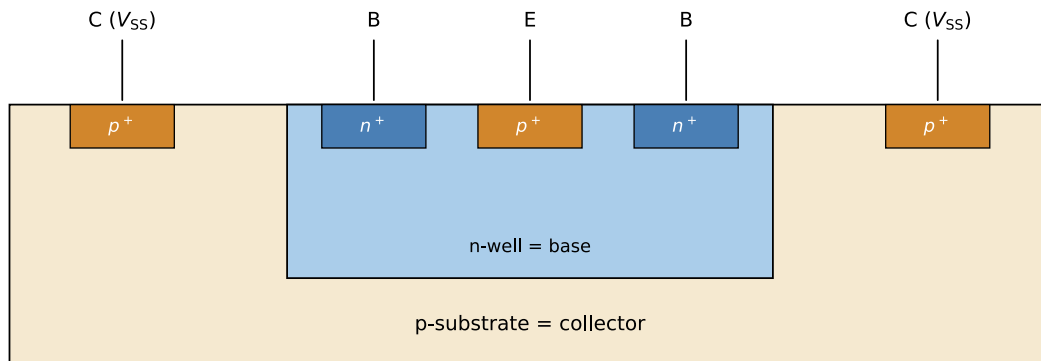


Figure 45: Cross section of the vertical substrate PNP available in any CMOS technology: the p^+ diffusion in the n-well is the emitter, the n-well is the base, and the common p -substrate is the collector (contacted at the surface and tied to V_{SS}).

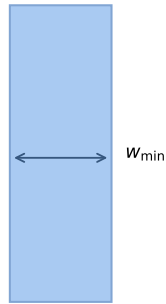
4.4 Design Rules

The **design rules (DRs)** are the interface between the mask designer (layout engineer) and the foundry. Complying with the layout design rules ensures manufacturability with a high yield:

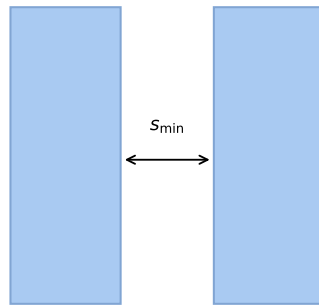
- The design rules are provided as a specification document and as a run set for layout verification tools (e.g., KLayout, Magic, Cadence Pegasus, or Siemens Calibre).
- The rules are either given in relative dimensions (“lambda rules”) for easy scaling, or in absolute dimensions (“micron rules”).
- Modern CMOS technologies typically involve several hundred to thousands of layout design rules.
- Consider the relatively large spacing rule between **RX** and **NW**: it pays off to **group PMOS and NMOS transistors** together, respectively!

The four basic geometric rule types—minimum width, minimum spacing, minimum enclosure, and minimum extension—are illustrated in Figure 46.

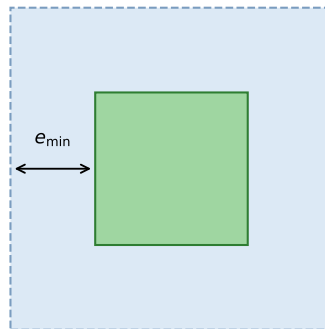
(a) width



(b) spacing



(c) enclosure



(d) extension



Figure 46: The four basic geometric design-rule types: (a) minimum width of a shape, (b) minimum spacing between shapes, (c) minimum enclosure of one shape by another (e.g., **R**X in **N**W), and (d) minimum extension of one shape beyond another (e.g., the poly endcap over **R**X).

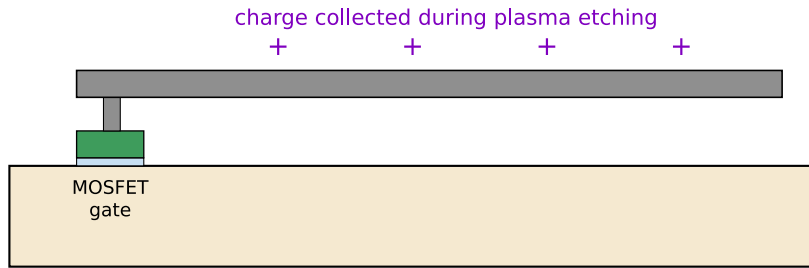
4.4.1 Density Rules

Density rules require a minimum and a maximum layer density within a specified area; this is needed to avoid over-polishing during CMP and to achieve uniform etch rates. For example, M1 might require 30 % minimum and 70 % maximum fill in any 1 mm × 1 mm window. For digital chips, density rules are satisfied automatically by fill programs. For analog chips, layout engineers have to add dummy structures manually (effort!) or with a fill program after the layout is finished; note that fill structures can cause asymmetries or reduced quality factor in inductors. Metal fill can also be put to good use as a ground plane.

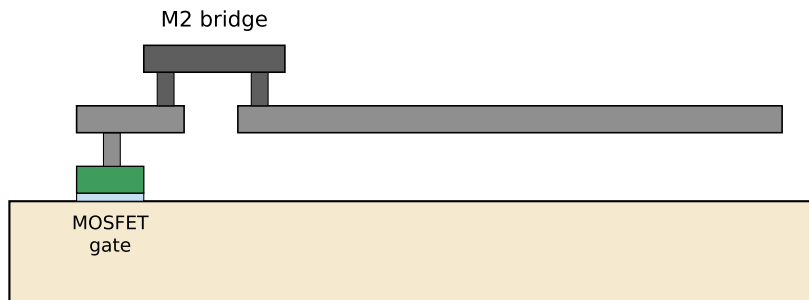
4.4.2 Antenna Rules

If a metal interconnect with a large area is tied to the small gate of a MOSFET, the metal area acts as an “antenna” during plasma etching: it collects ions and can rise to a high potential, and the gate oxide may break down! The remedies, shown in Figure 47, are either a **discontinuity** in the charge-collecting metal (routing briefly through a higher metal layer, which is patterned only after the lower layer has been etched) or an **antenna diode** connected to the gate node (the processing is done at high temperatures, where the diode is quite conductive in either polarity, harmlessly draining the collected charge).

(a) antenna violation: large metal area tied to the gate



(b) fix: break M1 and bridge with M2



(c) fix: antenna diode drains the charge

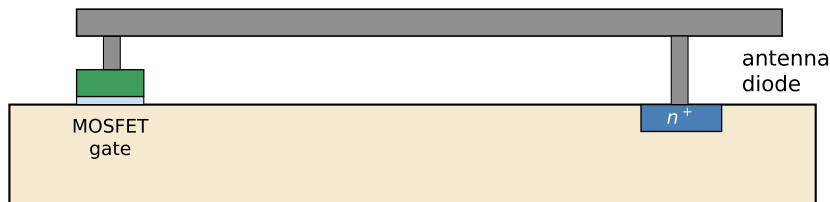


Figure 47: The antenna effect and its fixes (cross sections): (a) a long **M1** wire connected to a gate collects charge during plasma etching and can destroy the gate oxide; (b) breaking the **M1** wire and bridging with **M2** limits the charge collected while **M1** is etched; (c) an antenna diode drains the collected charge harmlessly into the substrate.

The complete layout rules of the open-source IHP SG13 PDK can be found in the SG13G2 layout rules document and serve as a practical example of a production DRC rule set.

4.5 Analog Layout Techniques

Analog circuits, unlike digital ICs, do not primarily aim to maximize density, but demand the minimization of effects such as crosstalk, mismatch, and noise. Useful techniques are multi-finger transistors, matching and symmetry, and careful floor planning; further treatments of analog layout can be found in [15], [16], and an exemplary analog IC design employing these techniques is described in [17].

4.5.1 Multi-Finger Transistors

A transistor with a large W should be split into multiple parallel devices (“fingers”) to make the layout structure compact, as shown in Figure 48. This splitting also reduces the gate resistance (see Equation 32; for RF, keep the resistance of each gate finger below $1/g_m$) and can lower the junction capacitance at the drain or source, since neighboring fingers share their diffusion regions.

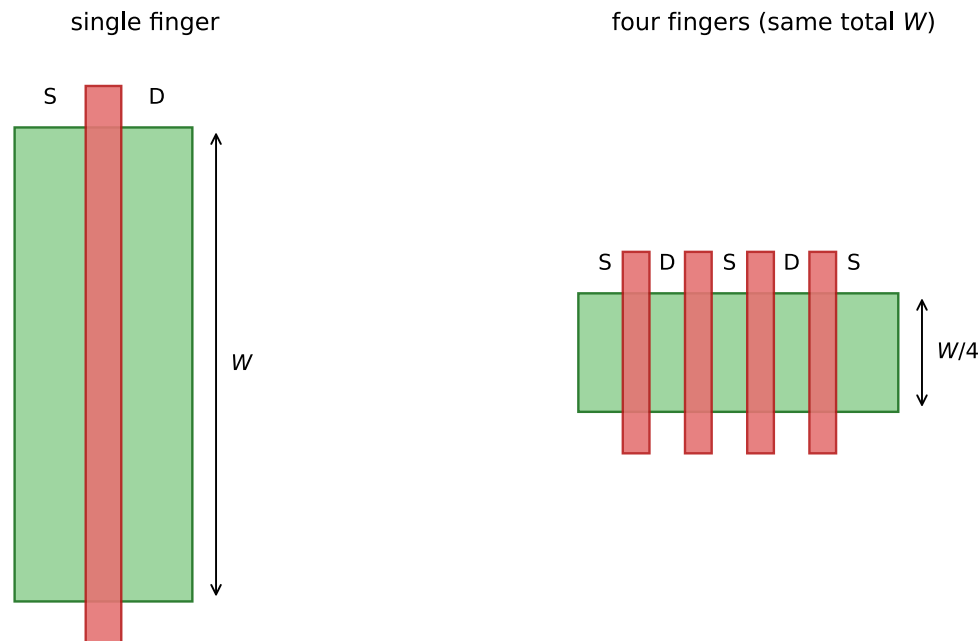


Figure 48: Multi-finger transistor: instead of a single device with a very large width (left), the transistor is split into four parallel fingers (right), which share their source/drain diffusions. The layout gets compact, the gate resistance drops, and the junction capacitance is reduced.

The layout of a cascode circuit (or any series connection of MOSFETs, e.g., in NAND/NOR gates) can be simplified if the stacked devices have equal widths: the drain of the bottom device and the source of the top device can then **share the same diffusion region**, which is small because it needs no contacts, as shown in Figure 49.

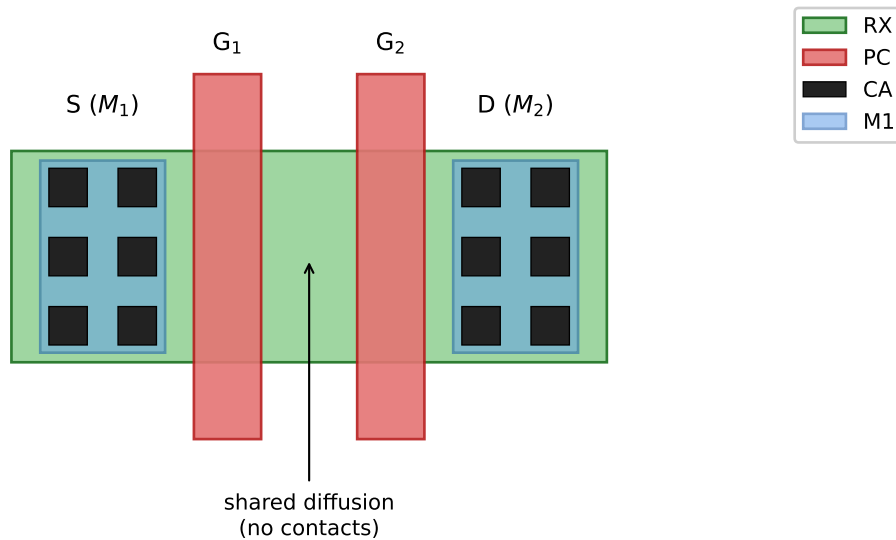


Figure 49: Merged diffusion in a series connection of two equally sized MOSFETs (e.g., a cascode): the drain of M_1 and the source of M_2 share the same diffusion region, which can be kept minimal because it carries no contacts—saving area and junction capacitance.

4.5.2 Matching

Asymmetries lead to mismatch, which introduces input-referred offsets and limits the minimum processable signal level; common-mode and power-supply rejection are degraded as well. Good matching of integrated devices can only be achieved if the following conditions are met:

- The devices are constructed from **identical unit devices**: a MOSFET with width $4W$ matches a transistor with width $2W$ only if both are constructed from unit transistors of equal width (and equal L —remember that $V_{th} = f(L)$ due to SCE/RSCE)!
- The devices are oriented in the **same direction**, and the **current flow** in all matched devices has the same direction—avoid mirroring or “snaking”!
- The devices are **surrounded by similar structures**; to achieve this for edge devices, spend **dummy elements** (see Figure 50).
- To combat process gradients, place the matched devices on the symmetry line of the gradient (if known), or use **common-centroid** arrangements (see Figure 52).
- For ac symmetry, consider the wiring of R and C , and make the **wiring parasitics** as symmetrical as possible.
- Consider secondary effects: gate shadowing (implant variation due to topographical differences and tilted implant angle), mechanical stress from shallow-trench isolation, and non-uniform doping due to the **well-proximity effect** (dopants scattering off the well-mask edges cause doping variations, especially impacting PMOS).

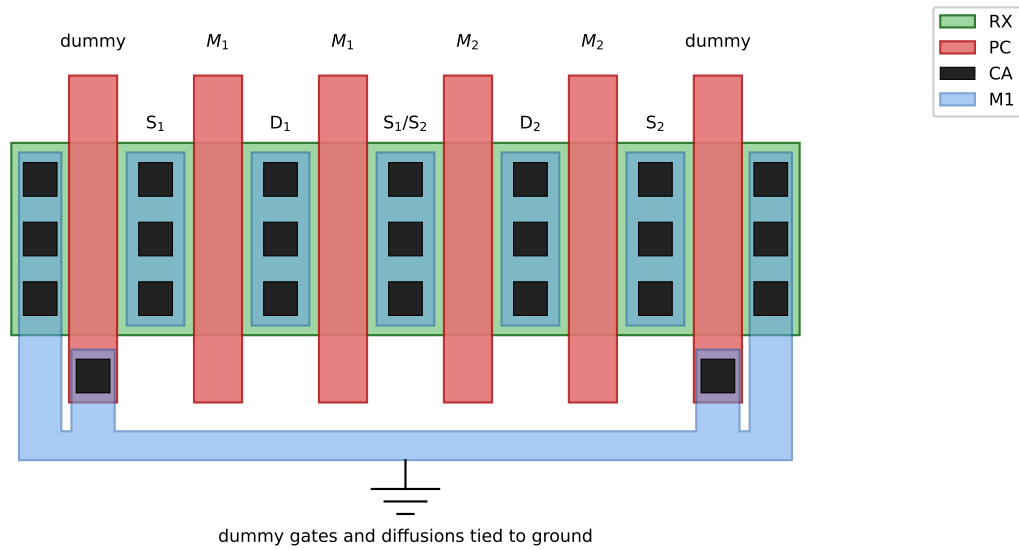


Figure 50: Dummy fingers for matching: the two matched transistors M_1 and M_2 are surrounded by unused dummy gate fingers, so that every active finger sees identical neighbors; this equalizes etch loading and STI stress. The dummy gates and the outer dummy diffusions are tied to ground so that they stay off and well defined. M_1 and M_2 share the center source diffusion, i.e., all source regions form one common node, which makes this arrangement directly usable as a differential pair or a current mirror.

The same reasoning applies to passive devices. Figure 51 shows an array of four matched unit resistors: identical width and length, identical orientation, and a constant pitch, so that R_1 and R_4 at the edges see the same surroundings as R_2 and R_3 in the middle. Note that the dummy strips are not left floating, but are shorted and tied to ground to keep them at a defined potential.

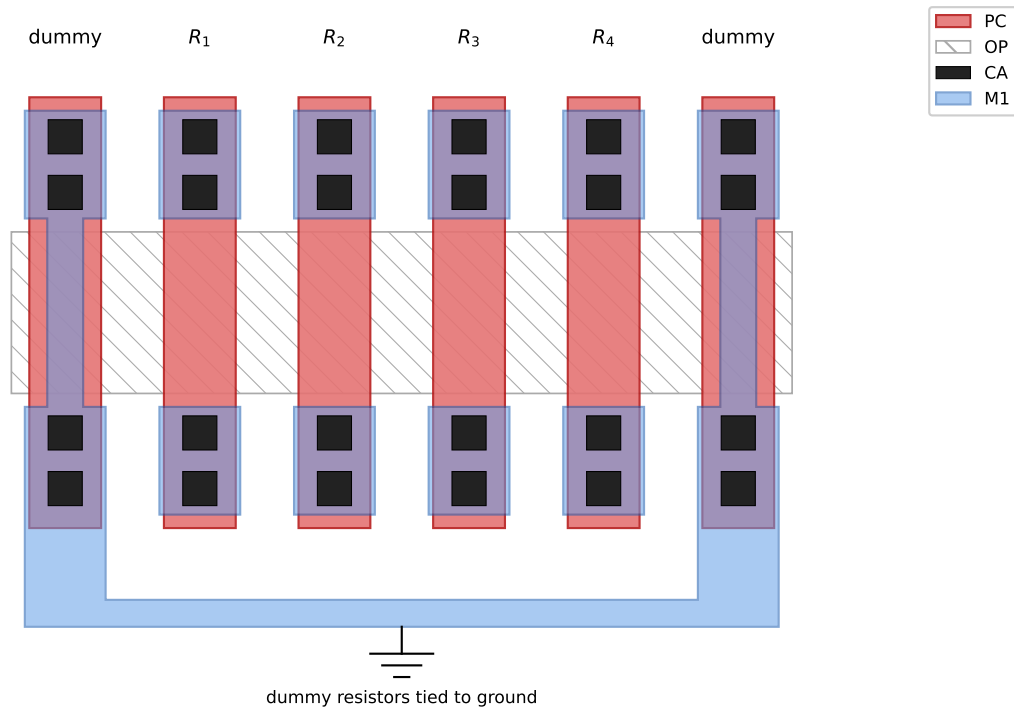


Figure 51: Array of four matched unit resistors $R_1 \dots R_4$: all strips have identical width and length, the same orientation, and the same pitch, and the silicide block **OP** runs across the whole array so that every resistor body sees the same edge. A dummy strip at each end gives the outer resistors the same neighborhood as the inner ones. The dummies are shorted by metal and tied to ground, so they sit at a defined potential instead of floating.

Gradients across a die can be caused by doping and film-thickness variations (well approximated by a linear gradient for devices in proximity), by temperature gradients originating from areas with large power dissipation, and by mechanical stress from packaging or metallization (avoid low-level metal routing over sensitive devices). A common-centroid arrangement cancels linear gradients to first order, as shown in Figure 52: each component is decomposed into two or more parts that are placed diagonally opposite each other, so that all component centroids coincide.

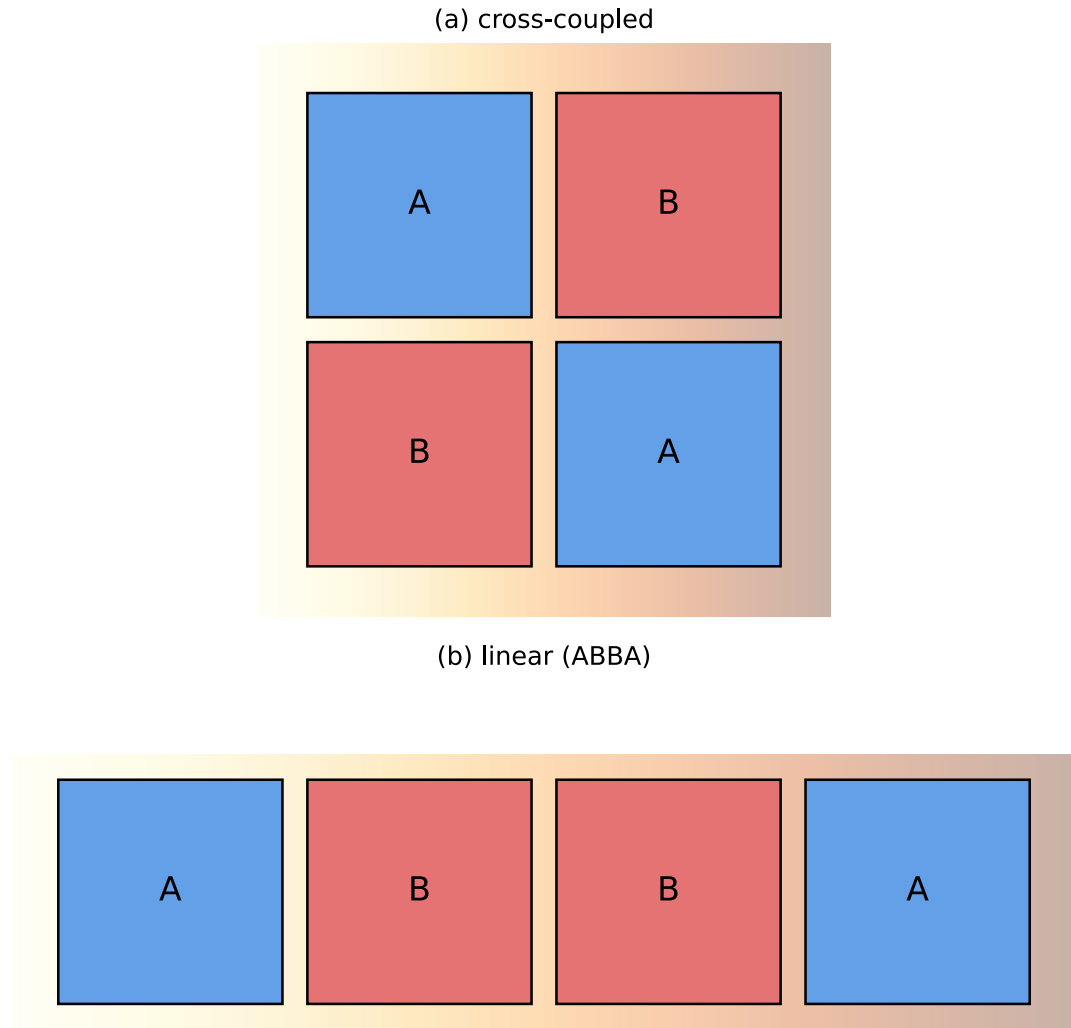


Figure 52: Common-centroid arrangements of two matched devices A and B, each split into two halves: (a) cross-coupled 2×2 arrangement and (b) linear ABBA arrangement. In both cases, the centroids of A and B coincide, so a linear process gradient (illustrated by the color wedge) affects both devices equally.

The same principle applies to capacitors: Figure 53 shows a matched capacitor pair with a ratio of $C_1 = 8C_2$, realized from identical unit capacitors with a common centroid and surrounded by dummy capacitors. If C_1 and C_2 are used, e.g., in a switched-capacitor amplifier, the voltage gain can be precisely set to $A_v = C_1/C_2 = 8$.

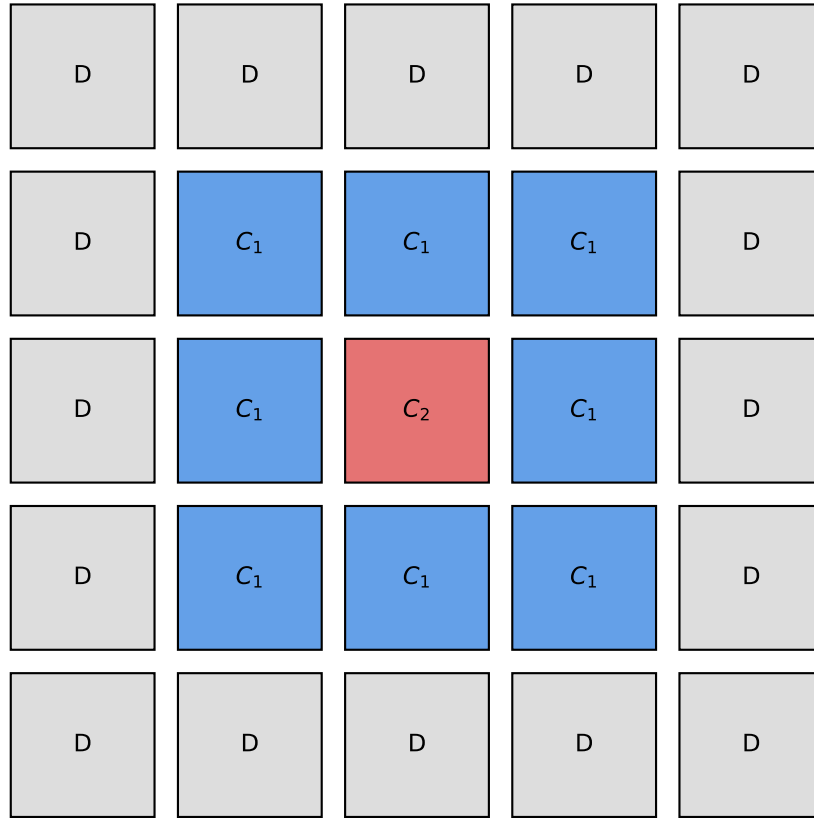


Figure 53: Precisely matched capacitor pair with $C_1 = 8C_2$, built from identical unit capacitors: the single C_2 unit sits at the common centroid of the eight C_1 units, and a ring of grounded dummy capacitors (D) ensures that every active unit sees identical surroundings.

4.5.3 Floor Planning

Floor planning is an important task when constructing the overall IC layout. Things to consider are:

- Locations of pins and interfaces; distribution of V_{DD} and V_{SS} pins; isolation of sensitive I/O signals.
- Cell sizes and aspect ratios (allowing an area-efficient placement of the macros).
- Isolation between different circuit blocks (e.g., keep sensitive analog blocks away from noisy digital blocks).
- Power and clock distribution; wiring channels for signal buses.
- The package type used (wire-bond or flip-chip).

A common mistake: a great job laying out lots of cells, then running out of time, resulting in a big mess when placing and connecting them—**start with a top-down plan!**

4.6 Additional Layout and Mask Structures

In addition to the final chip layout, further structures are placed on the masks (usually by the wafer fab):

- **Seal ring:** The seal ring runs along the circumference of the chip and seals the singulated die against moisture and other contamination. It also acts as a crack-stop, protecting the inner die from cracks formed while sawing the wafer into individual dice. Since the seal ring forms a closed metal loop around the chip, crosstalk considerations may suggest connecting it to a dedicated pin.
- **Scribe line (kerf):** This is where the chip is cut with a diamond saw or a laser.
- **Alignment marks:** Used to align the different masks during processing.
- **Critical-dimension structures:** Measured after processing to verify proper polysilicon or metal etching, enabling closed-loop process control.
- **Vernier structures (“Nonius”):** Closely spaced parallel lines on two different layers that allow judging the inter-layer alignment.
- **Test structures:** Chains of contacts and vias, as well as test transistors, are used to evaluate contact resistance, transistor parameters, and other important process parameters. They are usually placed in the scribe lines or on dedicated **process control monitor (PCM)** test chips; the PCM measurement reports can often be obtained from the foundry.

4.7 Layout Checks

Three automated checks safeguard the layout before tapeout:

- **DRC (design rule check):** a program/run set that automatically checks the layout data for compliance with the design rules of the wafer fab. All flagged errors must be corrected; otherwise, manufacturing issues can occur.
- **LVS (layout versus schematic):** a netlist extracted from the layout data is compared against the schematic netlist, checking connectivity of signals and power as well as device sizes (W and L of transistors, dimensions of R and C). Usually, all flagged errors must be corrected.
- **ERC (electrical rule check):** additional checks not covered by DRC or LVS, such as floating MOSFET gates, floating wells, or antenna rules. Sometimes common mistakes (e.g., high-resistance connections) are covered in the ERC run set and checked automatically.

4.8 Layout Extraction

The layout implementation has a large effect on the performance of the final circuit:

- Wiring adds delays (R , L , C) and coupling (k , C) to the circuits.
- **Layout-dependent effects (LDE)**, like well proximity, can only be modeled with information about the actual layout of the devices and their surroundings.
- Substrate effects (like crosstalk) can only be extracted from the actual placement.

Considering layout effects in simulation is paramount and usually requires iteration between circuit design and layout: circuit design and simulation $\rightarrow$ layout creation $\rightarrow$ parasitic extraction $\rightarrow$ re-simulation $\rightarrow$ circuit and/or layout refinement $\rightarrow$ repeat. In the digital flow, wiring delays extracted from the layout are added to the gate-level netlist for timing analysis, and voltage drops in the power network (V_{DD}/V_{SS} droop and ripple) are extracted and considered as additional timing derating. For analog and high-performance circuits, simulation including layout parasitics is **mandatory** to guarantee the performance and functionality of the fabricated circuit!

Different levels of extraction realism trade off accuracy against simulation speed:

- **C-decoupled:** The parasitic capacitance of each node's wiring is added as a lumped capacitor to ground. There is almost no impact on simulation speed (no new nodes are introduced), but no crosstalk between nodes is captured. This extraction can be used on fairly large circuits to add a first level of realism.
- **C-coupled:** Parasitic capacitances to the substrate *and between nodes* are added; capacitive crosstalk is modeled, but no inductive effects. Moderate impact on simulation speed (more elements in the circuit matrix).
- **RC-extracted:** Parasitic R and C of the wiring are extracted and added to the netlist, yielding a realistic model of performance and crosstalk without inductive effects. This has a large impact on simulation speed, as many new nodes are introduced; this mode should be used for circuit blocks of moderate complexity.
- **RLCK-extracted:** Parasitic R , C , L , and k of the wiring are extracted, resulting in a very accurate simulation including the main crosstalk effects—but the circuit network gets much larger, impacting simulation time and convergence.
- **EM simulation:** Highest accuracy; the resulting S -parameters are either used directly in frequency-domain simulations or transformed into equivalent lumped-element circuits. Huge simulation-time impact, but valid to very high frequencies (> 100 GHz); usually only suitable for individual circuit blocks.

i Parasitic Extraction with KLayout-PEX

Our open-source parasitic extraction tool **KLayout-PEX** (invoked as `kpex`) implements several of these extraction levels for KLayout, using different engines (an analytical 2.5D engine, a wrapper around MAGIC, and the FasterCap 3D field solver) that trade off speed against accuracy. Supported PDKs include IHP SG13G2 and SkyWater sky130A. Further information can be found in the online documentation.

5 IC Packaging

The finished wafer is not yet a usable product: the individual dice have to be separated, connected, and protected. This chapter covers the path from the processed wafer to the packaged integrated circuit, the most important package families, thermal design, and the electrical parasitics that packages introduce.

5.1 From Wafer to Packaged IC

After wafer processing (see Section 3), the wafer is usually first **thinned** by grinding from the backside, depending on the package technology (typical final thickness 100 to 300 μm). After thinning, the wafer is mounted on a carrier tape and split into individual dice—this process is called **dicing**. A saw blade is the standard tool; alternatively, a laser can be used to (pre-)cut the wafer. An exception is the wafer-level package (WLP), where the wafer is further processed *before* sawing.

The IC package then provides the following important functions:

- Electrical connection of signals and power from the IC to the PCB.
- Interconnects adding little delay or distortion (small parasitic R , L , C , and k).
- Mechanical connection of the IC to the PCB.
- Removal of the heat produced in the IC.

- Protection of the IC from mechanical damage and the environment.
- Compatibility with the thermal expansion (CTE) of the PCB.
- Inexpensive manufacturing and testing.

5.2 Chip-to-Package Connection

Traditionally, an IC is surrounded by a **pad frame**: metal pads on a pitch of 80 to 200 μm along the die circumference. **Bond wires** attach the pads to the package, a **lead frame** distributes the signals within the package, and a metal heat spreader (paddle) helps with cooling. In **flip-chip** packages or WLPs, the pads can instead be distributed arbitrarily across the die area (close to the relevant circuit blocks). Figure 54 shows the wire-bond arrangement.

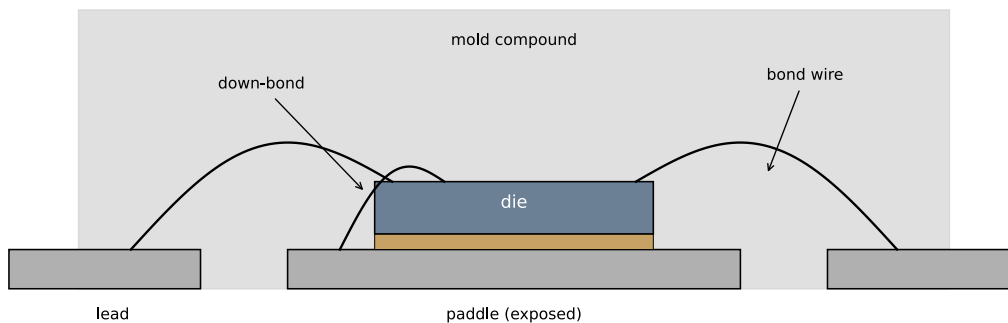


Figure 54: Cross section of a wire-bonded lead-frame package: the die is glued to the paddle, bond wires connect the die pads to the leads (a down-bond connects directly to the grounded paddle), and the epoxy mold compound protects the assembly. The exposed paddle is soldered to the PCB for grounding and heat removal.

A few notes on the bond pads themselves:

- A simple pad consisting only of a square of top metal may “**lift off**” during bonding due to insufficient adherence; a mechanically robust pad is formed from the two topmost metal layers, connected by many small vias, at the price of somewhat larger capacitance.
- Pads carrying RF signals can be shaped as octagons to reduce their capacitance (about 20 % reduction).
- In some processes, it is allowed to place circuitry (e.g., ESD protection) below the pads to save chip area—called pad-over-active-area (PoAA).

5.3 Package Types

Simple lead-frame-based packages using bond wires are cost-effective but have a limited number of I/Os, and the bond wires cause parasitic inductance and mutual (inductive and capacitive) coupling. Laminate-based packages can have several signal and power layers (2–4), like tiny PCBs; they are based either on bond wires or on **flip-chip** assembly:

- Pads can be placed across the whole surface of the die rather than only at the periphery.
- The top-level metal pads are covered with solder balls.
- The IC is mounted upside down and must be carefully aligned to the package (done blind!); then heat is applied to melt the solder balls.
- This process is also called controlled collapse chip connection (C4).

Figure 55 shows cross sections of the most important package families.

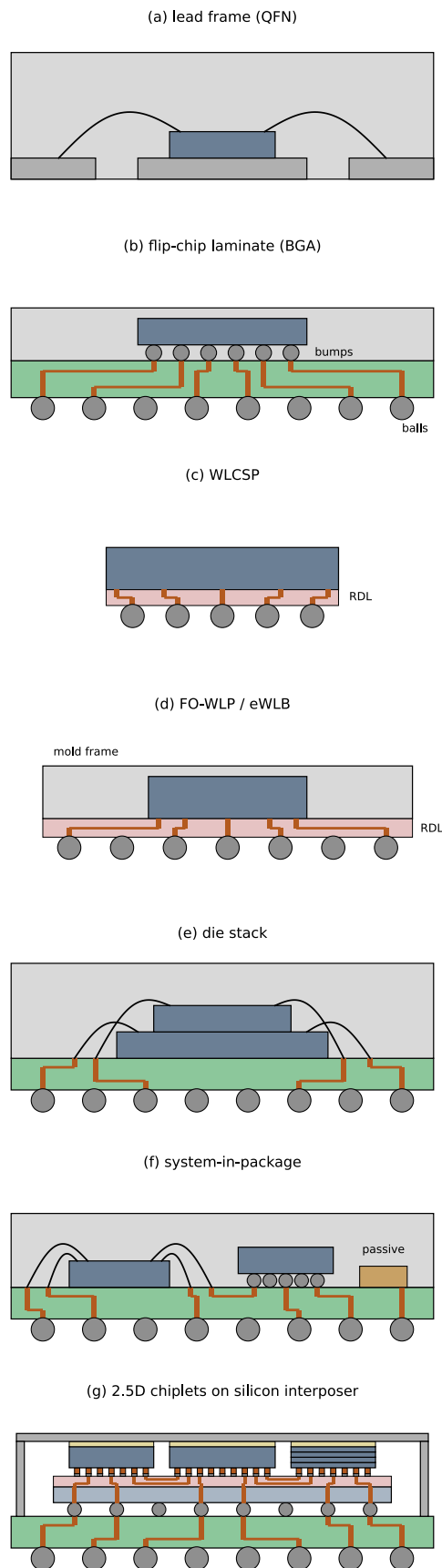


Figure 55: Cross sections of important package families: (a) wire-bonded lead-frame package (e.g., QFN), (b) flip-chip laminate package (BGA) with solder bumps and balls, (c) wafer-level chip-scale package (WLCSP), where the redistribution layer (RDL) and the balls are processed on the wafer, (d) fan-out wafer-level package (FO-WLP), where the fan-out area is created by a mold frame, (e) die stack with bond wires, and (f) system-in-package with a wire-bonded die, a flip-chip die, and a passive component on one laminate substrate, and (g) 2.5D integration,

5.3.1 Lead-Frame Package

A lead-frame package is available with or without leads (e.g., QFP with gull-wing leads versus leadless QFN). The die is glued to the lead-frame paddle and bonded to the leads using Au, Al, Cu, or Ag wires; a “down-bond” to the paddle is usually possible and provides low inductance due to the short wire. The whole assembly is covered and protected by an epoxy mold compound. The lead-frame paddle can be exposed and soldered to the PCB for best grounding and thermal performance.

5.3.2 Laminate Package

A laminate package can be seen as a small PCB with fine design rules; the die is either wire-bonded (cheaper) or soldered face-down using solder bumps (C4) or Cu pillars (better electrical performance). The connection to the main PCB is made with solder balls (BGA), pins, or exposed lands. The mold compound protects the assembly; for flip-chip assembly it can be omitted and potentially replaced by a metallic heat spreader.

5.3.3 Chip-Scale and Wafer-Level Packages

The **wafer-level package (WLP)** offers very low cost, excellent electrical properties, and the smallest possible size (package size = die size), but provides little protection of the die. The processing happens on the wafer level: a redistribution layer (RDL) re-routes the pads, solder balls are applied, and only then are the dice singulated. If the I/O count is too large to fit onto the die area, a **fan-out WLP (FO-WLP/eWLB)** can be used, where the die is extended in size by a mold frame: the dice are first singulated, then re-assembled on an artificial wafer where the surrounding area is filled with mold compound, and then processed like a WLP.

5.3.4 Stacked Packages

For dense systems, several dice can be combined in one package: as a **die stack** (dice stacked on top of each other, connected by bond wires or through-silicon vias), as **package-on-package (PoP)** (e.g., a memory package soldered on top of a processor package), or as a **system-in-package (SiP)**, where several dice and passives are integrated on a common laminate substrate.

5.3.5 2.5D Integration with Chiplets and Silicon Interposer

High-end systems like GPUs, AI accelerators, and server processors increasingly split a large SoC into several smaller dice, called **chiplets**, which are placed side by side on a **silicon interposer** (Figure 55 (g)); a well-known example is TSMC’s CoWoS (chip-on-wafer-on-substrate). This approach is motivated by several factors:

- A single monolithic die is limited by the reticle size of the lithography tool, and the yield drops rapidly with die area; several smaller dice yield better and can exceed the reticle limit in total.
- Each chiplet can be fabricated in the most suitable technology (**heterogeneous integration**), e.g., the compute logic in a leading-edge node, while I/O, analog, or RF functions stay in a mature, cheaper node.
- Memory can be placed very close to the processor; a typical example is high-bandwidth memory (HBM), a stack of DRAM dice connected by through-silicon vias.
- Proven chiplets can be reused across products, reducing design effort.

The interposer is a passive silicon die (or a large part of a wafer) that is fabricated with a back-end-of-line process. Its fine-pitch wiring (line width and spacing in the μm range) allows thousands of short, dense connections between the chiplets, which is not possible with laminate substrates. The chiplets are flipped and attached to the interposer with **Cu pillars** with a thin solder cap (sometimes called microbumps, pitch typically 40 to 50 μm). **Through-silicon vias (TSVs)** lead the supply and external signals through the thinned interposer (thickness ca. 100 μm) to C4 bumps (pitch ca. 150 μm), which connect the interposer to a laminate substrate. The laminate then fans out to the BGA balls, just like a regular flip-chip package. Instead of a mold compound, such packages are typically covered by a metal lid, which is attached to the backside of the chiplets with a thermal interface material (TIM) and acts as a heat spreader.

As the pitch grows by roughly an order of magnitude at each level (chiplet, interposer, laminate, PCB), this arrangement is a hierarchy of increasingly coarse wiring. Several variants exist to reduce cost: an organic RDL interposer instead of silicon, or small silicon bridges embedded in the laminate only below the chiplet edges (e.g., Intel EMIB). In **3D integration**, chiplets are stacked directly on top of each other using TSVs and hybrid bonding (Cu-to-Cu bonding without solder, pitch below 10 μm).

Designing chiplet-based systems brings new challenges: the die-to-die interfaces need to be standardized to allow combining chiplets from different sources (e.g., UCIE, Universal Chiplet Interconnect Express), each chiplet must be tested before assembly (known-good die, as a single faulty chiplet ruins the whole expensive package), power delivery and heat removal must be co-designed with the package, and the mechanical stress and warpage of the large assembly must be controlled.

5.4 Thermal Design

Managing the die temperature is an important aspect of package selection. Depending on the application, different ambient temperature ranges apply, which result in different on-die temperature ranges (due to the self-heating of the die caused by its power dissipation):

- Consumer ambient temperature range: -30 to $+85$ $^{\circ}\text{C}$.
- Industrial/automotive ambient temperature range: -40 to $+85$ $^{\circ}\text{C}$.
- Usual IC technology junction temperature limits: -40 to $+125$ $^{\circ}\text{C}$.
- Automotive IC junction temperature limits: -40 to $+150$ $^{\circ}\text{C}$.

5.4.1 Thermal Calculations

Thermal calculations behave analogously to Ohm's law, as thermal paths can be modeled as series or parallel connections, using these equivalences: temperature difference $\Delta T \hat{=}$ voltage V , heat flow $Q \hat{=}$ current I , thermal resistance $R_{\text{th}} \hat{=}$ resistance R :

$$\Delta T = Q \cdot R_{\text{th}} \quad (49)$$

$$R_{\text{th,series}} = \sum_i R_{\text{th},i} \quad \frac{1}{R_{\text{th,parallel}}} = \sum_i \frac{1}{R_{\text{th},i}} \quad (50)$$

An important property of an IC package is its ability to transport the heat generated on the die to the outside, limiting the die's self-heating. This property is the **thermal resistance**, usually a function of the package, the die, and the PCB design. Using

$$\Delta T = R_{\text{th,ja}} \cdot P_{\text{diss}} \quad (51)$$

with ΔT the on-chip temperature rise versus the ambient temperature in K, $R_{th,ja}$ the junction-to-ambient thermal resistance in K/W, and P_{diss} the on-chip power dissipation in W, the die temperature can be estimated. A typical $R_{th,ja}$ value is provided by the package manufacturer; for high accuracy, a thermal simulation has to be performed. Strictly speaking, $R_{th,ja}$ is defined for a specific PCB design and environment (standardized in, e.g., JEDEC JESD51) and needs re-evaluation for specific cases (PCB size, number of layers, copper and via density, casing, airflow, etc.). Figure 56 shows how a complete thermal path is modeled as a resistor network.

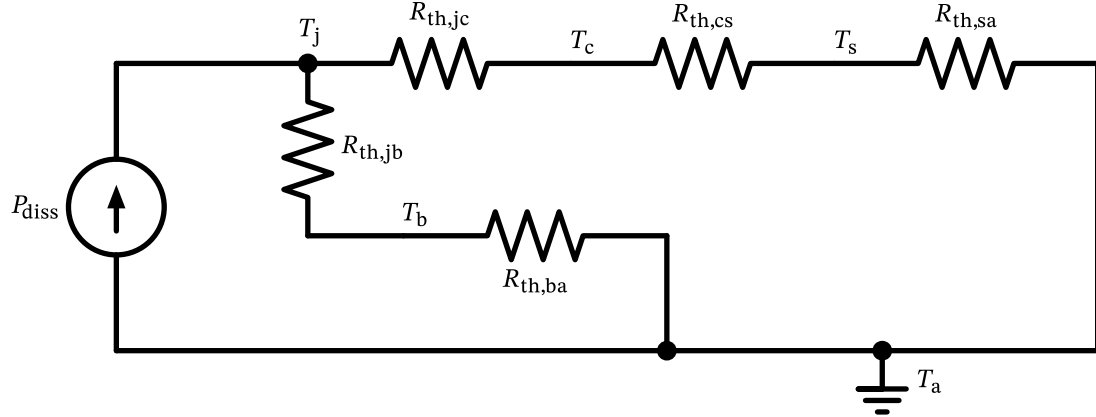


Figure 56: Thermal equivalent network of a packaged IC: the dissipated power P_{diss} acts as a current source injecting heat into the junction node. The heat flows through the parallel paths junction→case→heatsink→ambient and junction→board→ambient; every interface is modeled by a thermal resistance, and the node ‘voltages’ are the local temperatures (T_j junction, T_c case, T_s heatsink, T_b board). The bottom rail is the ambient temperature T_a .

5.5 Package Parasitics

The parasitics associated with the package and the connections to the chip introduce difficulties in high-speed or high-accuracy designs: trace self-inductance, mutual inductance, and capacitance limit the achievable performance. Packaging continues to limit the performance of high-end designs, which is why advanced packages like WLCSP with minimal L and C are used for high-performance ICs. Figure 57 shows the equivalent circuit of a single package connection.

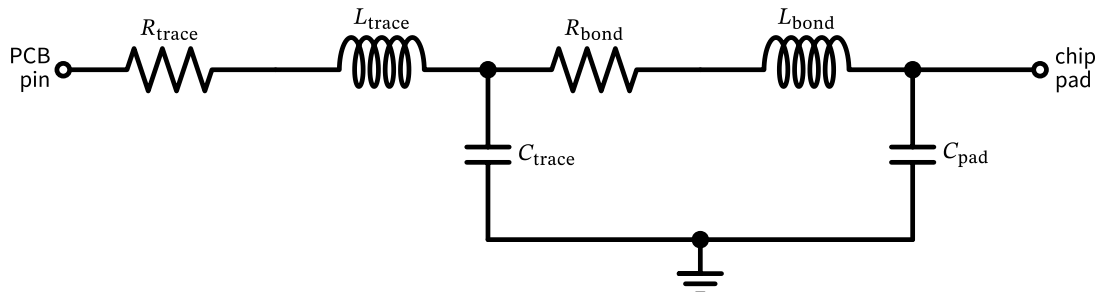


Figure 57: Equivalent circuit of a single package connection from the PCB to the on-chip pad: the bond wire and package trace contribute a series inductance and resistance, while the package trace and the bond pad add shunt capacitances. These parasitics limit bandwidth and cause supply bounce ($V = L di/dt$).

Simple approximations for the inductance of common package geometries are given below (higher accuracy requires EM simulation; the underlying transmission-line theory is covered in [18]). For a **round wire** at height h above a ground plane (wire radius r):

$$L' \approx 0.2 \ln\left(\frac{2h}{r}\right) \text{nH/mm} \quad (52)$$

For a **flat trace** of width W at height d above a ground plane:

$$L' \approx \frac{1.6}{0.72 + W/d} \text{nH/mm} \quad (53)$$

The **mutual inductance of two round wires** at distance d and height h above the ground plane:

$$L'_m \approx 0.1 \ln\left[1 + \left(\frac{2h}{d}\right)^2\right] \text{nH/mm} \quad (54)$$

A typical bond wire has a radius of 12.5 μm and larger, and an inductance of roughly 1 nH/mm.

i Note 4: Example Calculation of the Bond-Wire Inductance

Let us assume a bond wire with a diameter of $25 \mu\text{m}$ (i.e., $r = 12.5 \mu\text{m}$) and a length of $l = 2 \text{ mm}$, running at a height of $h = 300 \mu\text{m}$ above the ground plane. With Equation 52, the inductance per length is

$$L' \approx 0.2 \ln\left(\frac{2 \cdot 300 \mu\text{m}}{12.5 \mu\text{m}}\right) \text{nH/mm} = 0.2 \ln(48) \text{nH/mm} = 0.77 \text{nH/mm}$$

and the total inductance of the bond wire is

$$L = L' \cdot l = 0.77 \text{nH/mm} \cdot 2 \text{ mm} = 1.55 \text{nH}$$

Note that the height enters only logarithmically: doubling h to $600 \mu\text{m}$ increases the inductance by just 18 % to 1.83 nH. At a frequency of $f = 1 \text{ GHz}$, this bond wire already shows a reactance of

$$X_L = 2\pi f L = 9.7 \Omega$$

If this bond wire carries the supply current of a digital block, a current step of $\Delta i = 10 \text{ mA}$ within $\Delta t = 100 \text{ ps}$ causes a supply bounce of

$$V = L \frac{\Delta i}{\Delta t} = 1.55 \text{nH} \cdot \frac{10 \text{ mA}}{100 \text{ ps}} = 155 \text{ mV}$$

which is significant for a core supply voltage of about 1.2 V.

To reduce the supply bounce, a second bond wire of the same dimensions is placed in parallel at a distance of $d = 300 \mu\text{m}$. With Equation 54, the mutual inductance per length is

$$L'_m \approx 0.1 \ln\left[1 + \left(\frac{2 \cdot 300 \mu\text{m}}{300 \mu\text{m}}\right)^2\right] \text{nH/mm} = 0.1 \ln(5) \text{nH/mm} = 0.16 \text{nH/mm}$$

resulting in a mutual inductance of $L_m = L'_m \cdot l = 0.32 \text{ nH}$ between the two bond wires. The magnetic coupling factor is

$$k = \frac{L_m}{\sqrt{L_1 L_2}} = \frac{L_m}{L} = \frac{0.32 \text{ nH}}{1.55 \text{ nH}} = 0.21$$

As both bond wires carry currents in the same direction, the mutual inductance adds to the self-inductance of each wire, and the effective inductance of the parallel connection is

$$L_{\text{eff}} = \frac{L + L_m}{2} = \frac{L}{2}(1 + k) = 0.94 \text{ nH}$$

Instead of halving the inductance to 0.77 nH, the second bond wire reduces it only by 40 %, and the supply bounce for the same current step drops from 155 mV to 94 mV. A larger distance between the bond wires reduces k and brings L_{eff} closer to $L/2$. Conversely, if the two bond wires carry unrelated signals, the same coupling causes crosstalk: the current step of 10 mA in 100 ps in one bond wire induces a voltage of $L_m \Delta i / \Delta t = 32 \text{ mV}$ in the other.

5.5.1 Multiple Pads, Bond Wires, and Pins

Where a single connection to the chip would sustain a prohibitively large transient voltage drop ($V = L di/dt$), multiple pads, bond wires, and package pins are used in parallel: N parallel connections reduce the effective inductance by roughly N (somewhat less due to the mutual coupling between the parallel wires). This is standard practice for power supplies and high-current outputs.

5.5.2 Exposed Paddle and Down-Bonds

Some packages contain a metal plane (the paddle) to which the die is attached with conductive epoxy. Such packages avoid long, narrow traces in the ground connection: the ground pads are “**down-bonded**” directly to the paddle, minimizing the inductance, and the paddle is soldered to the board ground. Sometimes the paddle has to be plated in the areas where a down-bond lands, adding cost.

5.5.3 Mutual Inductance and Crosstalk

Supply noise and other disturbances can couple through the mutual inductance (and capacitance) of adjacent bond wires and package traces. The design of the pad frame and the position of the bond wires play a critical role in isolating sensitive connections! Two methods reduce the mutual coupling, illustrated in Figure 58:

1. Arrange critical wires **perpendicular** to each other (mutual inductance requires parallel current paths).
2. Interpose V_{SS} or V_{DD} wires between critical bond wires; the return current in the interposed wire partially cancels the coupled flux.

Mutual inductance can even be used to advantage: routing a supply and its return current in adjacent, parallel bond wires carrying **opposite currents** reduces the effective loop inductance.

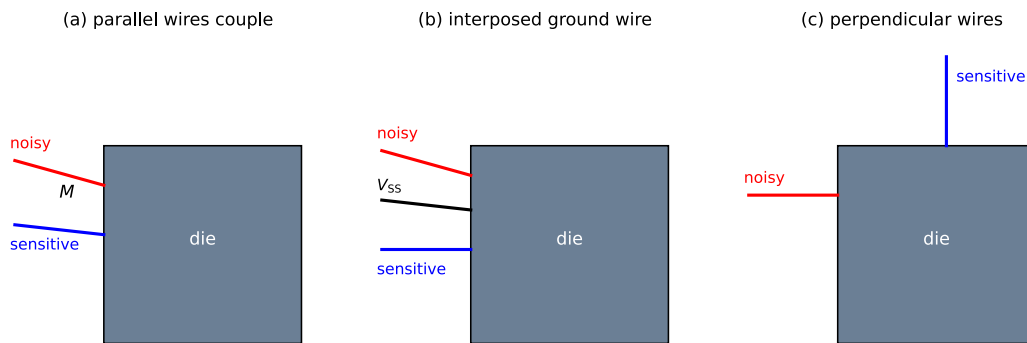


Figure 58: Reducing mutual coupling between bond wires (top view of a chip corner): the noisy output wire couples into the parallel sensitive wire via the mutual inductance M (left); interposing a grounded V_{SS} wire (middle) or arranging the wires perpendicular to each other at a corner (right) strongly reduces the coupling.

5.5.4 Bond Wires as Circuit Elements

Since a bond wire is a relatively high- Q inductor (inductance ca. 1 nH/mm, low resistivity of Au or Al), some designs use it as a circuit element, e.g., in high-frequency LC oscillators [19]. However, the mechanical tolerances of the bond-wire geometry translate into inductance

tolerance, which must be taken into account. More generally, any conducting structure in a package can be used to form inductors, capacitors, or even antennas (e.g., in CSP or laminate packages).

5.6 Package Qualification

In addition to the qualification of the silicon technology (see Section 3.10), the package must be qualified to prove that the assembly of die, bond wires or bumps, substrate, and mold compound survives soldering, handling, and the thermal and environmental stress of its application over the intended lifetime. The framework is again set by JEDEC JESD47, and for automotive products by AEC-Q100, which groups these tests as *accelerated environment stress tests* and *package assembly integrity tests*. As for the silicon, a typical sample consists of three assembly lots of 77 parts per test, and zero failures are accepted. Table 10 lists typical tests.

Table 10: Typical package qualification tests

Test (examples)	Standard	Motivation & check
Pre-conditioning (PC)	JESD22-A113	Moisture soak and reflow simulate storage and board assembly before the other stress tests.
Moisture sensitivity level (MSL)	J-STD-020	Classification of how long a package may be exposed to ambient moisture before soldering.
Temperature cycling (TC)	JESD22-A104	Repeated heating and cooling must not cause failures (thermal expansion mismatch, see CTE).
Temperature humidity bias (THB) / biased HAST	JESD22-A101 / A110	Corrosion and leakage under humidity, temperature, and applied voltage.
Unbiased highly accelerated stress test (UHAST)	JESD22-A118	Exposure of package and die to high temperature and humidity without bias.
High-temperature storage life (HTSL)	JESD22-A103	Storage at high temperature for thousands of hours (intermetallic growth).
Wire bond pull and shear	MIL-STD-883 (2011) / JESD22-B116	Mechanical strength of the bond wires and their connections.
Solder ball shear	JESD22-B117	Mechanical strength of the solder balls of BGA and WLCSP packages.
Drop test	JESD22-B111	The assembled PCB is dropped repeatedly under standardized conditions.
Temperature cycling on board (TCoB)	IPC-9701	The assembled PCB is temperature-cycled to check the solder-joint reliability.

These tests fall into three groups: **component-level stress tests** (PC, TC, THB/HAST, UHAST, HTSL) that accelerate the degradation of the package, **assembly integrity tests** that check the mechanical quality of the interconnects directly, and **board-level tests** (drop, TCoB) that stress the solder joints between package and PCB. After each stress test, the parts are electrically tested against the datasheet at the read points; in addition, non-destructive inspection like scanning acoustic microscopy (C-SAM) reveals delamination and cracks inside the package, and X-ray imaging shows deformed bond wires. Failing parts are analyzed further, e.g., by cross-sectioning.

Pre-conditioning (PC) is not a test on its own but the mandatory first step before TC, THB/HAST, and UHAST. The mold compound and the laminate are polymers that absorb moisture from the ambient air during storage. When the package is then soldered to the PCB, it is heated within seconds to a peak temperature of up to 260 °C for lead-free solder, and the absorbed water evaporates explosively. The vapor pressure can delaminate the mold compound from the die or the lead frame, crack the package, or break bond wires—the so-called **popcorn effect**. Pre-conditioning therefore reproduces the life of a part before it reaches the application: a bake to dry the parts, a moisture soak according to the targeted MSL, and three reflow cycles to simulate board assembly including a rework.

The **moisture sensitivity level (MSL)** is determined according to J-STD-020 and states how long a package may be exposed to a factory environment (≤ 30 °C, 60 % RH) after opening its dry pack before it must be soldered. It ranges from MSL 1 (unlimited floor life, soaked at 85 °C and 85 % RH for 168 h) and MSL 3 (floor life of 168 h, a common target for plastic packages) to MSL 6 (parts must be baked right before soldering). Packages with MSL 2 and higher are shipped in sealed moisture-barrier bags with desiccant, and parts that exceeded their floor life have to be baked before use (J-STD-033).

Temperature cycling (TC) addresses the thermo-mechanical stress caused by the different coefficients of thermal expansion (CTE) of the package materials: silicon expands by about 2.6 ppm/K, the copper lead frame by about 17 ppm/K, the laminate by about 14 to 17 ppm/K, and the mold compound by roughly 8 to 15 ppm/K (below its glass transition temperature). Every temperature change therefore causes shear stress at the interfaces. The parts are cycled between typically -55 °C and $+125$ °C (or -40 °C and $+150$ °C for automotive), with several hundred to 1000 cycles. Typical failure mechanisms are cracked wire-bond heels, lifted ball bonds, delamination, cracked dice or passivation layers, and fatigue of flip-chip bumps (which is why flip-chip dice are usually underfilled). The acceleration factor to the milder temperature swings of the application follows the empirical Coffin–Manson relation, where the number of cycles to failure scales with a power of the temperature swing.

Temperature humidity bias (THB) and its accelerated variant, **biased HAST**, combine moisture with an electric field. In the classic 85/85 test, the parts are operated for 1000 h at 85 °C and 85 % RH with a static bias applied to the pins; HAST reaches a similar stress in about 96 h at 130 °C and 85 % RH in a pressurized chamber. Moisture diffuses through the mold compound and along interfaces to the die surface, where it dissolves ionic contaminations (e.g., chlorine from the mold compound); together with the applied voltage this causes electrochemical **corrosion** of the Al bond pads and metal lines, or leakage currents between adjacent pins. A dense passivation layer and a well-adhering mold compound are the main countermeasures. **Unbiased HAST (UHAST)** applies the same humidity stress without bias

(typically 96 h at 130 °C and 85 % RH), which isolates the purely chemical and mechanical effects of moisture, like galvanic corrosion between dissimilar metals and delamination.

High-temperature storage life (HTSL) stores unbiased parts at typically 150 °C for 1000 h. Its main target is the bond interface between the wire and the Al pad: gold and aluminum form **intermetallic compounds** whose growth is thermally activated. Uneven diffusion of the two metals creates voids (Kirkendall voiding), which increase the contact resistance and weaken the bond until it lifts off; the purple color of one of these compounds (AuAl_2) gave this failure the name “purple plague.” Copper wires, which have widely replaced gold for cost reasons, form Cu–Al intermetallics much more slowly, but they are more sensitive to corrosion in the humidity tests. As for HTOL, an Arrhenius acceleration factor relates the storage test to the application temperature.

Wire bond pull and shear tests and **solder ball shear tests** are destructive mechanical tests of the interconnects themselves. In the pull test, a hook is placed below the bond wire and pulled upward until the wire breaks; both the force and the location of the break are recorded—a break in the wire is acceptable, while a lifted bond indicates a weak connection. In the shear test, a tool pushes the ball bond or the solder ball sideways off its pad. These tests are performed on a small sample per assembly lot and are also used for monitoring the ongoing production.

Board-level reliability tests assess the solder joints between package and PCB, which usually fail before the package itself. In the **drop test**, a standardized JEDEC test board with the mounted components is subjected to repeated mechanical shocks (e.g., 1500 g for 0.5 ms) while the daisy-chained solder joints are monitored for interruptions; this test is essential for handheld devices. In **temperature cycling on board (TCoB)**, the assembled board is cycled (e.g., between 0 °C and 100 °C) until the solder joints fail by fatigue. The CTE mismatch between silicon and the PCB is particularly critical for WLCSPs, where the silicon die is soldered directly to the board, and for large BGAs, where the distance to the neutral point of expansion is large; the achievable number of cycles therefore limits the maximum die size of a WLCSP.

A qualified package is not qualified forever. As a package family covers many die sizes and pin counts, qualification is usually performed on worst-case representatives (e.g., the largest die in the thinnest package), and other products are qualified by similarity. Any significant change of the materials (mold compound, wire material, die attach), the assembly site, or the process requires a requalification and a formal notification of the customers (JESD46).

6 Testing

Integrated circuits must be tested for functionality and specification compliance before they are shipped. Defects introduced during manufacturing, as well as parameter variations, are unavoidable—not every IC on a wafer is a good device (the yield is below 100 %). Typical defect causes are doping errors and fluctuations, under- or over-etching, misalignment of layers, and contamination by particles. A comprehensive treatment of IC test is given in [20]; the related field of hardware security and trust is covered in [21].

6.1 Post-Silicon Production Flow

The production flow after wafer manufacturing consists of the following steps:

1. **Wafer test** (optional): bad dice are inked or recorded in the automatic test equipment (ATE) to avoid wasting packages on them. The cost of packaging a bad die has to be weighed against the cost of the wafer test.
2. **Wafer dicing**: the wafer is cut into separate chips, and bad chips are discarded.
3. **Packaging**.
4. **Final test**: ensures that no parts damaged during dicing or packaging are shipped; some parameters might only be testable on the packaged device.

The ATE is typically used in two distinct modes:

- **Prototype/characterization testing** is very extensive: it looks for worst-case behavior across conditions, establishes **guard bands** (margins for temperature and measurement uncertainty), is statistically based across hundreds or thousands of engineering samples, and test time is not the main concern.
- **Production testing** is very time-critical: the tester is expensive, so minimum test time matters; the tests are based on the worst-case conditions established during characterization, and each IC is tested individually.

6.2 Test Equipment

Mixed-signal IC test equipment (manufacturers include Teradyne, Advantest, Cohu, and National Instruments) consists of the parts shown in Figure 59:

1. The **workstation**: the user computer of the test engineer, used for debugging and day-to-day operation of the tester.
2. The **mainframe**: houses the power supplies and measurement instruments and contains the tester computer (including DSP hardware for fast signal analysis).
3. The **test head**: contains the most sensitive measurement electronics, and carries the **device interface board (DIB)** connecting to the device under test (DUT).

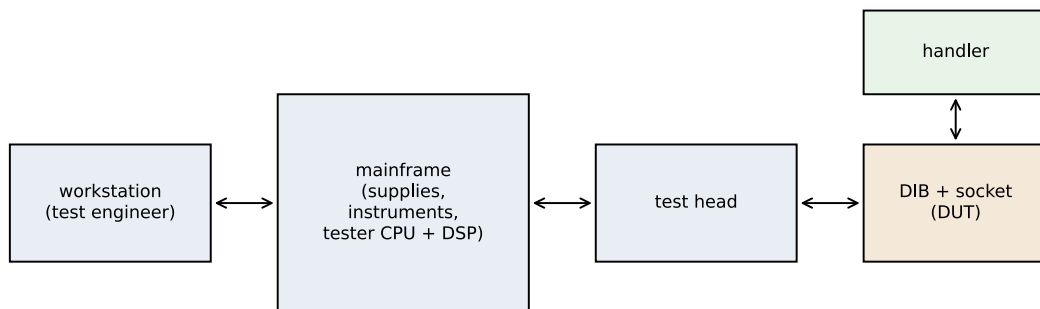


Figure 59: Structure of a mixed-signal automatic test equipment (ATE): the workstation controls the mainframe (power supplies, instruments, tester computer with DSP), which connects to the test head carrying the sensitive measurement frontends and the device interface board (DIB) with the DUT socket; a handler feeds the devices.

The **DIB** is a custom-designed PCB that hosts the package socket (for testing packaged ICs) or connects to the probe card (for wafer testing). It also contains interface circuits that cannot be handled by the tester hardware resources (e.g., special components required by the DUT). Contact to the packaged DUT is made with spring-loaded “pogo pins” in the socket. For wafer probing, different probe-card types are used: **cantilever** probe cards (needles coming in at

an angle), **vertical** (“buckling-beam” or “cobra”) probe cards, and **membrane** probe cards for high pin counts and fine pitches.

Handlers (robotic pick-and-place or gravity-fed) insert the DUT into the test socket on the DIB. After the test, the ICs are sorted into **bins** (good = pass, bad = fail, possibly multiple pass grades). The handler can also provide a temperature-controlled environment for hot/cold testing in addition to room-temperature testing (the latter is preferred for cost reasons).

6.3 Mixed-Signal Testing Challenges

- **Time is money:** A tester plus handler can cost several million dollars, and one second of test time costs 3 to 5 cents. The test program needs a certain time (up to minutes) to cover all relevant functional blocks and critical parameters. Multi-site testing (testing 2/4/8 devices in parallel) reduces the effective test time per device.
- **The test socket introduces parasitics** (R , L , C) which impact performance and make precision measurements difficult.
- **Accuracy** is limited by interference, calibration, and measurement time.
- **Repeatability & reproducibility (R&R)** are required to obtain the same test results across different testers, DIBs, and test sites.

The classification of measurement errors is summarized in Figure 60: accuracy relates the measurement to the true value (via instrument traceability to standards), while precision splits into repeatability and reproducibility, illustrated in Figure 61.

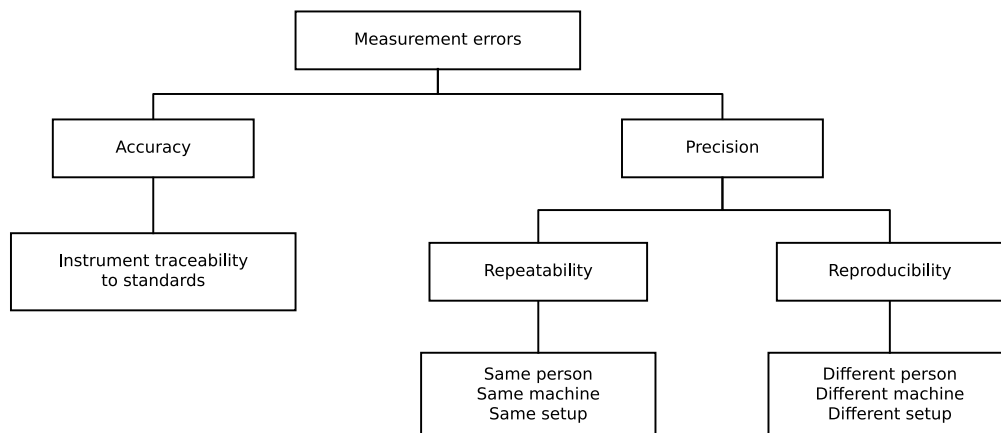


Figure 60: Classification of measurement errors: accuracy is ensured by instrument traceability to standards, while precision splits into repeatability (same person, machine, and setup) and reproducibility (different person, machine, or setup).

Repeatability is the dispersion of measurements when a quantity is measured several times under the same conditions: same measuring instrument, same parts, same operator, within a short time period. It represents the inherent variation of the measurement system—note that measurements may show good repeatability without being accurate! **Reproducibility** is the variation between the *mean* measurements on the same parts when one condition is changed per trial: a different operator, a different instrument, or a different setup.

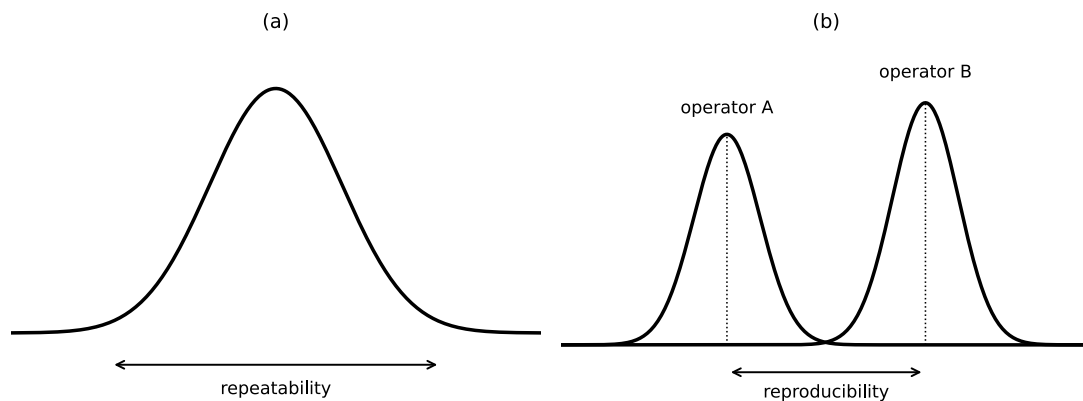


Figure 61: Repeatability versus reproducibility: (a) repeatability is the spread of a single measurement setup; (b) reproducibility is the difference between the mean values obtained by different operators, machines, or setups.

6.4 The Test Plan

The **test plan** is the governing document for generating the test program and contains:

- Device **background information** (in addition to the datasheet).
- Special **test requirements**.
- An explanation of the **purpose of each test**.
- **Assumptions** made for particular tests, and why.
- A **hardware setup** diagram for each test.
- Identification of **critical test parameters**: not all parameters from the datasheet need to be (or can be) tested in production, due to test time or measurement error; such parameters are guaranteed by design and marked accordingly in the datasheet.
- **Binning information**: devices might be binned into pass/fail or different pass grades (e.g., by maximum clock frequency). Continuity fails are binned separately, because they could be caused by contact issues of the handler or socket.

6.5 Types of Tests

6.5.1 Continuity Test

The **continuity test** verifies the contact between the DIB and all pins of the DUT. As essentially all pins have some circuitry connected (at least the ESD protection diodes), a small current is forced into or out of each pin (with a voltage limit to protect the device in case of opens), and the resulting voltage is measured, as sketched in Figure 62:

- A high voltage (running into the voltage limit) means no contact to the IC pin → **open**.
- A very small voltage means a short is detected at the IC pin → **short**.

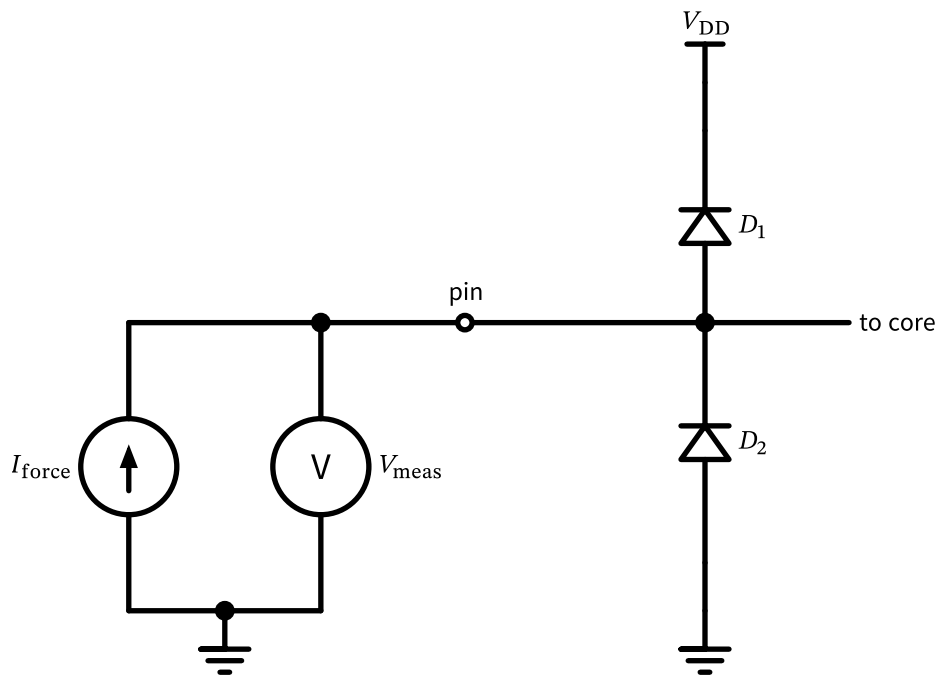


Figure 62: Continuity test: the tester forces a small current into the pin and measures the resulting voltage across the ESD protection diodes of the DUT. A voltage near one diode drop confirms contact; a voltage running into the limit indicates an open, a near-zero voltage a short.

6.5.2 Leakage-Current Test

A non-defective device draws only a small **leakage current** at the power-supply and I/O pins when in off-mode, so a leakage test is a **fast method for detecting failing devices** due to shorts and other misprocessing:

- A voltage is applied to all digital and analog inputs, and the current is measured.
- Digital outputs are set to high-Z (if possible), and the leakage current is measured as for the inputs.
- A voltage is applied to the V_{DD} pins and the leakage current is measured; the device must be put into off-mode by applying the proper initialization (reset or power-on sequence).

6.5.3 Memory Test

Memory cells have to be tested for the following fault types caused by production defects:

- **Stuck-at faults** (stuck-at-0, stuck-at-1) and **transition faults** ($0 \rightarrow 1$ or $1 \rightarrow 0$ not possible).
- **Coupling faults** (writing to one cell changes the content of another cell).
- **Neighborhood pattern-sensitive faults.**

Since on-chip memories can be large, **memory testing can be costly**, requiring a trade-off between fault coverage and test time. Example test methods:

- Zero-one (write all zeros, read all zeros, write all ones, read all ones)— $\mathcal{O}(n)$.
- Checkerboard (write alternating 0101..., read, write the complement 1010..., read)— $\mathcal{O}(n)$.
- Walking 1/0 (consecutively fill the memory with ones/zeros and check)— $\mathcal{O}(n^2)$.
- ROM: read out and compute a checksum/CRC or hash— $\mathcal{O}(n)$.

Applying these RAM tests can be done by external reads/writes (slow), by the internal CPU (faster), or by dedicated **built-in self-test (BIST)** hardware (fastest, at the cost of extra silicon area). Figure 63 shows the structure of a RAM BIST: a controller with pattern and address generators exercises the RAM, and a comparator checks the read-back data.

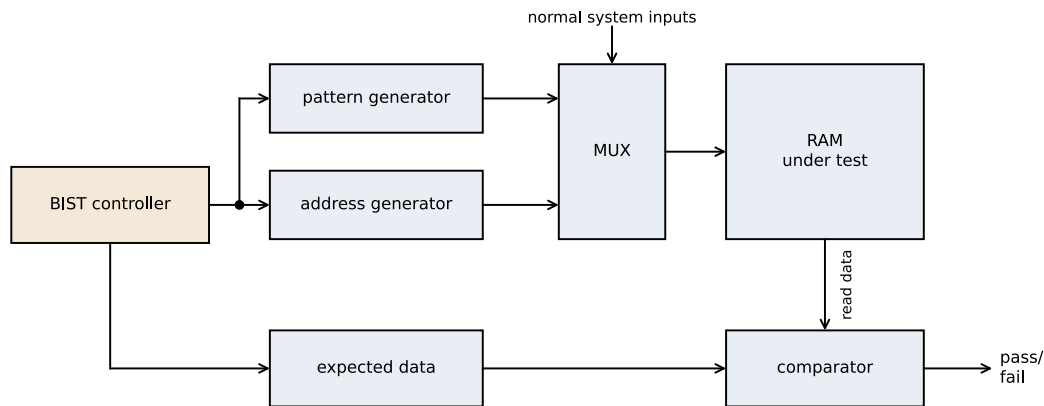


Figure 63: RAM built-in self-test (BIST): a controller sequences a test-pattern generator and an address generator that exercise the RAM through multiplexers; the read-back data is checked in a comparator, yielding a pass/fail result. In mission mode, the multiplexers connect the RAM to the normal system inputs.

6.5.4 JTAG Test and Debug Port

JTAG (IEEE Standard 1149.1) is a standardized test and debug port using serial communication to access IC-internal functional blocks (like the RAM BIST). JTAG uses four or five additional signals: *TDI* (test data in), *TDO* (test data out), *TCK* (test clock), *TMS* (test mode select), and optionally *TRST* (test reset). A two-pin version (cJTAG, IEEE 1149.7) uses only *TMSC* (test serial data) and *TCKC* (test clock).

JTAG was originally introduced to check **interconnection faults on the PCB level**, as solder joints are the main cause of failing boards: all IC pins can be forced to 0 or 1 (or read) via the boundary-scan chain, so all intended PCB connections between ICs can be tested after board manufacturing (see Figure 64). Meanwhile, JTAG is also extensively used to **access internal components of an IC for debugging and test**.

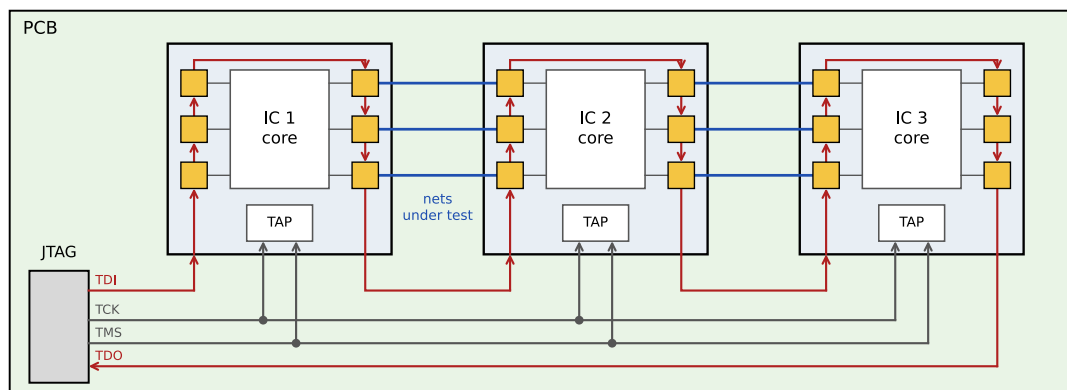


Figure 64: JTAG boundary scan on a PCB: the boundary-scan cells (small squares) at every pin of each IC form a serial shift register; the chain runs from TDI through all devices to TDO, controlled by TCK and TMS. This allows forcing and observing all pins, and thus testing the PCB interconnects between the ICs.

6.5.5 Trimming

If a property of the DUT needs to be **trimmed** (e.g., a reference voltage, an RC product, or an SRAM repair map), this property is measured on the tester. This might require a special test mode, where an internal signal is multiplexed to one of the pins, or an internal calibration circuit is triggered and the result is read out digitally. The resulting **trimming information** is **permanently stored** inside the DUT:

- If EEPROM bits are available, these can be used.
- **Fuses** (e.g., polysilicon or metal wires or minimum-size vias melted open) or **anti-fuses** (e.g., diodes melted into a short) can permanently store a bit by forcing a high current during programming.
- For analog precision circuits, **laser trimming** on the wafer can be used, e.g., to adjust the value of an integrated resistor.

6.5.6 Scan-Path Test

Digital circuits consist (besides RAM and ROM) of combinational logic and state machines. For testing, two properties of a circuit node matter:

- **Observability**: the ease of observing a node by watching the external output pins of the IC.
- **Controllability**: the ease of forcing a node to 0 or 1 by driving the input pins of the IC.

Combinational circuits are controllable and observable, so it is relatively easy to determine test patterns. Sequential circuits, however, have **states**! Finite state machines can be very difficult to test, requiring many cycles to enter a desired state. Beyond a certain size, **exhaustive testing** with all possible input vectors becomes impractical: for a circuit with N inputs and M state bits, 2^{N+M} patterns would be needed. For example, $N = 40$ inputs and 32 states ($M = 5$) at 10 ns per pattern results in roughly 100 hours of test time.

Chip failures (opens, weak or strong shorts to other nodes) can cause very complex failure behavior, which is too time-consuming to test exhaustively—so a simplified **stuck-at model** is used: all failures are assumed to make nodes stuck at 0 (shorted to V_{SS}) or stuck at 1 (shorted to V_{DD}). This simple model is not quite true in reality, but works well enough in practice. Its

limitation is demonstrated in Figure 65: an open in one pull-up branch of a CMOS NAND gate causes a *sequential* effect (the output keeps its previous value for one input combination), which needs two test vectors in the right order to be detected.

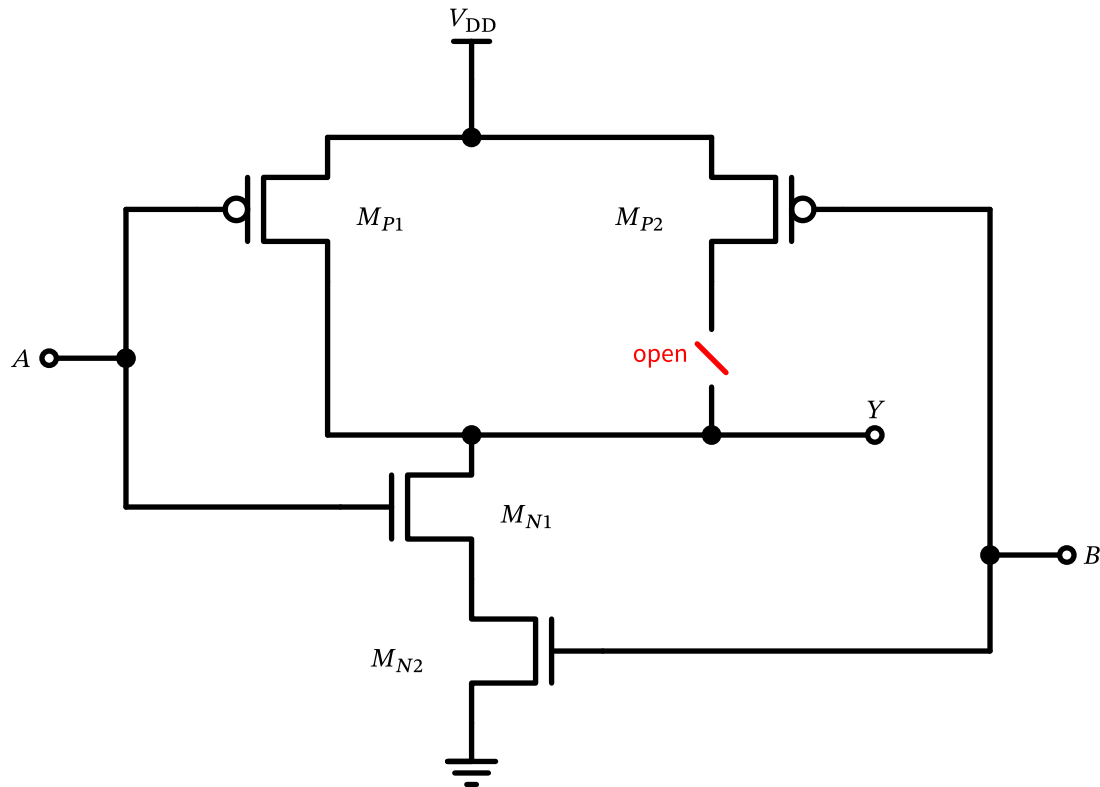


Figure 65: Limitation of the stuck-at fault model: a CMOS NAND gate with an open defect in the pull-up path of B . For $A = 1$, $B = 0$, neither the pull-up nor the pull-down network conducts, so the output floats and keeps its previous value Y_{n-1} —a sequential effect that requires two test vectors in the correct order to detect.

A manufacturing test would ideally check every node in the circuit, applying the smallest sequence of test vectors necessary to prove that each node is not stuck. To increase observability and controllability, i.e., **design for test (DFT)**, the **scan-path test** is introduced, shown in Figure 66:

- Each **flip-flop** is converted into a **scan register** (with a small area overhead): in **mission mode**, the flip-flops behave as usual; in **scan mode**, all flip-flops form a shift register, so their contents can be shifted in and out through dedicated pins.
- If each register can be observed and controlled this way, the test problem reduces to testing the combinational logic between the registers.
- During the production test, test patterns are shifted into the scan chains, one clock cycle is applied, and the captured result is shifted out and compared to the expected result ($\rightarrow$ pass/fail). Digital pattern generators in the ATE hardware accelerate this.
- The patterns are generated automatically by software (ATPG) from the design netlist, achieving high stuck-at fault coverage with a minimum number of patterns.
- To speed up testing further, multiple chains are operated in parallel (requiring test pins for the parallel chain inputs and outputs).

- An extension of scan is to use internal pattern generators (e.g., based on a pseudo-random number generator) as logic inputs, and to compress the outputs into a syndrome (e.g., a transition count or hash) that is compared to the expected syndrome → logic **BIST**.

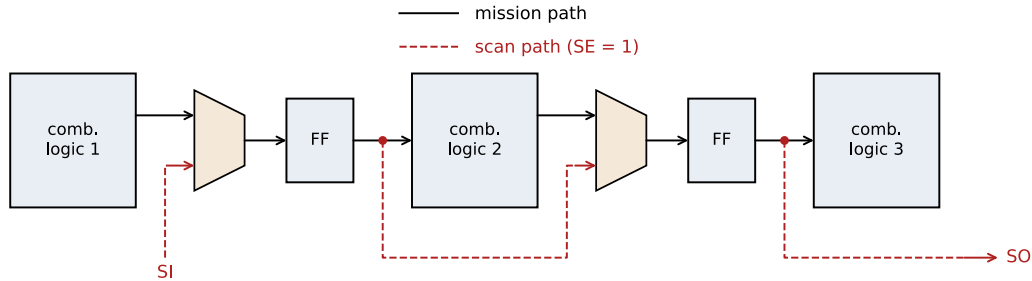


Figure 66: Scan-path test: every flip-flop is replaced by a scan flip-flop with an input multiplexer. In mission mode (scan enable $SE = 0$), the flip-flops capture the outputs of the combinational logic; in scan mode ($SE = 1$), all flip-flops form a shift register from scan-in (SI) to scan-out (SO), making every register controllable and observable.

6.5.7 AC Parametric Tests

Analog and mixed-signal parametric tests are highly dependent on the given function; examples are gain, phase, distortion, rejection (CMRR, PSRR), and noise. The DIB must be designed to reflect the conditions of the IC datasheet, e.g., applying the proper output loading (R and C); different loads for different tests can be switched in via relays. Noise tests can be very time-consuming due to the long averaging needed to detect small signals; additional circuitry on the DIB can help to increase repeatability and speed.

Modern mixed-signal testers use **arbitrary waveform generators (AWGs)** and **digitizers** to create and capture analog signals, with **DSPs** to speed up signal generation and analysis, as shown in Figure 67. Concepts from sampling theory (quantization noise, under/oversampling, aliasing and reconstruction, FFT, windowing, sampling-clock coherence, etc.) apply!

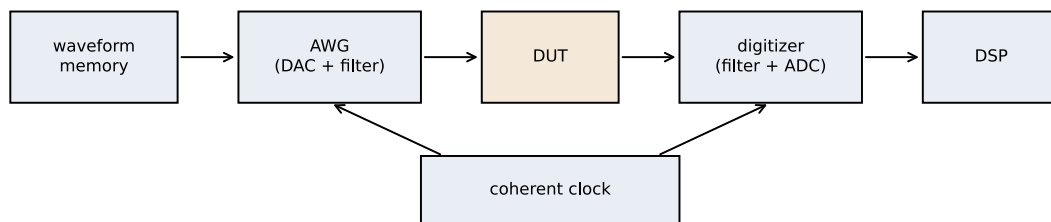


Figure 67: Mixed-signal AC parametric test setup: an arbitrary waveform generator (AWG) synthesizes the stimulus from a pattern memory, the DUT response is captured by a digitizer, and a DSP evaluates the records (FFT, filtering, averaging). A common, coherent clock for AWG and digitizer avoids spectral leakage.

For analog test parameters, the variation over production must stay inside the **test limits**, which are derived from the **datasheet limits minus guard bands**: the lower specification limit (LSL) and the upper specification limit (USL).

6.6 Process Capability Index

To manufacture within a specification, the difference between the USL and the LSL must be greater than the total process variation, and the distribution should be well centered between the limits. One way to check this is the C_p/C_{pk} analysis. Given a parameter distribution x_i from a sample of size N , the arithmetic mean and the standard deviation are

$$\bar{X} = \frac{1}{N} \sum_{i=1}^N x_i \quad \sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{X})^2}$$

Then, the process capability indices are calculated as

$$C_p = \frac{USL - LSL}{6\sigma} \quad (55)$$

$$C_{pk} = \min \left\{ \frac{USL - \bar{X}}{3\sigma}, \frac{\bar{X} - LSL}{3\sigma} \right\} \quad (56)$$

C_p measures whether the process spread fits between the limits at all, while C_{pk} additionally accounts for the centering of the distribution. Ideally, $C_p = C_{pk}$ (the process is well centered), and $C_{pk} > 2$ (which might not be achievable, so lower values like $C_{pk} > 1.5$ may have to be accepted). Figure 68 illustrates well-centered and off-center distributions.

During product qualification, the C_p/C_{pk} values are evaluated on a sufficient sample size, and critical tests are improved (change of datasheet limits, better centering of a parameter by redesign, process centering, etc.).

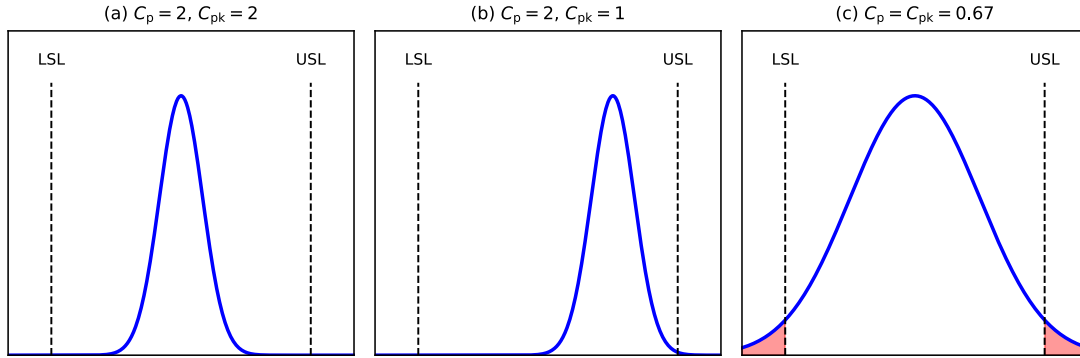


Figure 68: Process capability: (a) a well-centered distribution with small spread ($C_p = C_{pk} = 2$); (b) the same spread but off-center— C_p is unchanged, but C_{pk} drops to 1, indicating parts close to the upper specification limit; (c) a centered but wide distribution, where both indices are poor and yield is lost at both limits.

The advantage of using the C_p/C_{pk} analysis is that it provides a *quantitative* measure of both the process spread and the centering of the distribution relative to the specification limits. This helps in automatically identifying whether the process is capable of consistently producing parts within the desired specifications and whether any adjustments are needed to improve yield and quality with just two numbers: C_p and C_{pk} . Manual inspection of the distribution alone may not reveal subtle misalignments or variations that could impact production quality and can be extremely time consuming and error prone.

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